
UNITED STATES SECURITIES AND EXCHANGE COMMISSION

Washington, DC 20549

Form 10-K

(MARK ONE)

☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the fiscal year ended October 30, 2022

OR

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the transition period from to

Broadcom Inc.

Delaware
(State or Other Jurisdiction of
Incorporation or Organization)

1320 Ridder Park Drive
San Jose, CA 95131-2313
(408) 433-8000

001-38449
(Commission File
Number)

35-2617337
(I.R.S. Employer
Identification No.)

(Exact Name of Registrant as Specified in Its Charter
Address of Principal Executive Offices, Including Zip Code
Registrant's Telephone Number, Including Area Code)

Securities registered pursuant to Section 12(b) of the Act:

Title of Each Class	Trading Symbol(s)	Name of Each Exchange on Which Registered
Common Stock, \$0.001 par value	AVGO	The NASDAQ Global Select Market

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer ☒ Accelerated filer ☐ Non-accelerated filer ☐ Smaller reporting company ☐ Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

The aggregate market value of voting and non-voting common equity held by non-affiliates as of April 29, 2022, based upon the closing sale price of such shares on The Nasdaq Global Select Market on such date was approximately \$220.1 billion.

As of November 25, 2022, there were 417,886,140 shares of our common stock outstanding.

Documents Incorporated by Reference

Portions of the registrant's definitive Proxy Statement for its 2023 Annual Meeting of Stockholders are incorporated by reference into Part III of this Annual Report on Form 10-K.

BROADCOM INC.
2022 ANNUAL REPORT ON FORM 10-K

TABLE OF CONTENTS

	<u>Page</u>
<u>PART I.</u>	
ITEM 1. BUSINESS	3
ITEM 1A. RISK FACTORS	13
ITEM 1B. UNRESOLVED STAFF COMMENTS	32
ITEM 2. PROPERTIES	32
ITEM 3. LEGAL PROCEEDINGS	32
ITEM 4. MINE SAFETY DISCLOSURES	32
<u>PART II.</u>	
ITEM 5. MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES	33
ITEM 6. [RESERVED]	34
ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS	35
ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK	46
ITEM 8. FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA	47
ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE	89
ITEM 9A. CONTROLS AND PROCEDURES	89
ITEM 9B. OTHER INFORMATION	90
ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS	90
<u>PART III.</u>	
ITEM 10. DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE	91
ITEM 11. EXECUTIVE COMPENSATION	91
ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS	91
ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE	91
ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES	91
<u>PART IV.</u>	
ITEM 15. EXHIBITS AND FINANCIAL STATEMENT SCHEDULES	92
ITEM 16. FORM 10-K SUMMARY	100
SIGNATURES	101

PART I

The following discussion should be read in conjunction with the consolidated financial statements and notes thereto included elsewhere in this Annual Report on Form 10-K. This Annual Report on Form 10-K contains forward-looking statements within the meaning of the federal securities laws and particularly in Item 1: “Business,” Item 1A: “Risk Factors,” Item 3: “Legal Proceedings” and Item 7: “Management’s Discussion and Analysis of Financial Condition and Results of Operations” of this Annual Report on Form 10-K. These statements are indicated by words or phrases such as “anticipate,” “expect,” “estimate,” “seek,” “plan,” “believe,” “could,” “intend,” “will,” and similar words or phrases. These forward-looking statements may include projections of financial information; statements about historical results that may suggest trends for our business; statements of the plans, strategies, and objectives of management for future operations; statements of expectation or belief regarding future events (including any acquisitions we may make), technology developments, our products, product sales, expenses, liquidity, cash flow and growth rates, or enforceability of our intellectual property rights; any backlog; and the effects of seasonality on our business. Such statements are based on current expectations, estimates, forecasts and projections of our industry performance and macroeconomic conditions, based on management’s judgment, beliefs, current trends and market conditions, and involve risks and uncertainties that may cause actual results to differ materially from those contained in the forward-looking statements. We derive most of our forward-looking statements from our operating budgets and forecasts, which are based upon many detailed assumptions. While we believe that our assumptions are reasonable, we caution that it is very difficult to predict the impact of known factors, and it is impossible for us to anticipate all factors that could affect our actual results. Accordingly, we caution you not to place undue reliance on these statements. Material factors that could cause actual results to differ materially from our expectations are summarized and disclosed under “Risk Factors” in Part I, Item 1A of this Annual Report on Form 10-K.

Unless stated otherwise or the context otherwise requires, references to “Broadcom,” “we,” “our,” and “us” mean Broadcom Inc. and its consolidated subsidiaries. Our fiscal year ends on the Sunday closest to October 31 in a 52-week year and the first Sunday in November in a 53-week year. We refer to our fiscal years by the calendar year in which they end. For example, the fiscal year ended October 30, 2022 was a 52-week year.

ITEM 1. BUSINESS

Overview

We are a global technology leader that designs, develops and supplies a broad range of semiconductor and infrastructure software solutions. Our over 50-year history of innovation dates back to our diverse origins from Hewlett-Packard Company, AT&T, LSI Corporation, Broadcom Corporation, Brocade Communications Systems LLC, CA, Inc. and Symantec Enterprise Security. Over the years, we have assembled a large team of semiconductor and software design engineers around the world. We maintain design, product and software development engineering resources at locations in the U.S., Asia, Europe and Israel, providing us with engineering expertise worldwide. We strategically focus our research and development resources to address niche opportunities in our target markets and leverage our extensive portfolio of U.S. and other patents, and other intellectual property (“IP”) to integrate multiple technologies and create system-on-chip (“SoC”) component and software solutions that target growth opportunities. We design products and software that deliver high-performance and provide mission critical functionality.

We develop semiconductor devices with a focus on complex digital and mixed signal complementary metal oxide semiconductor (“CMOS”) based devices and analog III-V based products. We have a history of innovation in the semiconductor industry and offer thousands of products that are used in end products such as enterprise and data center networking, home connectivity, set-top boxes (“STB”), broadband access, telecommunication equipment, smartphones and base stations, data center servers and storage systems, factory automation, power generation and alternative energy systems, and electronic displays. We differentiate ourselves through our high performance design and integration capabilities and focus on developing products for target markets where we believe we can earn attractive margins.

Our infrastructure software solutions enable customers to plan, develop, automate, manage, and secure applications across mainframe, distributed, mobile, and cloud platforms. Many of the largest companies in the world, including most of the Fortune 500, and many government agencies rely on our software solutions to help manage and secure their on-premise and hybrid cloud environments. Our portfolio of industry-leading infrastructure and security software is designed to modernize, optimize, and secure the most complex hybrid environments, enabling scalability, agility, automation, insights, resiliency and security. We also offer mission critical fibre channel storage area networking (“FC SAN”) products and related software in the form of modules, switches and subsystems incorporating multiple semiconductor products.

Business Strategy

Our strategy is to combine best-of-breed technology leadership in semiconductor and infrastructure software solutions, with unmatched scale, on a common sales and administrative platform to deliver a comprehensive suite of infrastructure technology products to the world’s leading business and government customers. We seek to achieve this through responsibly financed acquisitions of category-leading businesses and technologies, as well as investing extensively in research and development, to ensure our products retain their technology leadership. This strategy results in a robust business model designed to drive diversified and sustainable operating and financial results.

Products and Markets

Semiconductor Solutions

Semiconductors are made by imprinting a network of electronic components onto a semiconductor wafer. These devices are designed to perform various functions such as processing, amplifying and selectively filtering electronic signals, controlling electronic system functions and processing, and transmitting and storing data. Our digital and mixed signal products are based on silicon wafers with CMOS transistors offering fast switching speeds and low power consumption, which are both critical design factors for the markets we serve. We also offer analog products, which are based on III-V semiconductor materials that have higher electrical conductivity than silicon, and thus tend to have better performance characteristics in radio frequency (“RF”), and optoelectronic applications. III-V refers to elements from the 3rd and 5th groups in the periodic table of chemical elements. Examples of these materials used in our products are gallium arsenide (“GaAs”) and indium phosphide (“InP”).

We provide semiconductor solutions for managing the movement of data in data center, service provider, enterprise and embedded networking applications. We provide a broad variety of RF semiconductor devices, wireless connectivity solutions, custom touch controllers and inductive charging solutions for the wireless market. We also provide semiconductor solutions for enabling the STB and broadband access applications and for enabling secure movement of digital data to and from host machines, such as servers, personal computers and storage systems, to the underlying storage devices, such as hard disk drives (“HDD”) and solid-state drives (“SSD”).

Our product portfolio ranges from discrete devices to complex sub-systems that include multiple device types and may also incorporate firmware for interfacing between analog and digital systems. In some cases, our products include mechanical hardware that interfaces with optoelectronic or capacitive sensors. We focus on markets that require high quality and the

[Table of Contents](#)

technology leadership and integrated performance characteristic of our products. The table below presents our material semiconductor product families and their major end markets and applications during fiscal year 2022.

<u>Major End Markets</u>	<u>Major Applications</u>	<u>Material Product Families</u>
Broadband	STB and Broadband Access	- STB SoCs - DSL/PON gateways - DOCSIS cable modem - DSLAM/PON optical line termination - Wi-Fi access point SoCs
Networking	Data Center, Service Provider, Enterprise and Embedded Networking	- Ethernet switching and routing merchant silicon - Embedded processors and controllers - Custom silicon solutions - Optical and copper PHYs - Fiber optic transmitter and receiver components
Wireless	Mobile Device Connectivity	- RF front end modules and filters - Wi-Fi, Bluetooth, GPS/GNSS SoCs - Custom touch controllers - Inductive charging ASICs
Storage	- Servers and Storage Systems - HDD and SSD	- SAS and RAID controllers and adapters - PCIe switches - Fibre channel host bus adapters - Ethernet NIC - Read channel based SoCs; Custom flash controllers - Preamplifiers
Industrial	Factory Automation, Renewable Energy and Automotive Electronics	- Optocouplers - Industrial fiber optics - Motion control encoders and subsystems - Light emitting diode - Ethernet PHYs, switch ICs and camera microcontrollers

Set-Top Box Solutions: We offer complete SoC platform solutions for cable, satellite, Internet Protocol television, over-the-top and terrestrial STBs. Our products enable global service providers to introduce new and enhanced technologies and services in STBs, including transcoding, digital video recording functionality, higher definition video processing, increased networking capabilities, and more tuners to enable faster channel change and more simultaneous recordings. We are also enabling service providers in deploying High Efficiency Video Coding (“HEVC”), a video compression format that is a successor to the H.264/MPEG-4 format. HEVC enables ultra-high definition (“Ultra HD”), services by effectively doubling the capacity of existing networks to deploy new or existing content. Our families of STB solutions support the complete range of resolutions, from standard definition, to high definition, and Ultra HD.

Broadband Access Solutions: We offer complete SoC platform solutions for digital subscriber line (“DSL”), cable, passive optical networking (“PON”) and wireless local area network for both consumer premise equipment (“CPE”) and central office (“CO”) deployments. Our CPE devices are used in broadband modems, residential gateways and Wi-Fi access points and routers. Our CO devices, including DSL Access Multiplexer (“DSLAM”), cable modem termination systems and PON optical line termination medium access controller, are empowering modern operator broadband infrastructure. Our products enable global service providers to continue to deploy next generation broadband access technologies across multiple standards, including G.fast, Data Over Cable Service Interface Specifications (“DOCSIS”), PON and Wi-Fi to provide more bandwidth and faster speeds to consumers.

Ethernet Switching & Routing: Ethernet is a ubiquitous interconnection technology that enables high performance and cost effective networking infrastructure. We offer a broad set of Ethernet switching and routing products that are optimized for data center, service provider network, enterprise network, and embedded network applications. In the data center market, our high capacity, low latency, switching silicon supports advanced protocols around virtualization and multi-pathing. Our Ethernet switching fabric technologies provide the ability to build highly scalable flat networks supporting tens of thousands of servers. Our service provider switch portfolio enables carrier/service provider networks to support a large number of services in the wireless backhaul, access, aggregation and core of their networks. For enterprise networks and embedded Ethernet applications, we offer product families that combine multi-layer switching capabilities and support lower power modes that comply with industry standards around energy efficient Ethernet.

Embedded Processors & Controllers: Our embedded processors leverage our ARM central processing unit and Ethernet switching technology to deliver SoCs for high performance embedded applications in a wide range of communication products such as voice-over-internet-protocol, telephony, point-of-sale devices and enterprise and retail access points and gateways. We offer a range of knowledge-based processors to enable high-performance decision-making for packet processing in a variety of advanced devices in the enterprise, metro, access, edge and core networking spaces. We also offer a range of Ethernet controllers for servers and storage systems supporting multiple generations of Ethernet technology.

Custom Silicon Solutions: We provide advanced technology and IP platforms for customers to design and develop application specific integrated circuits ("ASICs"), targeting data center compute offload, legacy and new 5G radio infrastructure, and wired communication networks. Our custom silicon provides the platform to integrate embedded logic, memory, serializer/deserializer ("SerDes") technology, IP cores and processor cores. The ASICs are custom products built to individual customers specifications.

Physical Layer Devices: These devices, also referred to as PHYs, are transceivers that enable the reception and transmission of Ethernet data packets over a physical medium such as copper wire or optical fibers. Our high performance Ethernet transceivers are built upon a proprietary digital signal processing communication architecture optimized for high-speed network connections and support the latest standards and advanced features, such as energy efficient Ethernet, data encryption and time synchronization. We also offer a range of automotive Ethernet products, including PHYs, switches and camera microcontrollers, to meet growing consumer demand for in-vehicle connectivity and smart vision.

Fiber Optic Components: We supply a wide array of optical components to the Ethernet networking, storage, and access, metro- and long-haul telecommunication markets. Our optical components enable the high speed reception and transmission of data through optical fibers.

RF Semiconductor Devices: Our RF semiconductor devices selectively filter, as well as amplify and route, RF signals. Filters enable modern wireless communication systems to support a large number of subscribers simultaneously by ensuring that the multiple transmissions and receptions of voice and data streams do not interfere with each other. We were among the first to deliver commercial film bulk acoustic resonator ("FBAR") filters that offer technological advantages over competing filter technologies, to allow mobile handsets to function more efficiently in today's congested RF spectrum. FBAR technology has a significant market share within the cellular handset market. Our RF products include multi-chip module front-end modules that integrate transmit/receive switching and filtering functions for multiple frequency bands, filter modules and discrete filters, all using our proprietary FBAR technology.

Our expertise in FBAR technology, amplifier design, and module integration enables us to offer industry-leading performance in cellular RF transceiver applications.

Connectivity Solutions: Our connectivity solutions include discrete and integrated Wi-Fi and Bluetooth solutions, and global positioning system/global navigation satellite system ("GPS/GNSS") receivers, designed for use in mobile devices including smartphones, tablets and wearable products.

Wi-Fi allows devices on a local area network to communicate wirelessly, adding the convenience of mobility to the utility of high-speed data networks. We offer a family of high performance, low power Wi-Fi chipsets. Bluetooth is a low power technology that enables direct connectivity between devices. We offer a complete family of Bluetooth silicon and software solutions that enable manufacturers to easily and cost-effectively add Bluetooth functionality to virtually any device. These solutions include combination chips that offer integrated Wi-Fi and Bluetooth functionality, which provides significant performance advantages over discrete solutions.

We also offer a family of GPS, assisted-GPS and GNSS semiconductor products, software and data services. These products are part of a broader location platform that leverages a broad range of communications technologies, including Wi-Fi, Bluetooth and GPS, to provide more accurate location and navigation capabilities.

Custom Touch Controllers: Our touch controllers process signals from touch screens in mobile handsets and tablets.

Inductive Charging ASICs: Our custom inductive charging ASIC devices offer high efficiency and are highly integrated solutions for mobile and wearable devices.

SAS, RAID & PCIe Products: We provide serial attached small computer system interface ("SAS") and redundant array of independent disks ("RAID") controller and adapter solutions to server and storage system original equipment manufacturers ("OEMs"). These solutions enable secure and high speed data transmission between a host computer, such as a server, and storage peripheral devices, such as HDD, SSD and optical disk drives and disk and tape-based storage systems. Some of these solutions are delivered as stand-alone semiconductors, typically as a controller. Other solutions are delivered as circuit boards, known as adapter products, which incorporate our semiconductors onto a circuit board with other features. RAID technology is a critical part of our server storage connectivity solutions as it provides protection against the loss of critical data resulting from HDD failures.

We also provide interconnect semiconductors that support the peripheral component interconnect express (“PCIe”) communication standards. PCIe is the primary interconnection mechanism inside computing systems today.

Fibre Channel Products: We provide fibre channel host bus adapters, which connect host computers such as servers to FC SANs.

Ethernet Network Interface Card (“NIC”) Controllers: Our Ethernet NIC controllers are designed for high-performance virtualization, intelligent flow processing, secure data center connectivity, and machine learning.

HDD & SSD Products: We provide read channel-based SoCs and preamplifiers to HDD OEMs. These are the critical chips required to read, write and protect data. An HDD SoC is an integrated circuit (“IC”) that combines the functionality of a read channel, serial interface, memory and a hard disk controller in a small, high-performance, low-power and cost-effective package. Read channels convert analog signals that are generated by reading the stored data on the physical media into digital signals.

In addition, we sell preamplifiers, which are complex, high speed, mixed signal devices that enable writing and reading data to and from the HDD heads. The preamplifier interfaces with the SoC to provide the electronics data path in a HDD.

We also provide custom flash controllers to SSD OEMs. An SSD stores data in flash memory instead of on a hard disk, providing high speed access to the data. Flash controllers manage the underlying flash memory in SSDs, performing critical functions such as reading and writing data to and from the flash memory and performing error correction, wear leveling and bad block management.

Industrial End Markets: We also provide a broad variety of products for the general industrial and automotive markets, including optocouplers, industrial fiber optics, motion encoders, light emitting diode devices, and Ethernet ICs. Our industrial products are used in a diverse set of applications, spanning industrial automation, power generation and distribution systems, medical systems and equipment, defense and aerospace, and vehicle subsystems including those used in electric vehicle powertrain, infotainment and advanced driver assistance system.

Infrastructure Software

Our infrastructure software solutions enables customers greater choice and flexibility to build, run, manage, connect and protect applications at scale across diversified and distributed environments.

Our mainframe software provides market-leading DevOps, AIOps, Security and Data Management Systems solutions. We help enterprises embrace open tools and technologies, integrate their mainframe into their cloud infrastructures, and amplify the value of their mainframe investments. By partnering with our customers and providing creative value-added programs, we help customers overcome challenges related to skills development, technical education, strategy and planning, and the need for cloud-like pricing flexibility to support their overall business success with the platform.

Our distributed software solutions enable global enterprises to optimize the planning, development and delivery of software, powering their business critical digital services. Our solutions are designed to enable customers to innovate, improve customer experience, and drive profitability by aligning business, development, and operational teams. Our products, organized in the domains of ValueOps, DevOps, and AIOps, deliver end-to-end visibility across all stages of the digital lifecycle and help our customers realize better business outcomes and better experiences for their customers.

Our Symantec cyber security software solutions help organizations and governments secure against threats and compliance risks by protecting their users and data on any app, device, or network. Our integrated cyber defense approach simplifies cyber security with comprehensive solutions designed to secure critical business assets across on-premises and cloud infrastructures. Our Symantec solutions utilize rich threat intelligence from a global network of security engineers, threat analyst and researchers, as well as advanced artificial intelligence (“AI”) and machine-learning engines, enabling customers to protect data, connect authorized users with trusted applications, and detect and respond to the most advanced targeted attacks.

We also offer mission critical FC SAN products designed to help customers reduce the cost and complexity of managing business information within a shared data storage environment, enabling high levels of availability of mission critical applications in the form of modules, switches and subsystems incorporating multiple semiconductor products. We deliver reliable and simplified management of these FC SAN products through our software-based management tools designed to maximize uptime, dramatically simplify storage area networking deployment and management, and provide high levels of visibility and insight into the storage network.

The table below presents our software portfolios and their material offerings during fiscal year 2022.

<u>Software Portfolio</u>	<u>Portfolio Description</u>	<u>Major Portfolio Offerings</u>
Mainframe Software	Solutions for DevOps, AIOps, Security and Database Management Systems	- Operational Analytics & Management - Automation - Database & Database Management - Application Development & Testing - Identity & Access Management - Compliance & Data Protection - Security Insights
Distributed Software	Solutions that optimize the planning, development and delivery of business critical services	- ValueOps - DevOps - AIOps
Symantec Cyber Security	Comprehensive threat protection and compliance solutions that secure against threats and compliance risks by protecting users and data on any app, device, or network	- Endpoint Security - Network Security - Information Security - Identity Security
FC SAN Management	Solutions that transforms current storage networks with autonomous SAN capabilities	Fibre Channel Switch
Payment Security	Arcot payment authentication network powered by 3-D Secure	Payment Security Suite

Operational Analytics & Management: These solutions combine big data, machine learning and AI with mainframe expertise to deliver meaningful and actionable insights to augment and automate day-to-day operations and deliver exceptional customer experiences.

Automation: These solutions reduce manual effort by enabling customers to proactively optimize resources and orchestrate automation across enterprise applications and systems.

Databases & Database Management: These high-performance databases and management tools store, organize, and manage mainframe data to ensure optimal performance, efficient administration, and reliability of critical systems.

Application Development & Testing: These solutions enable customers to accelerate software delivery while increasing code quality through the use of our agile processes and tools, and DevOps solutions. Our open-first strategy helps customers modernize their mainframe environment through the use of open source and open application programming technologies across people, process, tooling and applications, resulting in greater synergy and alignment with their corporate information technology ("IT").

Identity & Access Management: These solutions manage mainframe access and elevate it with modern practices such as multi-factor authentication, managing access for privileged users, and supporting all external security managers.

Compliance & Data Protection: These solutions locate and protect sensitive mainframe data to ensure compliance and identify risk, identify and proactively respond to potential risks and bad actors, and reduce risk and lighten security management load with automated identification and authorization cleanup.

Security Insights Platform: This solution helps ensure a trusted environment for customers and their employees by quickly interpreting and assessing mainframe security posture, identifying risks and developing remediation steps on an ongoing and ad hoc basis. This data is available for use with in-house tools for security information and event management.

ValueOps: This solution delivers value stream management capabilities that enable customers to schedule, track, and manage work throughout its lifecycle from investment planning to execution. It aligns business and development teams across the enterprise, increasing transparency, reducing inefficiencies, and improving time to value.

DevOps: This solution offers capabilities that empower users of our agile processes and tools to track development progress and deploy releases confidently with assurance of feature completeness, high-quality and reduced risk. Key stakeholders have a single view of key insights into release progress, health, quality, and defect trends, and metrics that drive focus, gauge readiness, and help to ensure successful, quality releases.

AIOps: This solution combines application, infrastructure and network monitoring and correlation with intelligent remediation capabilities to help customers create more resilient production environments and improve customer experience.

Endpoint Security: Endpoints are the critical last line of defense against cyber attackers. Our Symantec endpoint security solutions prevent, detect and respond to emerging threats across all devices and operating systems including laptops, desktops, tablets, mobile phones, servers and cloud workloads through an intelligent AI driven security console and single agent.

Network Security: Email and web access are the lifeblood and essential communication means for every modern organization. We have a full array of network security solutions, as well as a shared set of advanced threat protection technologies to stop inbound and outbound threats targeting end users, information and key infrastructure.

Information Security: Information protection and compliance is critical to managing risk. We offer integrated information security solutions, based on an efficient, single-policy that can be applied across the entire environment, to help organizations identify and protect risky users, applications and their most sensitive data everywhere across endpoints, on-premises networks, cloud services and private applications.

Identity Security: User identities are under attack by cyber criminals hoping to exploit their access and privileges and do harm. We mitigate these attacks by positively identifying legitimate users, enforcing granular access control policies, and streamlining access governance to prevent unauthorized access to sensitive resources and data.

Fibre Channel Switch Products: Our Brocade Fibre Channel switch products provide interconnection, bandwidth and high-speed switching between servers and storage devices which are in a FC SAN. FC SANs are networks dedicated to mission critical storage traffic, and enable simultaneous high speed and secure connections among multiple host computers and multiple storage arrays.

Payment Security Suite: This is a software as a service ("SaaS")-based payment authentication service to help banks and merchants protect against fraud and ensure a hassle-free online shopping experience for their customers.

Research and Development

We are committed to continuous investment in product development and enhancement, with a focus on rapidly introducing new, proprietary products and releases. Many of our products have grown out of our own research and development efforts, and have given us competitive advantages in certain target markets due to performance differentiation. However, we opportunistically seek to enhance our capabilities through the acquisition of engineers with complementary research and development skills and complementary technologies and businesses. We focus our research and development efforts on the development of mission critical, innovative, sustainable and higher value product platforms and those that improve the quality and stability in our broadly deployed products. We leverage our design capabilities in markets where we believe our innovation and reputation will allow us to earn attractive margins by developing high value-add products.

We plan to continue investing in product development, both organically and through acquisitions, to drive growth in our business. We also invest in process development and improvements to product features and functions, as well as fabrication capabilities to optimize processes for devices that are manufactured internally. Our field application engineers, design engineers, and product and software development engineers are located in many places around the world, and in many cases near our top customers. This enhances our customer reach and our visibility into new product opportunities and, in the case of our semiconductor customers, enables us to support our customers in each stage of their product development cycle, from the early stages of production design to volume manufacturing and future growth. By collaborating with our customers, we have opportunities to develop high value-added, customized products for them that leverage our existing technologies. We anticipate that we will continue to make significant research and development expenditures in order to maintain our competitive position, and to ensure a continuous flow of innovative and sustainable product platforms.

Customers, Sales and Distribution

We sell our products through our direct sales force and a select network of distributors and channel partners globally. Distributors and OEMs, or their contract manufacturers, typically account for the substantial majority of our semiconductor sales. A relatively small number of customers account for a significant portion of our net revenue. Sales to distributors accounted for 56% and 53% of our net revenue for fiscal years 2022 and 2021, respectively. We believe aggregate sales to our top five end customers, through all channels, accounted for approximately 35% of our net revenue for each of our fiscal years 2022 and 2021. We believe aggregate sales to Apple Inc., through all channels, accounted for approximately 20% of our net revenue for each of fiscal years 2022 and 2021. We expect to continue to experience significant customer concentration in future periods. The loss of, or significant decrease in demand from, any of our top five end customers could have a material adverse effect on our business, results of operations and financial condition.

Many of our semiconductor customers design products in North America or Europe that are then manufactured in Asia. To serve customers around the world, we have strategically developed relationships with large global electronic component distributors, complemented by a number of regional distributors with customer relationships based on their respective product ranges. We also sell our products to a wide variety of OEMs or their contract manufacturers. We have established

strong relationships with leading OEM customers across multiple target markets. Our direct sales force focuses on supporting our large OEM customers, and has specialized product and service knowledge that enables us to sell specific offerings at key levels throughout a customer's organization. Certain customers require us to contract with them directly and with specified intermediaries, such as contract manufacturers. Many of our major customer relationships have been in place for many years and are often the result of years of collaborative product development. This has enabled us to build our extensive IP portfolio and develop critical expertise regarding our customers' requirements, including substantial system-level knowledge. This collaboration has provided us with key insights into our customers' businesses and has enabled us to be more efficient and productive and to better serve our target markets and customers. Many of our customers and their contract manufacturers often require time critical delivery of our products to multiple locations around the world. With sales offices located in various countries, our primary warehouse in Malaysia, and dedicated regional customer support call centers, where we address customer issues and handle logistics and other order fulfillment requirements, we believe we are well-positioned to support our customers throughout the design, technology transfer and manufacturing stages across all geographies.

Our software customers are in most major industries worldwide, including banks, insurance companies, other financial services providers, government agencies, global IT service providers, telecommunication providers, transportation companies, manufacturers, technology companies, retailers, educational organizations and health care institutions. Our customers generally consist of large enterprises that have computing environments from multiple vendors and are highly complex. We remain focused on strengthening relationships and increasing penetration within our existing core, mainframe-centric, and Symantec endpoint customers and expanding the adoption of our enterprise software offerings with these customers. We believe our enterprise-wide license model will continue to offer our customers reduced complexity, more flexibility and an easier renewal process that will help drive revenue growth.

Manufacturing Operations

We focus on maintaining an efficient global supply chain and a variable, low-cost operating model. Accordingly, we outsource a majority of our manufacturing operations, utilizing third-party foundry and assembly and test capabilities, as well as some of our corporate infrastructure functions. The majority of our front-end wafer manufacturing operations is outsourced to external foundries, including Taiwan Semiconductor Manufacturing Company Limited ("TSMC"). We use third-party contract manufacturers for a significant majority of our assembly and test operations, including Advanced Semiconductor Engineering, Inc., Foxconn Technology Group, Amkor Technology, Inc. and Siliconware Precision Industries Co., Ltd. We use our internal fabrication facilities for products utilizing our innovative and proprietary processes, such as our FBAR filters for wireless communications and our vertical-cavity surface emitting laser and side emitting lasers-based on GaAs and InP lasers for fiber optic communications, while outsourcing commodity processes such as standard CMOS. By doing so, we can protect our IP and accelerate time to market for our products. The majority of our internal III-V semiconductor wafer fabrication is done in the U.S. and Singapore.

We also have a long history of operating in Asia where we manufacture and source the majority of our products and materials. We store the majority of our product inventory in our warehouse in Malaysia and our presence in Asia places us in close proximity to many of our customers' manufacturing facilities.

Manufacturing Materials and Suppliers

Our manufacturing operations employ a wide variety of semiconductors, electromechanical components and assemblies and raw materials. We purchase materials from hundreds of suppliers on a global basis. These purchases are generally on a purchase order basis and some parts are not readily available from alternate suppliers due to their unique design or the length of time and cost necessary for re-design or qualification. To address the potential disruption in our supply chain, we may use a number of techniques, including redesigning products for alternative components, making incremental or "lifetime" purchases, or qualifying more than one source of supply. Our long-term relationships with our suppliers allow us to proactively manage our technology development and product discontinuance plans, and to monitor our suppliers' financial health. Some suppliers may, nonetheless, extend their lead times, limit supplies, increase prices or cease to produce necessary parts for our products. If these are unique or highly specialized components, we may not be able to find a substitute quickly, or at all.

Competition

The markets in which we participate are highly competitive. Our competitors range from large, international companies offering a wide range of products to smaller companies specializing in narrow markets. The competitive landscape is changing as a result of a trend toward consolidation within many industries, as some of our competitors have merged with or been acquired by other competitors, while others have begun collaborating with each other. We expect this consolidation trend to continue. We expect competition in the markets in which we participate to continue to increase as existing competitors improve or expand their product offerings and as new companies enter the market. Additionally, our ability to compete effectively depends on a number of factors, including: quality, technical performance, price, product features, product system

[Table of Contents](#)

compatibility, system-level design capability, engineering expertise, responsiveness to customers, new product innovation, product availability, delivery timing and reliability, and customer sales and technical support.

In the semiconductor market, we compete with integrated device manufacturers, fabless semiconductor companies, as well as the internal resources of large, integrated OEMs. Our primary competitors are Amlogic Inc., Analog Devices, Inc., Advanced Micro Devices, Inc., Cisco Systems, Inc., Wolfspeed, Inc. (f/k/a Cree, Inc.), GlobalFoundries Inc., Hamamatsu Photonics K.K., Heidenhain Corporation, HiSilicon Technologies Co. Ltd., iC-Haus GmbH, Intel Corporation, Lumentum Holdings Inc., MACOM Technology Solutions Holdings, Inc., MaxLinear, Inc., Marvell Technology, Inc., MediaTek Inc., NVIDIA Corporation, Microchip Technology Incorporated, Mitsubishi Electric Corporation, Murata Manufacturing Co., Ltd., NXP Semiconductors N.V., ON Semiconductor Corporation, OSRAM Licht AG, Qorvo, Inc., Qualcomm Inc., Realtek Semiconductor Corp., Renesas Electronics Corporation, Skyworks Solutions, Inc., STMicroelectronics N.V., Sumitomo Corporation, Synaptics Incorporated, TDK-EPC Corporation, Toshiba Corporation, Texas Instruments, Inc. and II-VI Incorporated. We compete based on the strength and expertise of our high speed proprietary design expertise, FBAR technology, amplifier design, module integration, proprietary materials processes, multiple storage protocols and mixed-signal design, our broad product portfolio, support of key industry standards, reputation for quality products, and our customer relationships.

In the infrastructure software market, we compete with large enterprise software vendors who continue to expand their product and service offerings and consolidate offerings into broad product lines, and smaller, niche players focused on specific markets. Our primary competitors are Atlassian Corporation, Plc, BMC Software Inc., BeyondTrust Corporation, Cisco Systems, Inc., CrowdStrike Holdings, Inc., CyberArk Software, Ltd., International Business Machines Corporation, Micro Focus International plc, Microsoft Corporation, New Relic, Inc., Oracle Corporation, Proofpoint, Inc., Rocket Software, Inc., SailPoint Technologies Holdings, Inc., Salesforce.com, Inc., ServiceNow, Inc., SolarWinds Corporation, Splunk, Inc. and Zscaler, Inc. We compete based on our breadth of portfolio of enterprise management tools, breadth and synergy of offerings, our platform and hardware independence, our global reach, and our deep customer relationships and industry experience.

Intellectual Property

Our success depends in part upon our ability to protect our IP. To accomplish this, we rely on a combination of IP rights, including patents, copyrights, trademarks, service marks, trade secrets and similar IP, as well as customary contractual protections with our customers, suppliers, employees and consultants, and through security measures to protect our trade secrets. We believe our current product expertise, key engineering talent and IP portfolio provide us with a strong platform from which to develop application specific products in key target markets.

As of October 30, 2022, we had 17,035 U.S. and other patents and 618 U.S. and other pending patent applications. The expiration dates of our patents range from 2022 to 2041, with a small number of patents expiring in the near future, none of which are expected to be material to our IP portfolio. We are not substantially dependent on any single patent or group of related patents.

We focus our patent application program to a greater extent on those inventions and improvements that we believe are likely to be incorporated into our products, as contrasted with more basic research. However, we do not know how many of our pending patent applications will result in the issuance of patents or the extent to which the examination process could require us to narrow our claims.

We and our predecessors have also entered into a variety of IP licensing and cross-licensing arrangements that have both benefited our business and enabled some of our competitors. A portion of our revenue comes from IP licensing royalty payments and from litigation settlements relating to such IP. We also license third-party technologies that are incorporated into some elements of our design activities, products and manufacturing processes. Historically, licenses of the third-party technologies used by us have generally been available to us on acceptable terms.

The industries in which we compete are characterized by the existence of a large number of patents, copyrights, trademarks and trade secrets and by the vigorous pursuit, protection and enforcement of IP rights, including by patent holding companies that do not make or sell products. Some of our customer agreements require us to indemnify our customers for third-party IP infringement claims arising from our products. Claims of this sort could harm our relationships with our customers and might deter future customers from doing business with us. With respect to any IP rights claims against us or our customers or distributors, we may be required to defend ourselves or our customers or distributors in litigation, cease manufacturing the infringing products, pay damages, expend resources to develop non-infringing technology, seek a license which may not be available on commercially reasonable terms or at all, or relinquish patents or other IP rights.

With respect to our infrastructure software, the proprietary portions of our source code for our products are protected both as a trade secret and as copyrighted works. Except with respect to software components that are subject to open source licenses, our customers do not generally have access to the source code for our products. Rather, on-premise customers typically access only the executable code for our products, and SaaS customers access only the functionality of our SaaS offerings. Under certain contingent circumstances, some of our customers are beneficiaries of a source code escrow

arrangement that would enable them to obtain a limited right to access and use our source code if specific conditions are met.

Employees

Our success depends on our continued ability to attract, motivate and retain our workforce. As the source of our technological and product innovations, our engineering and technical personnel are a significant asset. Competition for these and other talented employees is significant in many locations where we operate, such as Silicon Valley and Southeast Asia.

We measure our employees' engagement by our voluntary attrition rate and employee feedback. Our global voluntary attrition rate in fiscal year 2022 was approximately 6.5%, below the technology industry benchmark (AON, 2022 Salary Increase and Turnover Study — Second Edition, September 2022).

We also track the portion of our workforce in research and development roles. As of October 30, 2022, we had approximately 20,000 employees worldwide, with approximately 63% in research and development roles. By geography, approximately 54% of our employees are located in North America, 35% in Asia, and 11% in Europe, the Middle East and Africa.

Governmental Regulation

Our semiconductor manufacturing operations and research and development involve the use of hazardous substances and are regulated under international, federal, state and local laws governing health, safety and the environment. These regulations include limitations on discharge of pollutants to air, water, and soil; remediation requirements; product chemical content limitations; manufacturing chemical use and handling restrictions; pollution control requirements; waste minimization considerations; and treatment, transport, storage and disposal of solid and hazardous wastes. We are also subject to regulation by the U.S. Occupational Safety and Health Administration and similar health and safety laws in other jurisdictions.

We believe that our properties and operations at our facilities comply in all material respects with applicable environmental laws and worker health and safety laws. However, the risk of environmental liabilities cannot be completely eliminated and there can be no assurance that the application of environmental, health and safety laws to our business will not require us to incur significant expenditures.

We are also regulated under a number of international, federal, state and local laws regarding recycling, product packaging and product content requirements, including legislation enacted in the U.S., European Union and China, among a growing number of jurisdictions, which have placed greater restrictions on the use of lead, among other restricted substances, in electronic products, which affects materials composition and semiconductor packaging. In addition, our business is subject to various import/export regulations, such as the U.S. Export Administration Regulations, and applicable executive orders, and rules of industrial standards bodies, like the International Organization for Standardization, as well as regulation by other agencies, such as the U.S. Federal Trade Commission ("FTC"). These laws, regulations and orders are complex, may change frequently and with limited notice, have generally and may continue to become more stringent over time. We may incur significant expenditures in future periods as a result.

Seasonality

Historically, our net revenue has typically been higher in the second half of the fiscal year than in the first half, primarily due to seasonality in our wireless communications products. These products have historically experienced seasonality due to launches of new mobile devices manufactured by our OEM customers. However, from time to time, typical seasonality and industry cyclicalities are overshadowed by other factors such as wider macroeconomic effects, the timing of significant product transitions and launches by large OEMs, particularly with our wireless communications products. We have a diversified business portfolio and we believe that our overall revenue is less susceptible to seasonal variations as a result of this diversification.

Other Information

Our website is www.broadcom.com. You may access our annual reports on Form 10-K, quarterly reports on Form 10-Q, current reports on Form 8-K and other reports (and amendments thereto) filed or furnished pursuant to Section 13(a) or 15(d) of the Securities Exchange Act of 1934 (the "Exchange Act") with the Securities and Exchange Commission (the "SEC"), as well as proxy statements filed by Broadcom, free of charge at the "Investor Center - SEC Filings" section of our website at www.broadcom.com, as soon as reasonably practicable after such material is electronically filed with, or furnished to, the SEC. Such periodic reports, proxy statements and other information are also available at the SEC's website at www.sec.gov. The reference to our website address does not constitute incorporation by reference of the information contained on or accessible through our website.

Information About Our Executive Officers

The following table provides information regarding our executive officers as of December 16, 2022:

<u>Name and Title</u>	<u>Age</u>	<u>Position and Offices</u>
Hock E. Tan	71	President, Chief Executive Officer and Director
Kirsten M. Spears	58	Chief Financial Officer and Chief Accounting Officer
Mark D. Brazeal	54	Chief Legal and Corporate Affairs Officer
Charlie B. Kawwas, Ph.D.	52	President, Semiconductor Solutions Group

Hock E. Tan has served as our President and Chief Executive Officer since March 2006. He held several executive leadership positions at Integrated Circuit Systems, Inc., a publicly traded timing solutions IC company, including President and Chief Executive Officer from 1999 until its acquisition by Integrated Device Technology, Inc. in 2005, Chief Operating Officer from 1996 to 1999 and Senior Vice President and Chief Financial Officer from 1995 to 1999. He was Vice President of Finance at Commodore International, Ltd. from 1992 to 1994, and held senior management positions at PepsiCo, Inc. and General Motors Corporation. He was also managing director of Pacven Investment, Ltd., a venture capital fund in Singapore, from 1988 to 1992, and was managing director of Hume Industries Ltd. in Malaysia from 1983 to 1988.

Kirsten M. Spears has served as our Chief Financial Officer and Chief Accounting Officer since December 2020. She served as our Principal Accounting Officer from March 2016 to December 2020 and Vice President and Corporate Controller from May 2014 to December 2020. She was Vice President and Corporate Controller at LSI Corporation from 2007 until its acquisition by us in 2014. She held several management positions in accounting and reporting at LSI from 1997 to 2007. She also worked for PriceWaterhouseCoopers prior to joining LSI.

Mark D. Brazeal has served as our Chief Legal and Corporate Affairs Officer since December 2021. He served as our Chief Legal Officer from March 2017 to December 2021. He was Chief Legal Officer and Senior Vice President, IP Licensing for SanDisk Corporation from 2014 until its acquisition by Western Digital Corporation in 2016. He held several senior legal positions at Broadcom Corporation from 2000 to 2014, including Senior Vice President and Senior Deputy General Counsel in charge of all commercial, operational, IP licensing and litigation matters. He was also an attorney in the transactional and IP groups at the law firms of Wilson Sonsini Goodrich & Rosati, Yuasa & Hara and Howrey & Simon prior to joining Broadcom Corporation.

Charlie B. Kawwas has served as our President, Semiconductor Solutions Group since July 2022. He served as our Chief Operating Officer from December 2020 to July 2022, Senior Vice President and Chief Sales Officer from June 2015 to December 2020 and Senior Vice President, Worldwide Sales from May 2014 to June 2015. He was head of worldwide sales at LSI Corporation from 2010 until its acquisition by us in 2014. He held several executive leadership positions at LSI from 2007 to 2010, including Vice President of Sales and Marketing for the networking division and Vice President of Marketing for the networking and storage products group. He was also the leader of Product Line Management for the Optical Ethernet and Multi-service Edge portfolio at Nortel Networks Corporation prior to joining LSI.

ITEM 1A. RISK FACTORS

Our business, operations and financial results are subject to various risks and uncertainties, including those described below, that could adversely affect our business, financial condition, results of operations, cash flows, and the trading price of our common stock. Many of the following risks and uncertainties are, and may continue to be, exacerbated by the COVID-19 pandemic. The following material factors, among others, could cause our actual results to differ materially from historical results and those expressed in forward-looking statements made by us or on our behalf in filings with the SEC, press releases, communications with investors and oral statements.

Risk Factors Summary

The following is a summary of the principal risks that could adversely affect our business, operations and financial results.

Risks Related to Our Business

- Adverse global economic conditions could have a negative effect on us.
- We operate in the highly cyclical semiconductor industry.
- The majority of our sales come from a small number of customers and a reduction in demand or loss of one or more of our significant customers may adversely affect our business.
- Dependence on contract manufacturing and suppliers of critical components within our supply chain may adversely affect our ability to bring products to market.
- We purchase a significant amount of the materials used in our products from a limited number of suppliers.
- Our business is subject to various governmental regulations and trade restrictions. Compliance with these regulations may cause us to incur significant expense and, if we fail to maintain compliance, we may be forced to cease manufacture and distribution of certain products or subjected to administrative proceedings and civil or criminal penalties.
- Global political and economic conditions and other factors related to our international operations could adversely affect us.
- The COVID-19 pandemic has disrupted normal business activity.
- We are subject to risks associated with our distributors and other channel partners, including product inventory levels and product sell-through.
- Our dependence on senior management and if we are unable to attract and retain qualified personnel, we may not be able to execute our business strategy effectively.
- The failure to complete or realize the expected benefits of our acquisition of VMware, Inc. ("VMware Merger") may adversely affect our business and our stock price.
- We may pursue acquisitions, investments, joint ventures and dispositions, which could adversely affect our results of operations.
- We may be involved in legal proceedings, including IP, securities litigation, and employee-related claims.
- Our operating results are subject to substantial quarterly and annual fluctuations.
- Failure to adjust our manufacturing and supply chain to accurately meet customer demand could adversely affect our results of operations.
- Winning business in the semiconductor solutions industry is subject to a lengthy process that often requires us to incur significant expense, from which we may ultimately generate no revenue.
- Competition in our industries could prevent us from growing our revenue.
- A prolonged disruption of our manufacturing facilities, research and development facilities, warehouses or other significant operations, or those of our suppliers, could have a material adverse effect on us.
- We may be unable to maintain appropriate manufacturing capacity or product yields at our own manufacturing facilities.
- An impairment of the confidentiality, integrity, or availability of our IT systems, or those of one or more of our corporate infrastructure vendors, could have a material adverse effect on our business.
- Our ability to maintain or improve gross margin.
- Our ability to protect the significant amount of IP in our business.
- Incompatibility of our software products with operating environments, platforms, or third-party products, demand for our products and services could decrease.
- Failure to enter into software license agreements on a satisfactory basis could adversely affect us.

[Table of Contents](#)

- Licensed third party software used in our products may not be available to us in the future, which may delay product development and production or cause us to incur additional expense.
- Use of open source code sources, which, under certain circumstances could materially adversely affect us.
- Failure of our software products to manage and secure IT infrastructures and environments could have a material adverse effect on our business.
- We are subject to warranty claims, product recalls and product liability.
- The complexity of our products could result in unforeseen delays or expense or undetected defects or bugs.
- We make substantial investments in research and development and unsuccessful investments could materially adversely affect our business, financial condition and results of operations.
- We collect, use, store, or otherwise process personal information, which subjects us to privacy and data security laws and contractual commitments, and our actual or perceived failure to comply with such laws and commitments could harm our business.
- We are subject to environmental, health and safety laws, which could increase our costs, restrict our operations and require expenditures.
- Social and environmental regulations, policies and provisions, as well as customer and investor demands, may make our supply chain more complex and may adversely affect our relationships with customers and investors.
- The average selling prices of semiconductor products in our markets have often decreased rapidly and may do so in the future.
- Fluctuations in foreign exchange rates could result in losses.

Risks Relating to Taxes

- Changes in tax legislation or policies could materially impact our financial position and results of operations.
- Our corporate income taxes could significantly increase if we are unable to maintain our tax concessions or if our assumptions and interpretations regarding tax laws and concessions prove to be incorrect.
- Our income taxes and overall cash tax costs are affected by a number of factors that could materially, adversely affect financial results.

Risks Relating to Our Indebtedness

- Our substantial indebtedness could adversely affect our financial health and our ability to execute our business strategy.
- The instruments governing our indebtedness impose certain restrictions on our business.
- Servicing our debt requires a significant amount of cash, and we may not have sufficient cash flows from our business to pay our substantial debt.

Risks Relating to Owning Our Common Stock

- Volatility of our stock price could result in substantial losses for our investors as well as class action litigation against us and our management.
- The amount and frequency of our stock repurchases may fluctuate.
- A substantial amount of our stock is held by a small number of large investors.
- There can be no assurance that we will continue to declare cash dividends.

For a more complete discussion of the material risks facing our business, see below.

Risks Related to Our Business

Adverse global economic conditions could have a negative effect on our business, results of operations and financial condition and liquidity.

A general slowdown in the global economy, including a recession, or in a particular region or industry, an increase in trade tensions with U.S. trading partners, inflation or a tightening of the credit markets could negatively impact our business, financial condition and liquidity. Adverse global economic conditions have from time to time caused or exacerbated significant slowdowns in the industries and markets in which we operate, which have adversely affected our business and results of operations. Macroeconomic weakness and uncertainty also make it more difficult for us to accurately forecast revenue, gross margin and expenses, and may make it more difficult to raise or refinance debt. An escalation of trade tensions between the U.S. and China has resulted in trade restrictions and increased tariffs that harm our ability to participate in Chinese markets or compete effectively with Chinese companies. Sustained uncertainty about, or worsening of, current global economic conditions and further escalation of trade tensions between the U.S. and its trading partners, especially China, and possible

decoupling of the U.S. and China economies, could result in a global economic slowdown and long-term changes to global trade. Such events may also (i) cause our customers and consumers to reduce, delay or forgo technology spending, (ii) result in customers sourcing products from other suppliers not subject to such restrictions or tariffs, (iii) lead to the insolvency or consolidation of key suppliers and customers, and (iv) intensify pricing pressures. Any or all of these factors could negatively affect demand for our products and our business, financial condition and results of operations.

We operate in the highly cyclical semiconductor industry.

The semiconductor industry is highly cyclical and is characterized by price erosion, wide fluctuations in product supply and demand, constant and rapid technological change, evolving technical standards, frequent new product introductions, and short product life cycles (for semiconductors and for many of the end products in which they are used). From time to time, these factors, together with changes in general economic conditions, cause significant upturns and downturns in the industry in general, and in our business in particular. The industry recently experienced a significant upturn due to the supply imbalance that resulted in record profitability and increases in average selling prices. It is possible that this recent industry up-cycle will be followed by a downturn, and historically, such down-cycles have been characterized by diminished demand for end-user products, high inventory levels and periods of inventory adjustment, under-utilization of manufacturing capacity, changes in revenue mix, accelerated erosion of average selling prices and elimination of expedite fees leading to reduced profitability and a decline in our stock price. The Creating Helpful Incentives to Produce Semiconductors for America Act could also result in an increase in supply leading to excess inventory and a decrease in average selling prices. We expect our business to continue to be subject to cyclical downturns even when overall economic conditions are relatively stable. If we cannot offset industry or market downturns, our net revenue may decline and our financial condition and results of operations may suffer.

The majority of our sales come from a small number of customers and a reduction in demand or loss of one or more of our significant customers may adversely affect our business.

We are dependent on a small number of end customers, OEMs, their respective contract manufacturers ("CMs"), and certain distributors for a majority of our business and revenue. For fiscal year 2022, sales to distributors accounted for 56% of our net revenue. We believe aggregate sales, through all channels, to Apple and our top five end customers, accounted for approximately 20% and 35% of our net revenue for fiscal year 2022, respectively. This customer concentration increases the risk of quarterly fluctuations in our operating results and our sensitivity to any material, adverse developments experienced by our significant customers.

Our semiconductor customers are not generally required to purchase specific quantities of products. Even when customers agree to source an agreed portion of their product needs from us, such arrangements often include pricing schedules or methodologies that apply regardless of the volume of products purchased, and those customers may not purchase the amount of product we expect. As a result, we may not generate the amount of revenue or achieve the level of profitability we expect under such arrangements. Moreover, our top customers' purchasing power has, in some cases, given them the ability to make greater demands on us with regard to pricing and contractual terms in general. We expect this trend to continue, which may adversely affect our gross margin on certain products and, should we fail to perform under these arrangements, we could also be liable for significant monetary damages.

The loss of, or any substantial reduction in sales to, any of our top customers could have a material adverse effect on our business, financial condition, results of operations and cash flows.

Dependence on contract manufacturing and suppliers of critical components within our supply chain may adversely affect our ability to bring products to market, damage our reputation and adversely affect our results of operations.

We operate a primarily outsourced manufacturing business model that principally utilizes CMs, such as third-party wafer foundries and module assembly and test capabilities. Our semiconductor products require wafer manufacturers with state-of-the-art fabrication equipment and techniques, and most of our products are designed to be manufactured in a specific process, typically at one particular fab or foundry, either our own or with a particular CM.

We depend on our CMs to allocate sufficient manufacturing capacity to meet our needs, to produce products of acceptable quality at acceptable yields, and to deliver those products to us on a timely basis. We do not generally have long-term capacity commitments with our CMs and substantially all of our manufacturing services are on a purchase order basis with no minimum quantities. Further, our CMs may fail to timely develop new, advanced manufacturing processes, including transitions to smaller geometry process technologies or, from time to time, will cease to, or will become unable to, manufacture a component for us. As lead times to identify, qualify and establish reliable production at acceptable yields with a new CM is typically lengthy, there is often no readily available alternative source and there may be other constraints on our ability to change CMs. In addition, qualifying new CMs is often expensive, and they may not produce products as cost-effectively as our current suppliers.

[Table of Contents](#)

TSMC, one of our CMs, manufactured approximately 90% of the wafers manufactured by our CMs during fiscal year 2022. We believe our wafer requirements represent a meaningful portion of TSMC's total production capacity. However, TSMC also fabricates wafers for other companies, including some of our competitors, and could choose or be required to prioritize capacity for other customers or reduce or eliminate deliveries to us on short notice. In addition, TSMC has, and may in the future, raise their prices to us.

If any of the foregoing circumstances occur, we may be unable to meet our customer demand, or to the same extent as our competitors, fail to meet our contractual obligations or forgo revenue opportunities. This has in the past damaged, and may in the future damage our relationships with our customers. This could also result in litigation for alleged failure to meet our obligations, payment of significant damages, and our net revenue could decline, adversely affecting our business, financial condition, results of operations and gross margin.

Further, any substantial disruption in the contract manufacturing services that we utilize, including TSMC's supply of wafers to us, as a result of a natural disaster, climate change, water shortages, political unrest, military conflicts, geopolitical turmoil, trade tensions, government orders, medical epidemics, such as the COVID-19 pandemic, economic instability, equipment failure or other cause, could materially harm our business, customer relationships and results of operations.

We purchase a significant amount of the materials used in our products from a limited number of suppliers.

Our manufacturing processes and those of our CMs rely on many materials, including silicon, GaAs and InP wafers, copper lead frames, precious and rare earth metals, mold compound, ceramic packages and various chemicals and gases. During fiscal year 2022, we purchased approximately two-thirds of our manufacturing materials from five materials providers, some of which are single source suppliers. As certain materials are highly specialized, the lead time needed to identify and qualify a new supplier is typically lengthy and there is often no readily available alternative source. We do not generally have long-term contracts with our materials providers and substantially all of our purchases are on a purchase order basis. Suppliers may extend lead times, limit supplies, place products on allocation or increase prices due to commodity price increases, capacity constraints, inflation, or other factors, any of which could lead to interruption of supply or increased demand in the industry. For example, due to the COVID-19 pandemic, we have experienced some supply constraints and increases in prices, including with respect to wafers and substrates. Additionally, the supply of these materials may be negatively impacted by increased trade tensions between the U.S. and its trading partners, particularly China. The supply constraints have had, and may continue to have, a negative impact on our customer relationships. Further, continued supply constraints for these or any other reasons could result in loss of revenue opportunities and adversely impact our business, financial condition and results of operations.

Our business is subject to various governmental regulations, and compliance with these regulations may cause us to incur significant expense. If we fail to maintain compliance with applicable regulations, we may be forced to cease the manufacture and distribution of certain products, and we could be subject to administrative proceedings and civil or criminal penalties.

Our business is subject to various domestic and international laws and other legal requirements, including anti-competition and import/export regulations, such as the U.S. Export Administration Regulations, and applicable executive orders. These laws, regulations and orders are complex, may change frequently and with limited notice, and have generally and may continue to become more stringent over time. We may be required to incur significant expense to comply with, or to remedy violations of, these regulations. In addition, if our customers fail to comply with these regulations, we may be required to suspend sales to these customers, which could damage our reputation and negatively impact our results of operations. The U.S. government may also add companies to its restricted entity list and/or technologies to its list of prohibited exports to specific countries, which have had and will continue to have an adverse effect on our ability to sell our products and our revenue. For example, Huawei Technologies Co., Ltd., one of our customers, is subject to certain U.S. export restrictions, which has required us to suspend sales to Huawei until we obtain licenses from the U.S. Department of Commerce. We may be unable to obtain or maintain the necessary licenses to allow us to export products to them. These restrictive governmental actions and any similar measures that may be imposed on U.S. companies by other governments, especially in light of ongoing trade tensions with China, will likely limit or prevent us from doing business with certain of our customers or suppliers and harm our ability to compete effectively or otherwise negatively affect our ability to sell our products, and adversely affect our business and results of operations.

Our products and operations are also subject to regulation by U.S. and non-U.S. regulatory agencies, such as the FTC. From time to time, we may also be involved or required to participate in regulatory investigations or inquiries, such as the ongoing investigation by the Korean Fair Trade Commission into certain of our contracting and business practices, which may evolve into legal or other administrative proceedings. Growing public concern over concentration of economic power in corporations is likely to result in increased anti-competition legislation, regulation, administrative rule making, and enforcement activity. Involvement in regulatory investigations or inquiries, can be costly, lengthy, complex and time consuming, diverting the attention and energies of our management and technical personnel.

If any pending or future governmental investigations result in an unfavorable resolution, we could be required to cease the manufacture and sale of the subject products or technology, pay fines or disgorge profits or other payments, and/or cease certain conduct and/or modify our contracting or business practices, which could have a material adverse effect on our business, financial condition and results of operations. We may be obligated to indemnify our current or former directors or employees, or former directors or employees of companies that we have acquired, in connection with regulatory investigations. These liabilities could be substantial and may include, among other things, the cost of government, law enforcement or regulatory investigations and civil or criminal fines and penalties.

In addition, the manufacture and distribution of our semiconductors must comply with various laws and adapt to changes in regulatory requirements as they occur. For example, if a country in which our products are manufactured or sold sets technical standards that are not widely shared, it may require us to stop distributing our products commercially until they comply with such new standards, lead certain of our customers to suspend imports of their products into that country, require manufacturers in that country to manufacture products with different technical standards and disrupt cross-border manufacturing relationships, any of which could have a material adverse effect on our business, financial condition and results of operations. If we fail to comply with these requirements, we could also be required to pay civil penalties or face criminal prosecution.

Global political and economic conditions and other factors related to our international operations could adversely affect our business, financial condition and results of operations.

A majority of our products are produced, sourced and sold internationally and our international revenue represents a significant percentage of our overall revenue. In addition, as of October 30, 2022, nearly 49% of our employees were located outside the U.S. Multiple factors relating to our international operations and to particular countries in which we operate could have a material adverse effect on our business, financial condition and results of operations. These factors include:

- changes in political, regulatory, legal or economic conditions or geopolitical turmoil (including China-Taiwan relations), including terrorism, war or political or military coups, state-sponsored or politically motivated cyber-attacks, or civil disturbances or political instability foreign and domestic;
- restrictive governmental actions, such as restrictions on the transfer or repatriation of funds and foreign investments, data privacy regulations, imposition of climate change regulations, and trade protection measures, including increasing protectionism, import/export restrictions (including with regards to advanced technologies), import/export duties and quotas, trade sanctions and customs duties and tariffs, all of which have increased in recent years;
- difficulty in obtaining product distribution and support, and transportation delays;
- potential inability to localize software products;
- difficulty in conducting due diligence with respect to business partners;
- public health or safety concerns, medical epidemics or pandemics, such as COVID-19, and other natural- or man-made disasters;
- nationalization of businesses and expropriation of assets; and
- changes in U.S. and foreign tax laws.

A significant legal risk associated with conducting business internationally is compliance with the various and differing laws and regulations of the many countries in which we do business. In addition, the laws in various countries are constantly evolving and may, in some cases, conflict with each other. Although our policies prohibit us, our employees and our agents from engaging in unethical business practices, there can be no assurance that all of our employees, distributors or other agents will refrain from acting in violation of our related anti-corruption or other policies and procedures. Any such violation could have a material adverse effect on our business.

The COVID-19 pandemic has disrupted normal business activity, which has impacted how we operate our business.

The COVID-19 pandemic and the efforts to control it disrupted, and reduced the efficiency of, normal business activities in much of the world. We experienced some disruption to parts of our global semiconductor supply chain, including procuring necessary components and inputs, such as wafers and substrates, in a timely fashion, with suppliers increasing lead times or placing products on allocation. As a result of these supply chain disruptions, we increased customer order lead times and placed some products on allocation. We are also largely building semiconductor products to order and this has limited and may continue to limit our ability to fulfill orders and satisfy all of the demand for our products.

The pandemic resulted in authorities around the world implementing numerous unprecedented measures, such as travel restrictions quarantines, shelter-in-place order, and factory and office shutdowns, that impacted our workforce and

operations, and those of our customers, CMs, suppliers and logistics providers. In addition, disruptions to commercial transportation infrastructure impacted delivery times for materials and components to our facilities, transfers of our products to our key suppliers and, in some cases, our ability to timely ship our products to customers. This resulted in significant logistical challenges and product delays, which could recur in the event of any future closures of, or periods of reduced operations at, our warehouse or the facilities of our suppliers and providers.

In response to the pandemic, we have taken extensive measures to protect the health and safety of our employees and contractors at our facilities. However, existing or new precautionary measures or modifications in our business practices and policies, may negatively impact our business or operations, especially if the spread of COVID-19 (including any variants) worsens significantly. In addition, any actions we take may not be sufficient to mitigate the risk of infection and could result in a significant number of COVID-19-related claims. If a significant number of our employees, or employees and third parties performing key functions, including our Chief Executive Officer and members of our Board of Directors, become ill, our business may be further adversely impacted. In addition, changes to state workers' compensation laws, such as those in California, may increase our potential liability for such claims. See also our risk factor *"If we are unable to attract and retain qualified personnel, especially our engineering and technical personnel, we may not be able to execute our business strategy effectively."*

The degree to which the pandemic ultimately impacts our business and results of operations will depend on future developments beyond our control, including the extent of actions to contain the virus (including any variants), availability and efficacy of the vaccines or other treatments, public acceptance of the vaccines (including boosters), and to what extent normal economic and operating conditions resume.

We are subject to risks associated with our distributors and other channel partners, including product inventory levels and product sell-through.

We sell our products through a direct sales force and a select network of distributors and other channel partners globally. Sales to distributors accounted for 56% of our net revenue in the fiscal year ended October 30, 2022 and are subject to a number of risks, including:

- fluctuations in demand based on our distributors' product inventory levels and end customer demand;
- our distributors and other channel partners are generally not subject to minimum sales requirements or any obligation to market our products to their customers;
- our distributors and other channel partners agreements are generally nonexclusive and may be terminated at any time without cause;
- our lack of control over the timing of delivery of our products to end customers; and
- our distributors and other channel partners may market and distribute competing products and may place greater emphasis on the sale of these products.

In addition, we are selling our semiconductor products through an increasingly limited number of distributors, which exposes us to additional customer concentration and related credit risks.

We do not always have a direct relationship with the end customers of our products. As a result, our semiconductor products may be used in applications for which they were not necessarily designed or tested, including, for example, medical devices, and they may not perform as anticipated in such applications. In such event, failure of even a small number of parts could result in significant liabilities to us, damage our reputation and harm our business and results of operations.

Our business would be adversely affected by the departure of existing members of our senior management team.

Our success depends, in large part, on the continued contributions of our senior management team, and in particular, the services of Hock E. Tan, our President and Chief Executive Officer. Effective succession planning is also important for our long-term success. Failure to ensure effective transfers of knowledge and smooth transitions involving senior management could hinder our strategic planning and execution. None of our senior management is bound by written employment contracts. In addition, we do not currently maintain key person life insurance covering our senior management. The loss of any of our senior management could harm our ability to implement our business strategy and respond to the rapidly changing market conditions in which we operate.

If we are unable to attract and retain qualified personnel, especially our engineering and technical personnel, we may not be able to execute our business strategy effectively.

Our future success depends on our ability to attract, retain and motivate qualified personnel. As the source of our technological and product innovations, our engineering and technical personnel (including cyber security experts) are a significant asset. Competition for these employees is significant in many areas of the world in which we operate, particularly

in Silicon Valley and Southeast Asia where qualified engineers are in high demand. In addition, current or future immigration laws may make it more difficult to hire or retain qualified engineers, further limiting the pool of available talent. We believe equity awards provide a powerful long-term retention incentive and have historically granted these awards to the substantial majority of our employees. However, the amendments to our 2012 Stock Incentive Plan approved by our stockholders in 2021 significantly reduced the number of shares available for equity awards. As a result, we may need to change our current equity granting philosophy, which could impair our efforts to attract and retain necessary personnel. Any inability to retain, attract or motivate such personnel and provide competitive employment benefits could have a material adverse effect on our business, financial condition and results of operations.

The failure to complete our acquisition of VMware, Inc. may adversely affect our business and our stock price.

Consummation of the VMware Merger is subject to the satisfaction or waiver of customary closing conditions, including (i) the expiration or termination of the waiting period under the Hart-Scott-Rodino Antitrust Improvement Act of 1976 and clearance under the antitrust laws of the European Union and certain other jurisdictions, (ii) the receipt by VMware of a tax opinion regarding the U.S. federal income tax treatment of certain aspects of the VMware Merger, (iii) the absence of certain orders or laws preventing consummation of the VMware Merger, (iv) authorization for listing additional shares of Broadcom common stock on Nasdaq, and (v) the absence of a material adverse effect with respect to either us or VMware. There can be no assurance that these or other closing conditions will be satisfied in a timely manner or at all. Any delay in completing the acquisition could cause us not to realize some or all of the anticipated benefits when expected, if at all. If the VMware Merger is not completed, our stock price could decline to the extent it reflects an assumption that we will complete the acquisition. Furthermore, if the VMware Merger is not completed, we may suffer other consequences that could adversely affect our business, results of operations and stock price, including incurring significant acquisition costs that we would be unable to recover, negative publicity and a negative impression of us in the investment community. Additionally, under certain specified circumstances, including the termination by either us or VMware because certain required regulatory clearances are not obtained, upon termination we would be required to pay VMware a termination fee of \$1.5 billion.

Failure to realize the benefits expected from the VMware Merger could adversely affect the value of our common stock.

Although we expect significant benefits to result from the VMware Merger, there can be no assurance that we will actually realize any of them, or realize them within the anticipated timeframe. Achieving these benefits will depend, in part, on our ability to integrate VMware's business successfully and efficiently. The challenges involved in this integration, which will be complex and time consuming, include the following:

- preserving customer and other important relationships of VMware and attracting new business and operational relationships;
- integrating financial forecasting and controls, procedures and reporting cycles;
- consolidating and integrating corporate, information technology, finance and administrative infrastructures;
- coordinating sales and marketing efforts to effectively position our capabilities;
- coordinating and integrating operations in countries in which we have not previously operated; and
- integrating employees and related HR systems and benefits, maintaining employee morale and retaining key employees.

If we do not successfully manage these issues and the other challenges inherent in integrating an acquired business, then we may not achieve the anticipated benefits of the VMware Merger on our anticipated timeframe or at all and our revenue, expenses, operating results, financial condition and stock price could be materially adversely affected. The successful integration of the VMware business will require significant management attention both before and after the completion of the VMware Merger, and may divert the attention of management from our business and operational issues.

We may pursue acquisitions, investments, joint ventures and dispositions, which could adversely affect our results of operations.

Our growth strategy includes acquiring or investing in businesses that offer complementary products, services and technologies, or enhance our market coverage or technological capabilities.

Any acquisitions we may undertake, including the VMware Merger, and their integration involve risks and uncertainties, such as:

- unexpected delays, challenges and related expenses, and disruption of our business;
- diversion of management's attention from daily operations and the pursuit of other opportunities;

[Table of Contents](#)

- incurring significant restructuring charges and amortization expense, assuming liabilities (some of which may be unexpected) and ongoing or new lawsuits, potential impairment of acquired goodwill and other intangible assets, and increasing our expenses and working capital requirements;
- the potential for deficiencies in internal controls at the acquired business, as well as implementing our own management information systems, operating systems and internal controls for the acquired operations;
- our due diligence process may fail to identify significant issues with the acquired business' products, financial disclosures, accounting practices, legal, tax and other contingencies, compliance with local laws and regulations (and interpretations thereof) in the U.S. and multiple international jurisdictions;
- additional acquisition-related debt, which could increase our leverage and potentially negatively affect our credit ratings resulting in more restrictive borrowing terms or increased borrowing costs thereby limiting our ability to borrow;
- dilution of stock ownership of existing stockholders;
- difficulties integrating the acquired business or company and in managing and retaining acquired employees, vendors and customers; and
- inaccuracies in our original estimates and assumptions used to assess a transaction, which may result in us not realizing the expected financial or strategic benefits of any such transaction.

In addition, current and future changes to the U.S. and foreign regulatory approval process and requirements related to acquisitions, including the VMware Merger, may cause approvals to take longer than anticipated, not be forthcoming or contain burdensome conditions, which may prevent the transaction or jeopardize, delay or reduce the anticipated benefits of the transaction, and impede the execution of our business strategy.

From time to time, we may also seek to divest or wind down portions of our business, either acquired or otherwise, or we may exit minority investments, any of which could materially affect our cash flows and results of operations. Such dispositions involve risks and uncertainties, including our ability to sell such businesses on terms acceptable to us, or at all, disruption to other parts of our business, potential loss of employees or customers, or exposure to unanticipated liabilities or ongoing obligations to us following any such dispositions. In addition, dispositions may include the transfer of technology and/or the licensing of certain IP rights to third-party purchasers, which could limit our ability to utilize such IP rights or assert these rights against such third-party purchasers or other third parties.

We may be involved in legal proceedings, including IP, securities litigation, and employee-related claims, which could, among other things, divert efforts of management and result in significant expense and loss of our IP rights.

We are often involved in legal proceedings, including cases involving our IP rights and those of others, commercial matters, acquisition-related suits, securities class action suits, employee-related claims and other actions. Litigation or settlement of such actions, regardless of their merit, can be costly, lengthy, complex and time consuming, diverting the attention and energies of our management and technical personnel.

The industries in which we operate are characterized by companies holding large numbers of patents, copyrights, trademarks and trade secrets and by the vigorous pursuit, protection and enforcement of IP rights, including actions by patent-holding companies that do not make or sell products. From time to time, third parties assert against us and our customers and distributors their IP rights to technologies that are important to our business. For example, in August 2020 judgment was entered against Broadcom and Apple for infringement of certain patents and California Institute of Technology was awarded past damages of \$270.2 million from Broadcom and \$837.8 million from Apple, for which Apple is seeking indemnification from Broadcom. Although the appellate court recently vacated these damages and ordered a new trial, there are no assurances that we will be successful or what, if any, damages we will be required to pay.

Many of our customer agreements, and in some cases our asset sale agreements, and/or the laws of certain jurisdictions may require us to indemnify our customers or purchasers for third-party IP infringement claims, including costs to defend those claims, and payment of damages in the case of adverse rulings. However, our CMs and suppliers may or may not be required to indemnify us should we or our customers be subject to such third-party claims. Claims of this sort could also harm our relationships with our customers and might deter future customers from doing business with us. If any pending or future proceedings result in an adverse outcome, we could be required to:

- cease the manufacture, use or sale of the infringing products, processes or technology and/or make changes to our processes or products;
- pay substantial damages for past, present and future use of the infringing technology, including up to treble damages if willful infringement is found;

[Table of Contents](#)

- expend significant resources to develop non-infringing technology;
- license technology from the third-party claiming infringement, which license may not be available on commercially reasonable terms, or at all;
- enter into cross-licenses with our competitors, which could weaken our overall IP portfolio and our ability to compete in particular product categories;
- pay substantial damages to our direct or end customers to discontinue use or replace infringing technology with non-infringing technology; or
- relinquish IP rights associated with one or more of our patent claims.

Any of the foregoing results could have a material adverse effect on our business, financial condition and results of operations.

In addition, we may be obligated to indemnify our current or former directors or employees, or former directors or employees of companies that we have acquired, in connection with such litigation. These liabilities could be substantial and may include, among other things, the cost of defending lawsuits against these individuals, as well as stockholder derivative suits; civil or criminal fines and penalties; legal and other expenses; and expenses associated with the remedial measure, if any, which may be imposed.

Our operating results are subject to substantial quarterly and annual fluctuations.

Our operating results have fluctuated in the past and are likely to fluctuate in the future. These fluctuations may occur on a quarterly and annual basis and are due to a number of factors, many of which are beyond our control. In addition to many of the risks described elsewhere in this “Risk Factors” section, these factors include, among others:

- the timing of launches by our customers of new product in which our products are included and changes in end-user demand for our customers’ the products;
- fluctuations in the levels of component or product inventories held by our customers;
- the shift to cloud-based IT solutions and services, such as hyperscale computing, which may adversely affect the timing and volume of sales of our products for use in traditional enterprise data centers;
- the timing of new software contracts and renewals, as well as the timing of any terminations of software contracts that require us to refund to customers any pre-paid amounts under the contract;
- our ability to timely develop, introduce and market new products and technologies;
- the timing and extent of our software license and subscription revenue, and other non-product revenue;
- new product announcements and introductions by us or our competitors;
- seasonality or other fluctuations in demand in our markets;
- timing and amount of research and development and related new product expenditures, and the timing of receipt of any research and development grant monies; and
- timing of any regulatory changes, particularly with respect to trade sanctions and customs duties and tariffs, and tax reform.

The foregoing factors are often difficult to predict, and these, as well as other factors, could materially adversely affect our quarterly or annual operating results. In addition, a significant amount of our operating expenses are relatively fixed in nature. Any failure to adjust spending quickly enough to compensate for a revenue shortfall could magnify the adverse impact of such revenue shortfall on our results of operations. As a result, we believe that quarter-to-quarter comparisons of our revenue and operating results may not be meaningful or a reliable indicator of our future performance. If our operating results in one or more future quarters fail to meet the expectations of securities analysts or investors, a significant decline in the trading price of our common stock may occur, which may happen immediately or over time.

Failure to adjust our manufacturing and supply chain to accurately meet customer demand could adversely affect our results of operations.

We make significant decisions, including determining the levels of business that we will seek and accept, production schedules, levels of reliance on contract manufacturing and outsourcing, internal fabrication utilization and other resource requirements, based on our estimates of customer requirements, which may not be accurate.

During the COVID-19 pandemic, we have moved largely to a build to order model and have extended customer lead times substantially in light of supply chain challenges. More typically, however, to ensure the availability of our semiconductor products we start manufacturing based on customer forecasts, which are not binding. As a result, we incur inventory and manufacturing costs in advance of anticipated sales that may be substantially lower than expected. If actual demand for our products is lower than forecast, we may also experience higher inventory carrying and operating costs and product obsolescence. Because certain of our sales, research and development, and internal manufacturing overhead expenses are relatively fixed, a reduction in customer demand may also decrease our gross margin and operating income.

Conversely, customers often require rapid increases in production on short notice. If we are unable to meet such increases in demand, this could damage our customer relationships, reduce revenue growth and margins, subject us to additional liabilities, harm our reputation, and prevent us from taking advantage of opportunities.

Winning business in the semiconductor solutions industry is subject to a lengthy process that often requires us to incur significant expense, from which we may ultimately generate no revenue.

Our semiconductor business is dependent on us winning competitive bid selection processes, known as “design wins”. These selection processes are typically lengthy and can require us to dedicate significant development expenditures and scarce engineering resources in pursuit of a single customer opportunity. Failure to obtain a particular design win may prevent us from obtaining design wins in subsequent generations of a particular product. This can result in lost revenue and can weaken our position in future selection processes.

Winning a product design does not guarantee sales to a customer. A delay or cancellation of a customer’s plans could materially and adversely affect our financial results, as we incur significant expense in the design process and may generate little or no revenue from it. In addition, the timing of design wins is unpredictable and implementing production for a major design win, or multiple design wins at the same time, may strain our resources and those of our CMs. In such event, we may be forced to dedicate significant additional resources and incur additional costs and expenses. Further, often customers will only purchase limited numbers of evaluation units until they qualify the products and/or the manufacturing line for those products. The qualification process can take significant time and resources. Delays in qualification or failure to qualify our products may cause a customer to discontinue use of our products and result in a significant loss of revenue. Finally, customers could choose at any time to stop using our products or could fail to successfully market and sell their products, which could reduce demand for our products, and cause us to hold excess inventory, materially adversely affecting our business, financial condition and results of operations. These risks are exacerbated by the fact that many of our products, and the end products into which our products are incorporated, often have very short life cycles.

Competition in our industries could prevent us from growing our revenue.

The industries in which we operate are highly competitive and characterized by rapid technological changes, evolving industry standards, changes in customer requirements, often aggressive pricing practices and, in some cases, new delivery methods. We expect competition in these industries to continue to increase as existing competitors improve or expand their product offerings or as new competitors enter our markets.

Some of our competitors have longer operating histories, greater name recognition, a larger installed customer base, larger technical staffs, more established relationships with vendors or suppliers, or greater manufacturing, distribution, financial, research and development, technical and marketing resources than us. We also face competition from numerous smaller companies that specialize in specific aspects of the highly fragmented software industry, open source authors who provide software and IP for free, competitors who offer their products through try-and-buy or freemium models, and customers who develop competing products.

In addition, the trend toward consolidation is changing the competitive landscape. We expect this trend to continue, which may result in combined competitors having greater resources than us. Some of our competitors may also receive financial and other support from their home country government or may have a greater presence in key markets, a larger customer base, a more comprehensive IP portfolio or better patent protection than us.

The actions of our competitors, in the areas of pricing and product bundling in particular, could have a substantial adverse impact on us. Further, competitors may leverage their superior market position, as well as IP or other proprietary information, including interface, interoperability or technical information, in new and emerging technologies and platforms that may inhibit our ability to compete effectively. If we are unable to compete successfully, we may lose market share for our products or incur significant reduction in our gross margins, either of which could have a material adverse effect on our business and results of operations.

A prolonged disruption of our manufacturing facilities, research and development facilities, warehouses or other significant operations, or those of our suppliers, could have a material adverse effect on our business, financial condition and results of operations.

Although we operate a primarily outsourced manufacturing business model, we also rely on our own manufacturing facilities, in particular in Fort Collins, Colorado, Singapore, and Breinigsville, Pennsylvania. We use these internal fabrication facilities for products utilizing our innovative and proprietary processes. Our Fort Collins and Breinigsville facilities are the sole sources for the FBAR components used in many of our wireless devices and for the indium phosphide-based wafers used in our fibre optics products, respectively. Many of our facilities, and those of our CMs and suppliers, are located in California and the Pacific Rim region, which have above average seismic activity and severe weather activity. In addition, a significant majority of our research and development personnel are located the Czech Republic, India, Israel and the U.S., with the expertise of the personnel at each such location tending to be focused on one or two specific areas, and our primary warehouse is in Malaysia.

A prolonged disruption at or shut-down of one or more of our manufacturing facilities or warehouses, especially our Colorado, Singapore, Malaysia and Pennsylvania facilities, or those of our CMs or suppliers, due to natural- or man-made disasters or other events outside of our control, such as equipment malfunction or widespread outbreaks of acute illness, including COVID-19, or for any other reason, would limit our capacity to meet customer demands and delay new product development until a replacement facility and equipment, if necessary, were found. To date, we have not experienced a material event, however such event could disrupt our operations, delay production, shipments and revenue, result in us being unable to timely satisfy customer demand, expose us to claims by our customers, result in significant expense to repair or replace our affected facilities, and, in some instances, could significantly curtail our research and development efforts in a particular product area or target market. As a result, we could forgo revenue opportunities, potentially lose market share, damage our customer relationships and be subject to litigation and additional liabilities, all of which could materially and adversely affect our business. Although we purchase insurance to mitigate certain losses, such insurance often carries a high deductible amount and any uninsured losses could negatively affect our operating results. In addition, even if we were able to promptly resume production of our affected products, if our customers cannot timely resume their own manufacturing following such an event, they may cancel or scale back their orders from us and this may in turn adversely affect our results of operations. Such events could also result in increased fixed costs relative to the revenue we generate and adversely affect our results of operations.

We may be unable to maintain appropriate manufacturing capacity or product yields at our own manufacturing facilities, which could adversely affect our relationships with our customers, and our business, financial condition and results of operations.

We must maintain appropriate capacity and product yields at our own manufacturing facilities to meet anticipated customer demand. From time to time, this requires us to invest in expansion or improvements of those facilities, which often involves substantial cost and other risks. Such expanded manufacturing capacity may still be insufficient, or may not come online soon enough, to meet customer demand and we may have to put customers on product allocation, forgo sales or lose customers as a result. Conversely, if we overestimate customer demand, we would experience excess capacity and fixed costs at these facilities will not be fully absorbed, all of which could adversely affect our results of operations. Similarly, reduced product yields, due to design or manufacturing issues or otherwise, may involve significant time and cost to remedy and cause delays in our ability to supply product to our customers, all of which could cause us to forgo sales, incur liabilities or lose customers, and harm our results of operations.

An impairment of the confidentiality, integrity, or availability of our IT systems, or those of one or more of our corporate infrastructure vendors could have a material adverse effect on our business.

Our business depends on various internally managed IT systems and outsourced IT services, including cloud-based and other critical corporate infrastructure services relating to, among other things, financial reporting, product orders and shipping, human resources, benefit plan administration, IT network development, network monitoring and electronic communication services, as well as third-party data centers. Any failure of these internal or third-party systems and services to operate effectively could disrupt our operations and could have a material adverse effect on our business, financial condition and results of operations. Our operations are dependent upon our ability to protect our IT infrastructure against damage from business continuity events that could have a significant disruptive effect. Although these systems are designed to protect and secure our customers', suppliers' and employees' confidential information, as well as our own proprietary information, we are, out of necessity, dependent on our vendors to adequately address cyber security threats to their own systems. In addition, software products we use (including technologies produced by us) have occasionally had in the past and may have in the future, vulnerabilities that, if left unmanaged, could reduce the overall level of security of the systems on which the software is installed.

Cyber-attacks are increasing in number and sophistication, are well-financed, in some cases supported by state actors, and are designed to not only attack, but also to evade detection. Since the techniques used to obtain unauthorized access to systems, or to otherwise sabotage them, change frequently and are often not recognized until launched against a target, we may be unable to anticipate these techniques or to implement adequate preventative measures. Geopolitical instability, such as Russia's invasion of Ukraine, may increase the likelihood that we will experience direct or collateral consequences from cyber conflicts between nation-states or other politically motivated actors targeting critical technology infrastructure. Accidental or willful security breaches or other unauthorized access to our information systems or the systems of our service providers, or the existence of computer viruses or malware (such as ransomware) in our or their data or software could expose us to a risk of information loss, business disruption, and misappropriation of proprietary and confidential information, including information relating to our products or customers and the personal information of our employees or third parties. Such an event could disrupt our business and result in, among other things, unfavorable publicity, damage to our reputation, loss of our trade secrets and other competitive information, litigation by affected parties and possible financial obligations for liabilities and damages related to the theft or misuse of such information, significant remediation costs, disruption of key business operations and significant diversion of our resources, as well as fines and other sanctions resulting from any related breaches of data privacy regulations (such as the General Data Protection Regulation), any of which could have a material adverse effect on our business, profitability and financial condition. While we may be entitled to damages if our vendors fail to perform under their agreements with us, any award may be insufficient to cover the actual costs incurred by us and, as a result of a vendor's failure to perform, we may be unable to collect any damages.

Despite our internal controls and investment in security measures, we have, from time to time, been subject to disruptive cyber-attacks or there have been attempts of unauthorized network intrusions and malware on our own IT networks. Although no such cyber security incidents have been material to Broadcom, we continue to devote resources to protect our systems and data from unauthorized access or misuse, and we may be required to expend greater resources in the future.

U.S. and foreign regulators have also increased their focus on cyber security vulnerabilities and risks. Compliance with laws and regulations concerning privacy, cyber security, data governance, and data protection could result in significant expense, and any failure to comply could result in proceedings against us by regulatory authorities or other third parties. Further, customers and service providers increasingly demand rigorous contractual, certification and audit provisions regarding privacy, cyber security, data governance, data protection, confidentiality, and IP, which may also increase our overall compliance burden.

Our gross margin is dependent on a number of factors, including our product mix, price erosion, acquisitions we may make, level of capacity utilization and commodity prices.

Our gross margin is highly dependent on product mix, which is susceptible to seasonal and other fluctuations in our markets. A shift in sales mix away from our higher margin products, as well as the timing and amount of our software licensing and non-product revenue, could adversely affect our future gross margin percentages. In addition, increased competition and the existence of product alternatives, more complex engineering requirements, lower demand, industry oversupply or reductions in our technological lead compared to our competitors, and other factors have in the past and may in the future lead to further price erosion, lower revenue and lower margin. Conversely, periods of robust demand that create a supply imbalance, as we have seen recently, can lead to higher gross margins that may not be sustainable over the longer term.

In addition, semiconductor manufacturing requires significant capital investment, leading to high fixed costs, including depreciation expense. If we are unable to utilize our owned manufacturing facilities at a high level, the fixed costs associated with these facilities will not be fully absorbed, resulting in higher average unit costs and a lower gross margin. Furthermore, we do not hedge our exposure to commodity prices, some of which are very volatile, and sudden or prolonged increases in commodities prices may adversely affect our gross margin.

Our gross margin may also be adversely affected if businesses or companies that we acquire have different gross margin profiles and by expenses related to such acquisitions.

We utilize a significant amount of IP in our business. If we are unable or fail to protect our IP, our business could be adversely affected.

Our success depends in part upon protecting our IP. To accomplish this, we rely on a combination of IP rights, including patents, copyrights, trademarks and trade secrets, as well as customary contractual protections with our customers, suppliers, employees and consultants. We spend significant resources to monitor and protect our IP rights, including the unauthorized use of our products, usage rates of the software seat licenses and subscriptions that we sell, and even with significant expenditures, we may not be able to protect the IP rights that are valuable to our business. We are unable to predict or assure that:

- our IP rights will not lapse or be invalidated, circumvented, challenged, or, in the case of third-party IP rights licensed to us, be licensed to others;

[Table of Contents](#)

- our IP rights will provide competitive advantages to us;
- rights previously granted by third parties to IP licensed or assigned to us, including portfolio cross-licenses, will not hamper our ability to assert our IP rights or hinder the settlement of currently pending or future disputes;
- any of our pending or future patent, trademark or copyright applications will be issued or have the coverage originally sought;
- our IP rights will be enforced in certain jurisdictions where competition is intense or where legal protection may be weak; or
- we have sufficient IP rights to protect our products or our business.

Effective IP protection may be unavailable or more limited in other jurisdictions, relative to those protections available in the U.S., and may not be applied for or may be abandoned in one or more relevant jurisdictions. In addition, when patents expire, we lose the protection and competitive advantages they provided to us.

We also generate revenue from licensing royalty payments and from technology claim settlements relating to certain of our IP. Licensing of our IP rights, particularly exclusive licenses, may limit our ability to assert those IP rights against third parties, including the licensee of those rights. In addition, we may acquire companies with IP that is subject to licensing obligations to other third parties. These licensing obligations may extend to our own IP following any such acquisition and may limit our ability to assert our IP rights. From time to time, we pursue litigation to assert our IP rights, including, in some cases, against our customers and suppliers. Claims of this sort could also harm our relationships with our customers and might deter future customers from doing business with us. Conversely, third parties have and may in the future pursue IP litigation against us, including as a result of our IP licensing business. An adverse decision in such types of legal action could limit our ability to assert our IP rights and limit the value of our technology, including the loss of opportunities to sell or license our technology to others or to collect royalty payments, which could otherwise negatively impact our business, financial condition and results of operations.

From time to time, we may need to obtain additional IP licenses or renew existing license agreements. We are unable to predict whether these license agreements can be obtained or renewed on acceptable terms or at all.

If our software products do not remain compatible with ever-changing operating environments, platforms, or third-party products, demand for our products and services could decrease, which could materially adversely affect our business.

We may be required to make substantial modification of our products to maintain compatibility with operating systems, systems software and computer hardware used by our customers or to provide our customers with desired features or capabilities. We must also continually address the challenges of dynamic and accelerating market trends and competitive developments, such as the emergence of advanced persistent threats in the security space to compete effectively. There can be no assurance that we will be able to adapt our products in response to these developments.

Further, our software solutions interact with a variety of software and hardware developed by third parties. If we lose access to third-party code and specifications for the development of code, this could negatively impact our ability to develop compatible software. In addition, if software providers and hardware manufacturers, including some of our largest vendors, adopt new policies restricting the use or availability of their code or technical documentation for their operating systems, applications, or hardware, or otherwise impose unfavorable terms and conditions for such access, this could result in higher research and development costs for the enhancement and modification of our existing products or development of new products. Any additional restrictions could materially adversely affect our business, financial condition and operating results and cash flow.

Failure to enter into software license agreements on a satisfactory basis could materially adversely affect our business.

Many of our existing customers have multi-year enterprise software license agreements, some of which involve substantial aggregate fee amounts. Customer renewal rates may decline or fluctuate as a result of a number of factors, including the level of customer satisfaction with our solutions or customer support, customer budgets and the pricing of our solutions as compared with the solutions offered by our competitors, any of which may cause our revenue to grow more slowly than expected, if at all. The failure to renew customer agreements of similar scope, on terms that are commercially attractive to us, could materially adversely affect our business, financial condition and operating results and cash flow.

Certain software that we use in our products is licensed from third parties and may not be available to us in the future, which may delay product development and production or cause us to incur additional expense.

Some of our solutions contain software licensed from third parties, some of which may not be available to us in the future on terms that are acceptable to us or allow our products to remain competitive. The loss of these licenses or the inability to

maintain any of them on commercially acceptable terms could delay development of future products or the enhancement of existing products.

Certain software we use is from open source code sources, which, under certain circumstances could materially adversely affect our business, financial condition, operating results and cash flow.

Some of our products contain software from open source code sources, the use of which may subject us to certain conditions, including the obligation to offer such products for no cost or to make the proprietary source code of those products publicly available. Further, although some open source vendors provide warranty and support agreements, it is common for such software to be available “as-is” with no warranty, indemnity or support. Although we monitor our use of such open source code to avoid subjecting our products to unintended conditions, such use, under certain circumstances, could materially adversely affect our business, financial condition and operating results and cash flow, including if we are required to take remedial action that may divert resources away from our development efforts.

Failure of our software products to manage and secure IT infrastructures and environments could have a material adverse effect on our business.

Certain aspects of our software products are intended to manage and secure IT infrastructures and environments, and as a result, we expect these products to be ongoing targets of cyber security attacks. Open source code or other third-party software used in these products could also be targeted. Although we continually seek to improve our countermeasures to prevent such incidents, we may be unable to anticipate every scenario and it is possible that certain cyber threats or vulnerabilities will be undetected or unmitigated in time to prevent an attack or an accidental incident on us and our customers. Additionally, efforts by malicious cyber actors or others could cause interruptions, delays or cessation of our product licensing, or modification of our software, which could cause us to lose existing or potential customers. A successful cyber security attack involving our products could cause customers and potential customers to believe our services are ineffective or unreliable and result in, among other things, the loss of customers, unfavorable publicity, damage to our reputation, difficulty in marketing our products, allegations by our customers that we have not performed our contractual obligations and give rise to significant costs, including costs related to developing solutions or indemnification obligations under our agreements. Any such event could adversely impact our revenue and results of operations. See also “*An impairment of the confidentiality, integrity, or availability of our IT systems, or those of one or more of our corporate infrastructure vendors, could have a material adverse effect on our business*”.

We are subject to warranty claims, product recalls and product liability.

From time to time, we may be subject to warranty or product liability claims that may lead to significant expense. Our customer contracts typically contain warranty and indemnification provisions, and in certain cases may also contain liquidated damages provisions, relating to product quality issues. The potential liabilities associated with such provisions are significant, and in some cases, including in agreements with some of our largest customers, are potentially unlimited. Any such liabilities may greatly exceed any revenue we receive from the relevant products. Costs, payments or damages incurred or paid by us in connection with warranty and product liability claims and product recalls could materially adversely affect our financial condition and results of operations. We may also be exposed to such claims as a result of any acquisition we may undertake in the future.

Product liability insurance is subject to significant deductibles and there is no guarantee that such insurance will be available or adequate to protect against all such claims, or we may elect to self-insure with respect to certain matters. For example, it is possible for one of our customers to recall a product containing one of our semiconductor devices. In such an event, we may incur significant costs and expenses, including among others, replacement costs, contract damage claims from our customers and reputational harm. Although we maintain reserves for reasonably estimable liabilities and purchase product liability insurance, our reserves may be inadequate to cover the uninsured portion of such claims. Conversely, in some cases, amounts we reserve may ultimately exceed our actual liability for particular claims and may need to be reversed.

The complexity of our products could result in unforeseen delays or expense or undetected defects or bugs, which could adversely affect the market acceptance of new products, damage our reputation with current or prospective customers, and materially and adversely affect our operating costs.

Highly complex products, such as those we offer, may contain defects and bugs when they are first introduced or as new versions, software documentation or enhancements are released, or their release may be delayed due to unforeseen difficulties during product development. If any of our products or third-party components used in our products, contain defects or bugs, or have reliability, quality or compatibility problems, we may not be able to successfully design workarounds. Furthermore, if any of these problems are not discovered until after we have commenced commercial production or deployment of a new product, we may be required to incur additional development costs and product recall, repair or replacement costs. Significant technical challenges also arise with our software products because our customers license and deploy our products across a variety of computer platforms and integrate them with a number of third-party software

applications and databases. As a result, if there is system-wide failure or an actual or perceived breach of information integrity, security or availability occurs in one of our end-user customer's system, it can be difficult to determine which product is at fault and we could ultimately be harmed by the failure of another supplier's product. Consequently, our reputation may be damaged and customers may be reluctant to buy our products, which could materially and adversely affect our ability to retain existing customers and attract new customers. To resolve these problems, we may have to invest significant capital and other resources and we would likely lose, or experience a delay in, market acceptance of the affected product or products. These problems may also result in claims against us by our customers or others. For example, if a delay in the manufacture and delivery of our products causes the delay of a customer's end-product delivery, we may be required, under the terms of our agreement with that customer, to compensate the customer for the adverse effects of such delays. As a result, our financial results could be materially adversely affected.

We make substantial investments in research and development and unsuccessful investments could materially adversely affect our business, financial condition and results of operations.

The industries in which we compete are characterized by rapid technological change, changes in customer requirements, frequent new product introductions and enhancements, short product cycles and evolving industry standards, and new delivery methods. In addition, semiconductor products transition over time to increasingly smaller line width geometries and failure to successfully transition to smaller geometry process technologies could impair our competitive position. In order to remain competitive, we have made, and expect to continue to make, significant investments in research and development. If we fail to develop new and enhanced products and technologies, if we focus on technologies that do not become widely adopted, or if new competitive technologies that we do not support become widely accepted, demand for our products may be reduced. Increased investments in research and development or unsuccessful research and development efforts could cause our cost structure to fall out of alignment with demand for our products, which would have a negative impact on our financial results.

We collect, use, store, or otherwise process personal information, which subjects us to privacy and data security laws and contractual commitments, and our actual or perceived failure to comply with such laws and commitments could harm our business.

We collect, use and store (collectively, "process") a high volume, variety and velocity of certain personal information in connection with the operation of our business. This creates various levels of privacy risks across different parts of our business, depending on the type of personal information, the jurisdiction in question and the purpose of their processing. The personal information we process is subject to an increasing number of federal, state, local, and foreign laws and regulations regarding privacy and data security, as well as contractual commitments. Privacy legislation and other data protection regulations, enforcement and policy activity in this area are expanding rapidly in many jurisdictions and creating a complex regulatory compliance environment. Sectoral legislation, certification requirements and technical standards applying to certain categories of our customers, such as those in the financial services or public sector, are likely to further exacerbate this trend. The cost of complying with and implementing these privacy-related and data governance measures could be significant as they may create additional burdensome security, business process, business record or data localization requirements. Concerns about government interference, sovereignty, expanding privacy, cyber security and data governance legislation could adversely affect our customers and our products and services, particularly in cloud computing, artificial intelligence and our own data management practices. The theft, loss or misuse of personal data collected, used, stored or transferred by us to run our business could result in significantly increased business and security costs or costs related to defending legal claims. Any inadvertent failure or perceived failure by us to comply with privacy, data governance or cyber security obligations may result in governmental enforcement actions, litigation, substantial fines and damages, and could cause our customers to lose trust in us, which could have an adverse effect on our reputation and business.

We are subject to environmental, health and safety laws, which could increase our costs, restrict our operations and require expenditures that could have a material adverse effect on our results of operations and financial condition.

We are subject to a variety of domestic and international laws and regulations relating to the use, disposal, clean-up of and human exposure to hazardous materials. Compliance with environmental, health and safety requirements could, among other things, require us to modify our manufacturing processes, restrict our ability to expand our facilities, or require us to acquire pollution control equipment, all of which can be very costly. Any failure by us to comply with such requirements could result in the limitation or suspension of the manufacture of our products and could result in litigation against us and the payment of significant fines and damages by us in the event of a significant adverse judgment. In addition, complying with any cleanup or remediation obligations for which we are or become responsible could be costly and have a material adverse effect on our business, financial condition and results of operations.

[Table of Contents](#)

Changing requirements relating to the materials composition of our semiconductor products, including the restrictions on lead and certain other substances in electronic products sold in various countries, including the U.S., China and Japan, and in the European Union, increase the complexity and costs of our product design and procurement operations and may require us to re-engineer our products. Such re-engineering may result in excess inventory or other additional costs and could have a material adverse effect on our results of operations. We may also experience claims from employees from time to time with regard to exposure to hazardous materials or other workplace related environmental claims.

Social and environmental regulations, policies and provisions, as well as customer and investor demands, may make our supply chain more complex and may adversely affect our relationships with customers and investors.

There is an increasing focus on corporate social and environmental responsibility in the semiconductor industry, particularly with OEMs that manufacture consumer electronics. A number of our customers have adopted, or may adopt, procurement policies that include social and environmental responsibility provisions or requirements that their suppliers should comply with, or they may seek to include such provisions or requirements in their procurement terms and conditions. An increasing number of investors are also requiring companies to disclose corporate social and environmental policies, practices and metrics. In addition, various jurisdictions are developing climate change-based laws or regulations that could cause us to incur additional direct costs for compliance, as well as indirect costs resulting from our customers, suppliers, or both incurring additional compliance costs that are passed on to us. These legal and regulatory requirements, as well as investor expectations, on corporate environmental and social responsibility practices and disclosure, are subject to change, can be unpredictable, and may be difficult and expensive for us to comply with, given the complexity of our supply chain and our significant outsourced manufacturing. If we are unable to comply, or are unable to cause our suppliers or CMs to comply, with such policies or provisions or meet the requirements of our customers and investors, a customer may stop purchasing products from us or an investor may sell their shares, and may take legal action against us, which could harm our reputation, revenue and results of operations.

In addition, as part of their corporate social and environmental responsibility programs, an increasing number of OEMs are seeking to source products that do not contain minerals sourced from areas where proceeds from the sale of such minerals are likely to be used to fund armed conflicts, such as in the Democratic Republic of Congo. This could adversely affect the sourcing, availability and pricing of minerals used in the manufacture of semiconductor devices, including our products. As a result, we may face difficulties in satisfying these customers' demands, which may harm our sales and operating results.

The average selling prices of semiconductor products in our markets have often decreased rapidly and may do so in the future, which could harm our revenue and gross profit.

The semiconductor products we develop and sell are used for high volume applications. As a result, the prices of those products have often decreased rapidly. Gross profit on our products may be negatively affected by, among other things, pricing pressures from our customers. In the past, we have reduced the average selling prices of our products in anticipation of future competitive pricing pressures, new product introductions by us or our competitors and other factors. In addition, some of our customer agreements provide for volume-based pricing and product pricing roadmaps, which can also reduce the average selling prices of our products over time. Our margins and financial results will suffer if we are unable to offset any reductions in our average selling prices by increasing our sales volumes, reducing manufacturing costs, or developing new and higher value-added products on a timely basis.

Fluctuations in foreign exchange rates could result in losses.

We operate global businesses and our consolidated financial results are reported in U.S. dollars. However, some of the revenue and expenses of our foreign subsidiaries are denominated in local currencies. Fluctuations in foreign exchange rates against the U.S. dollar could result in substantial changes in reported revenues and operating results due to the foreign exchange impact of translating these transactions into U.S. dollars.

In the normal course of business, we employ various hedging strategies to partially mitigate these risks, including the use of derivative instruments. These strategies may not be effective in protecting us against the effects of fluctuations in foreign exchange rates. As a result, fluctuations in foreign exchange rates could result in financial losses.

Risks Related to Our Taxes

Changes in tax legislation or policies could materially impact our financial position and results of operations.

Corporate tax reform, anti-base-erosion rules and tax transparency continue to be high priorities in many jurisdictions. As a result, policies regarding corporate income and other taxes in numerous jurisdictions are under heightened scrutiny and tax reform legislation has been, and will likely continue to be, proposed or enacted in a number of jurisdictions in which we operate.

[Table of Contents](#)

After enactment of the U.S. Tax Cuts and Jobs Act, most of our income is taxable in the U.S. with a significant portion taxable under the Global Intangible Low-Taxed Income (“GILTI”) regime. Beginning in fiscal year 2027, the deduction allowable under the GILTI regime will decrease from 50% to 37.5%, which will increase the effective tax rate imposed on our income. The U.S. also enacted the Inflation Reduction Act of 2022 (“IRA”) in August 2022, which creates a new book minimum tax of at least 15% of consolidated GAAP pre-tax income for corporations with average book income in excess of \$1 billion. The book minimum tax will first apply to our fiscal year 2024 and any increase in our effective tax rate or cash tax will depend on a number of factors, including any offsets for foreign tax credits or general business credits, or changes in book income following business combinations. The IRA also creates an excise tax of 1% of the value of any stock repurchased by us after December 31, 2022. We could be subject to this new excise tax, but the amount will vary depending on various factors, including the amount and frequency of any stock repurchases, applicability in certain business combination transactions, and any permitted reductions or exceptions to the amount subject to the tax. If (i) the U.S. tax rate increases, (ii) the deduction allowable under the GILTI regime is further reduced or eliminated, (iii) additional limitations are put on our ability to deduct interest expense, or (iv) the requirement for research and development costs to be capitalized beginning in fiscal year 2023 remains in effect, our provision for income taxes, net income, and cash flows would be adversely impacted.

In addition, many countries are implementing legislation and other guidance to align their international tax rules with the Organisation for Economic Co-operation and Development’s (“OECD”) Base Erosion and Profit Shifting recommendations and action plan that aim to standardize and modernize global corporate tax policy, including changes to cross-border tax, transfer pricing documentation rules, and nexus-based tax incentive practices. The OECD is also continuing discussions surrounding fundamental changes in allocation of profits among tax jurisdictions in which companies do business, as well as the implementation of a global minimum tax (namely the “Pillar One” and “Pillar Two” proposals). Some countries intend to implement laws based on Pillar Two proposals, which may adversely impact our provision for income taxes, net income and cash flows. As a result of this heightened scrutiny, prior decisions by tax authorities regarding treatments and positions of corporate income taxes could be subject to enforcement activities, and legislative investigation and inquiry, which could also result in changes in tax policies or prior tax rulings. Any such changes may also result in the taxes we previously paid being subject to change.

Any substantial changes in domestic or international corporate tax policies, regulations or guidance, enforcement activities or legislative initiatives may materially adversely affect our business, the amount of taxes we are required to pay and our financial condition and results of operations generally.

If the tax incentives or tax holiday arrangements we have negotiated change or cease to be in effect or applicable for any reason, or if our assumptions and interpretations regarding tax laws and incentives or holiday arrangements prove to be incorrect, our corporate income taxes could significantly increase.

Our operations are currently structured to benefit from the various tax incentives extended to us in various jurisdictions to encourage investment or employment. For example, absent our principal tax incentives from the Singapore Economic Development Board, which is scheduled to expire in 2025, the corporate income tax rate that would apply to our Singapore taxable income would be 17%. We also have a tax holiday on our qualifying income in Malaysia, which is scheduled to expire in fiscal year 2028. Each tax incentive and tax holiday is subject to our compliance with various operating and other conditions and may, in some instances, be amended or terminated prior to their scheduled termination date by the relevant governmental authority. If we cannot, or elect not to, comply with the operating conditions included in any particular tax incentive or tax holiday, we could, in some instances, be required to refund previously realized material tax benefits, or if such tax incentive or tax holiday is terminated prior to its expiration absent a new incentive applying, we will lose the related tax benefits earlier than scheduled. In addition, we may be required, or elect, to modify our operational structure and tax strategy in order to keep an incentive, which could result in a decrease in the benefits of the incentive. Our tax incentives could also be adversely impacted if the global minimum tax provisions (Pillar Two) are adopted in a country in which we have an existing tax incentive. Our tax incentives and tax holiday, before taking into consideration U.S. foreign tax credits, decreased the provision for income taxes by approximately \$1,821 million in the aggregate and increased diluted net income per share by \$4.31 for fiscal year 2022.

Our interpretations and conclusions regarding the tax incentives are not binding on any taxing authority, and if our assumptions about tax and other laws are incorrect or if these tax incentives are substantially modified or rescinded, we could suffer material adverse tax and other financial consequences, which would increase our expenses, reduce our profitability and adversely affect our cash flows.

Our income taxes and overall cash tax costs are affected by a number of factors that could materially, adversely affect financial results.

Significant judgment is required in determining our worldwide income taxes. In the ordinary course of our business, there are many transactions where the ultimate tax determination is uncertain. Additionally, our calculations of income taxes payable currently and on a deferred basis are based on our interpretations of applicable tax laws in the jurisdictions in which

[Table of Contents](#)

we are required to file tax returns. Although we believe our tax estimates are reasonable, there is no assurance that the final determination of our income tax liability will not be materially different than what is reflected in our income tax provisions and accruals.

Our income taxes are subject to volatility and could be adversely affected by numerous factors including:

- reorganization or restructuring of our businesses, tangible and intangible assets, outstanding indebtedness and corporate structure;
- jurisdictional mix of our income and assets;
- changes in the allocation of income and expenses, including adjustments related to changes in our corporate structure, acquisitions or tax law;
- changes in U.S. and foreign tax laws and regulations, changes to the taxation of earnings of foreign subsidiaries, taxation of U.S. income generated from foreign sources, the deductibility of expenses attributable to income and foreign tax credit rules;
- tax effects of increases in non-deductible employee compensation; and
- changes in tax accounting rules or principles and in the valuation of deferred tax assets and liabilities.

We have adopted transfer pricing policies that call for the provision of services, the sale of products, the arrangement of financing and the grant of licenses from one affiliate to another at prices that we believe are negotiated on an arm's length basis. Our taxable income is dependent upon acceptance by local authorities that our operational practices and intercompany transfer pricing are on an arm's length basis. Due to inconsistencies in application of the arm's length standard among taxing authorities, as well as lack of comprehensive treaty-based protection, transfer pricing challenges by tax authorities could, if successful, result in adjustments for prior or future years. The effects of any such changes could subject us to higher taxes and our earnings, results of operations and cash flow would be adversely affected.

In addition, we are subject to, and are under, tax audit in various jurisdictions, and such jurisdictions may assess additional income tax against us. Although we believe our tax positions are reasonable, the final determination of tax audits could be materially different from our income tax provisions and accruals. The ultimate result of an audit could have a material adverse effect on our results of operations and cash flows in the period or periods for which that determination is made.

Risks Related to Our Indebtedness

Our substantial indebtedness could adversely affect our financial health and our ability to execute our business strategy.

As of October 30, 2022, the aggregate indebtedness under our senior notes was \$41,218 million. This amount does not reflect any debt we expect to incur or assume in connection with the VMware Merger.

Our substantial indebtedness could have important consequences including:

- increasing our vulnerability to adverse general economic and industry conditions;
- exposing us to interest rate risk due to our variable rate term facilities, which we do not typically hedge against;
- limiting our flexibility in planning for, or reacting to, changes in the economy and the semiconductor industry;
- placing us at a competitive disadvantage compared to our competitors with less indebtedness;
- making it more difficult to borrow additional funds in the future to fund growth, acquisitions, working capital, capital expenditures and other purposes; and
- potentially requiring us to dedicate a substantial portion of our cash flow from operations to payments on our indebtedness, thereby reducing the availability of our cash flow to fund our other business needs.

We receive debt ratings from the major credit rating agencies in the U.S. Factors that may impact our credit ratings include debt levels, planned asset purchases or sales and near-term and long-term production growth opportunities. Liquidity, asset quality, cost structure, reserve mix and commodity pricing levels could also be considered by the rating agencies. While we are focused on maintaining investment grade ratings from these agencies, we may be unable to do so. Any downgrade in our credit rating or the ratings of our indebtedness, or adverse conditions in the debt capital markets, could:

- adversely affect the trading price of, or market for, our debt securities;
- increase interest expense under our term facilities;
- increase the cost of, and adversely affect our ability to refinance, our existing debt; and
- adversely affect our ability to raise additional debt.

The instruments governing our indebtedness impose certain restrictions on our business.

The instruments governing our indebtedness contain certain covenants imposing restrictions on our business. These restrictions may affect our ability to operate our business, to plan for, or react to, changes in the market conditions or our capital needs and may limit our ability to take advantage of potential business opportunities as they arise. The restrictions placed on us include maintenance of an interest coverage ratio and limitations on our ability to incur certain secured debt, enter into certain sale and lease-back transactions and consolidate, merge, sell or otherwise dispose of all or substantially all of our assets. In addition, the instruments contain customary events of default upon the occurrence of which, after any applicable grace period, the indebtedness could be declared immediately due and payable. In such event, we may not have sufficient available cash to repay such debt at the time it becomes due, or be able to refinance such debt on acceptable terms or at all. Any of the foregoing could materially adversely affect our business, financial condition and results of operations.

Servicing our debt requires a significant amount of cash, and we may not have sufficient cash flow from our business to pay our substantial debt.

Our ability to make scheduled payments of the principal of, to pay interest on, and to refinance our debt, depends on our future performance, which is subject to economic, financial, competitive and other factors. Our business may not continue to generate cash flow from operations in the future sufficient to satisfy our obligations under our current indebtedness and any future indebtedness we may incur and to make necessary capital expenditures. If we are unable to generate such cash flow, we may be required to adopt one or more alternatives, such as reducing or delaying investments or capital expenditures, selling assets, refinancing or obtaining additional equity capital on terms that may be onerous or highly dilutive. Our ability to refinance our outstanding indebtedness or future indebtedness will depend on the capital markets and our financial condition at such time. We may not be able to engage in any of these activities or engage in these activities on desirable terms when needed, which could result in a default on our indebtedness.

Risks Related to Owning Our Common Stock

At times, our stock price has been volatile and it may fluctuate substantially in the future, which could result in substantial losses for our investors as well as class action litigation against us and our management which could cause us to incur substantial costs and divert our management's attention and resources.

The trading price of our common stock has, at times, fluctuated significantly and could be subject to wide fluctuations in response to any of the risk factors listed in this "Risk Factors" section, and others, including:

- issuance of new or updated research or other reports by securities analysts;
- fluctuations in the valuation and results of operations of our significant customers as well as companies perceived by investors to be comparable to us;
- announcements of proposed acquisitions by us or our competitors;
- announcements of, or expectations of, additional debt or equity financing transactions;
- stock price and volume fluctuations attributable to inconsistent trading volume levels of our common stock;
- hedging or arbitrage trading activity involving our common stock; and
- unsubstantiated news reports or other inaccurate publicity regarding us or our business.

These fluctuations are often unrelated or disproportionate to our operating performance. Broad market and industry fluctuations, as well as general economic, political and market conditions such as recessions, interest rate changes or currency fluctuations, may negatively impact the market price of our common stock. You may not realize any return on your investment in us and may lose some or all of your investment. In the past, companies that have experienced volatility in the market price of their stock have been subject to securities class action litigation. We may be the target of this type of litigation in the future. In addition, we have been, and in the future we may be, subject to lawsuits stemming from our acquisitions, including the VMware Merger. Securities litigation against us, including the lawsuits related to such acquisitions, could result in substantial costs and divert our management's attention from other business concerns, which could seriously harm our business.

The amount and frequency of our stock repurchases may fluctuate.

The amount, timing and execution of our stock repurchase program may fluctuate based on our priorities for the use of cash for other purposes. These purposes include operational spending, capital spending, acquisitions, repayment of debt and returning cash to our stockholders as dividend payments. Changes in cash flows, tax laws and our stock price could also impact our stock repurchase program. We are not obligated to repurchase any specific amount of shares of common stock, and the stock repurchase program may be suspended or terminated at any time.

A substantial amount of our stock is held by a small number of large investors and significant sales of our common stock by one or more of these holders could cause our stock price to fall.

As of September 30, 2022, we believe 10 of our 20 largest holders of common stock were active institutional investors who held approximately 27% of our outstanding shares of common stock in the aggregate. These investors may sell their shares at any time for a variety of reasons and such sales could depress the market price of our common stock. In addition, any such sales of our common stock by these entities could also impair our ability to raise capital through the sale of additional equity securities.

There can be no assurance that we will continue to declare cash dividends.

Our Board of Directors has adopted a dividend policy pursuant to which we currently pay a cash dividend on our common stock on a quarterly basis. The declaration and payment of any dividend is subject to the approval of our Board of Directors and our dividend may be discontinued or reduced at any time. Because we are a holding company, our ability to pay cash dividends is also limited by restrictions or limitations on our ability to obtain sufficient funds through dividends from subsidiaries. There can be no assurance that we will declare cash dividends in the future in any particular amounts, or at all.

ITEM 1B. UNRESOLVED STAFF COMMENTS

None.

ITEM 2. PROPERTIES

We are headquartered in San Jose, California and our primary warehouse is located in Malaysia. We conduct our administration, manufacturing, research and development, sales and marketing in both owned and leased facilities. We believe that our owned and leased facilities are adequate for our present operations. We do not identify or allocate assets by operating segment.

As of October 30, 2022, our owned and leased facilities in excess of 100,000 square feet consisted of:

(In square feet)	United States	Other Countries	Total
Owned facilities ¹	2,586,368	928,888	3,515,256
Leased facilities ²	796,508	1,310,661	2,107,169
Total facilities	3,382,876	2,239,549	5,622,425

¹ Includes 318,000 square feet and 153,000 square feet of property owned in Malaysia subject to a 60-year land lease with the state authority expiring in May 2051 and March 2077, respectively, subject to renewal at our option.

² Building leases expire on varying dates through February 2046 and generally include renewals at our option.

ITEM 3. LEGAL PROCEEDINGS

The information set forth under Note 14. "Commitments and Contingencies" included in Part II, Item 8. of this Annual Report on Form 10-K, is incorporated herein by reference. For an additional discussion of certain risks associated with legal proceedings, see "Risk Factors" above.

ITEM 4. MINE SAFETY DISCLOSURES

None.

PART II

ITEM 5. MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES

Market Information

Broadcom common stock is listed on The Nasdaq Global Select Market under the symbol "AVGO".

Holders

As of November 25, 2022, there were 1,060 holders of record of our common stock. A substantially greater number of stockholders are "street name" or beneficial holders, whose shares are held of record by banks, brokers and other financial institutions.

Unregistered Sales of Equity Securities

On August 1, 2022, we issued 9,923 restricted shares of our common stock to one individual in connection with our acquisition of a company. The restrictions lapse over three years subject to the individual's continued employment. The issuance of these shares was exempt from registration under the Securities Act of 1933, as amended, in reliance upon Section 4(a)(2) thereof.

Issuer Purchases of Equity Securities

During the fiscal quarter ended October 30, 2022, we paid approximately \$274 million in employee withholding taxes due upon the vesting of net settled equity awards. We withheld approximately 1 million shares of common stock from employees in connection with such net share settlement at an average price of \$502.62 per share. These shares may be deemed to be "issuer purchases" of shares.

In December 2021, our Board of Directors authorized a stock repurchase program to repurchase up to \$10 billion of our common stock from time to time on or prior to December 31, 2022. During fiscal year 2022, we repurchased and retired approximately 12 million shares of our common stock for \$7 billion under this stock repurchase program.

In May 2022, our Board of Directors authorized another stock repurchase program to repurchase up to an additional \$10 billion of our common stock from time to time through December 31, 2023.

Repurchases under our stock repurchase programs may be effected through a variety of methods, including open market or privately negotiated purchases. The timing and amount of shares repurchased will depend on the stock price, business and market conditions, corporate and regulatory requirements, alternative investment opportunities, acquisition opportunities, and other factors. We are not obligated to repurchase any specific amount of shares of common stock, and the stock repurchase programs may be suspended or terminated at any time.

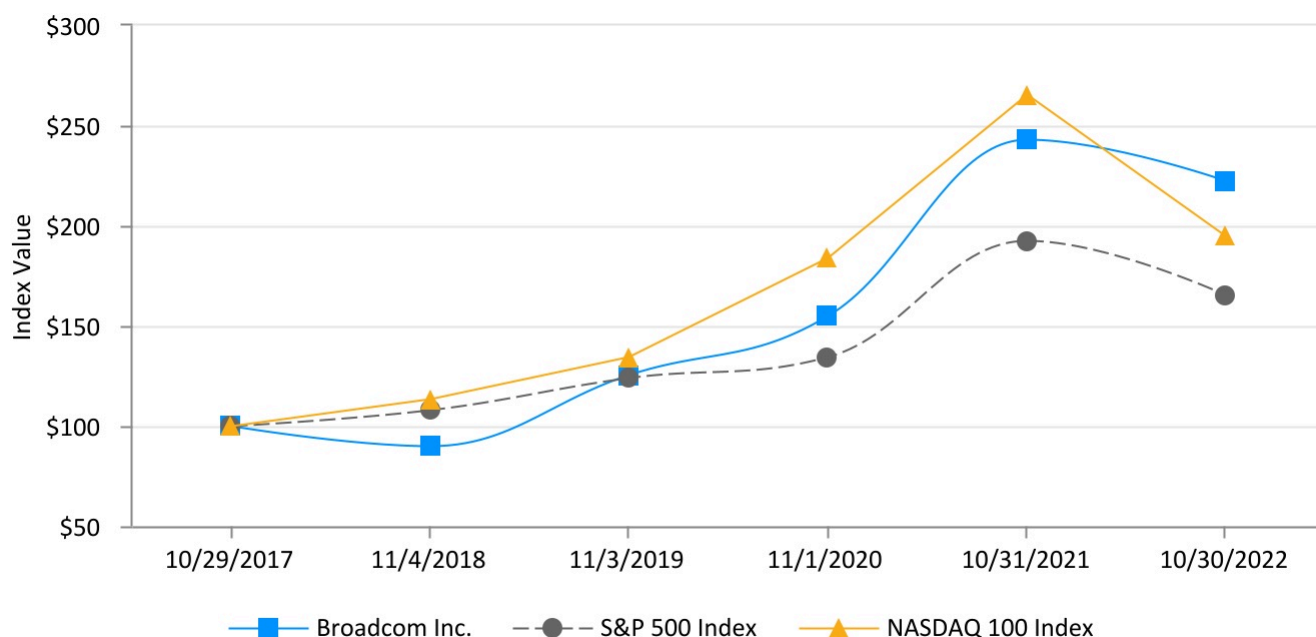
Stock Performance Graph

The following graph shows a comparison of cumulative total return for our common stock, the Standard & Poor's 500 Stock Index (the "S&P 500 Index") and the NASDAQ 100 Index for the five fiscal years ended October 30, 2022. The total return graph and table assume that \$100 was invested on October 27, 2017 (the last trading day of our fiscal year 2017) in each of Broadcom Inc. common stock, the S&P 500 Index and the NASDAQ 100 Index and assume that all dividends are reinvested. Indexes are calculated on a month-end basis.

The comparisons in the graph below are based on historical data and are not indicative of, or intended to forecast, the possible future performance of our common stock.

Comparison of Five Year Cumulative Total Return

Among Broadcom Inc., the S&P 500 Index and the NASDAQ 100 Index



	October 29, 2017	November 4, 2018	November 3, 2019	November 1, 2020	October 31, 2021	October 30, 2022
Broadcom Inc.	\$ 100.00	\$ 89.74	\$ 125.30	\$ 154.78	\$ 242.74	\$ 222.41
S&P 500 Index	\$ 100.00	\$ 107.58	\$ 123.66	\$ 134.35	\$ 192.01	\$ 165.18
NASDAQ 100 Index	\$ 100.00	\$ 113.29	\$ 134.24	\$ 183.52	\$ 265.05	\$ 194.63

The graph and the table above shall not be deemed "filed" with the SEC for the purposes of Section 18 of the Exchange Act or otherwise subject to the liabilities of that section, nor shall it be deemed incorporated by reference in any filing made by us with the SEC, regardless of any general incorporation language in such filing.

ITEM 6. [RESERVED]

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

This Management's Discussion and Analysis of Financial Condition and Results of Operations should be read in conjunction with our consolidated financial statements and notes thereto, which appear elsewhere in this Annual Report on Form 10-K. This discussion may contain forward-looking statements based upon current expectations that involve risks and uncertainties. Our actual results may differ materially from those anticipated in these forward-looking statements as a result of various factors, including those set forth under the caption "Risk Factors" or in other parts of this Annual Report on Form 10-K.

The following section generally discusses our financial condition and results of operations for our fiscal year ended October 30, 2022 ("fiscal year 2022") compared to our fiscal year ended October 31, 2021 ("fiscal year 2021"). A discussion regarding our financial condition and results of operations for fiscal year 2021 compared to our fiscal year ended November 1, 2020 ("fiscal year 2020") can be found in Part II, Item 7 of our Annual Report on Form 10-K for fiscal year 2021, filed with the Securities and Exchange Commission (the "SEC") on December 17, 2021.

Overview

We are a global technology leader that designs, develops and supplies a broad range of semiconductor and infrastructure software solutions. We develop semiconductor devices with a focus on complex digital and mixed signal complementary metal oxide semiconductor based devices and analog III-V based products. We have a history of innovation in the semiconductor industry and offer thousands of products that are used in end products such as enterprise and data center networking, home connectivity, set-top boxes, broadband access, telecommunication equipment, smartphones and base stations, data center servers and storage systems, factory automation, power generation and alternative energy systems, and electronic displays. Our infrastructure software solutions enable customers to plan, develop, automate, manage and secure applications across mainframe, distributed, mobile and cloud platforms. Our portfolio of industry-leading infrastructure and security software is designed to modernize, optimize, and secure the most complex hybrid environments, enabling scalability, agility, automation, insights, resiliency and security. We also offer mission critical fibre channel storage area networking ("FC SAN") products and related software in the form of modules, switches and subsystems incorporating multiple semiconductor products.

We have two reportable segments: semiconductor solutions and infrastructure software. Our semiconductor solutions segment includes all of our product lines and intellectual property ("IP") licensing. Our infrastructure software segment includes our mainframe, distributed and cyber security solutions, and our FC SAN business.

Our strategy is to combine best-of-breed technology leadership in semiconductor and infrastructure software solutions, with unmatched scale, on a common sales and administrative platform to deliver a comprehensive suite of infrastructure technology products to the world's leading business and government customers. We seek to achieve this through responsibly financed acquisitions of category-leading businesses and technologies, as well as investing extensively in research and development, to ensure our products retain their technology leadership. This strategy results in a robust business model designed to drive diversified and sustainable operating and financial results.

The demand for our products has been affected in the past, and is likely to continue to be affected in the future, by various factors, including the following:

- gain or loss of significant customers;
- general economic and market conditions in the industries and markets in which we compete;
- our distributors' product inventory and end customer demand;
- the rate at which our present and future customers and end-users adopt our products and technologies in our target markets, and the rate at which our customers' products that include our technology are accepted in their markets;
- the shift to cloud-based information technology solutions and services, such as hyperscale computing, which may adversely affect the timing and volume of sales of our products for use in traditional enterprise data centers; and
- the timing, rescheduling or cancellation of expected customer orders.

COVID-19 Update

The COVID-19 pandemic and the efforts to control it disrupted, and reduced the efficiency of, normal business activities in much of the world. The pandemic resulted in authorities around the world implementing numerous unprecedented measures that created supply chain and market disruption, impacting our workforce and operations, and those of our customers, contract manufacturers, suppliers and logistics providers.

[Table of Contents](#)

While the demand environment for our semiconductor products was consistent with our expectations for fiscal year 2022, with robust and increased profitability driven by the supply imbalance, the macroeconomic environment remains uncertain and it may not be sustainable over the longer term. We continue to experience various constraints in our supply chain, including with respect to wafers and substrates. Although supply lead times have stabilized, we continue to have difficulties in obtaining some necessary components and inputs in a timely manner to meet demand.

In response to the pandemic, we have taken extensive measures to protect the health and safety of our employees and contractors at our facilities. We continue to monitor the implications of the pandemic on our operations and may modify our business practices and policies from time to time.

Our ability to predict the impact of the pandemic on our business remains limited and its effects on our business are unlikely to be fully realized, or reflected in our financial results, until future periods.

Fiscal Year Highlights

Highlights during fiscal year 2022 include the following:

- We generated \$16,736 million of cash from operations.
- We paid \$7,032 million in cash dividends.
- We repurchased \$7,000 million of common stock.

Pending Acquisition of VMware, Inc.

On May 26, 2022, we entered into an Agreement and Plan of Merger (the “VMware Merger Agreement”) to acquire all of the outstanding shares of VMware, Inc. (“VMware”) in a cash-and-stock transaction (the “VMware Merger”) that values VMware at approximately \$61 billion, based on the closing price of Broadcom common stock on May 25, 2022. We will also assume VMware’s closing date outstanding debt, net of expected cash.

Under the terms of the VMware Merger Agreement, each share of VMware common stock issued and outstanding immediately prior to the effective time of the VMware Merger will be indirectly converted into the right to receive, at the election of the holder of such share of VMware common stock, either \$142.50 in cash, without interest, or 0.2520 shares of Broadcom common stock. The stockholder election will be subject to proration, such that the total number of shares of VMware common stock entitled to receive cash and the total number of shares of VMware common stock entitled to receive Broadcom common stock, will, in each case, be equal to 50% of the aggregate number of shares of VMware common stock issued and outstanding immediately prior to the effective time of the VMware Merger.

We will assume all outstanding VMware restricted stock unit (“RSU”) awards and performance stock unit awards held by continuing employees. The assumed awards will be converted into RSU awards for shares of Broadcom common stock. All outstanding in-the-money VMware stock options and RSU awards held by non-employee directors will be accelerated and converted into the right to receive cash and shares of Broadcom common stock, in equal parts.

Effective upon the effective time of the VMware Merger, one member of the VMware Board of Directors, to be mutually agreed by us and VMware, will be added to our Board of Directors.

In connection with the execution of the VMware Merger Agreement, we entered into a commitment letter on May 26, 2022, with certain financial institutions that committed to provide, subject to the terms and conditions of the commitment letter, a senior unsecured bridge facility in an aggregate principal amount of \$32 billion.

The VMware Merger, which is expected to be completed in our fiscal year ending October 29, 2023 (“fiscal year 2023”), is subject to satisfaction or waiver of customary closing conditions, including the expiration or termination of the waiting period under the Hart-Scott-Rodino Antitrust Improvement Act of 1976 and clearance under the antitrust laws of the European Union and certain other jurisdictions. On October 3, 2022, we registered approximately 59 million shares of our common stock. On November 4, 2022, VMware stockholders adopted the VMware Merger Agreement. We and VMware each have termination rights under the VMware Merger Agreement and, under specified circumstances, upon termination of the agreement, we and VMware would be required to pay the other a termination fee of \$1.5 billion.

Net Revenue

A majority of our net revenue is derived from sales of a broad range of semiconductor devices that are incorporated into electronic products, as well as from modules, switches and subsystems. Net revenue is also generated from the sale of software solutions that enable our customers to plan, develop, automate, manage, and secure applications across mainframe, distributed, mobile, and cloud platforms.

Our overall net revenue, as well as the percentage of total net revenue generated by sales in our semiconductor solutions and infrastructure software segments, have varied from quarter to quarter, due largely to fluctuations in end-

market demand, including the effects of seasonality, which are discussed in detail in Part I, Item 1. *Business* under “Seasonality” of this Annual Report on Form 10-K.

Distributors and original equipment manufacturers (“OEMs”), or their contract manufacturers, typically account for the substantial majority of our semiconductor sales. To serve customers around the world, we have strategically developed relationships with large global electronic component distributors, complemented by a number of regional distributors with customer relationships based on their respective product ranges. We have established strong relationships with leading OEM customers across multiple target markets. Our direct sales force focuses on supporting our large OEM customers and has specialized product and service knowledge that enables us to sell specific offerings at key levels throughout a customer’s organization. Certain customers require us to contract with them directly and with specified intermediaries, such as contract manufacturers. Many of our major customer relationships have been in place for many years and are often the result of years of collaborative product development. This has enabled us to build our extensive IP portfolio and develop critical expertise regarding our customers’ requirements, including substantial system-level knowledge. This collaboration has provided us with key insights into our customers’ businesses and has enabled us to be more efficient and productive and to better serve our target markets and customers. We recognize revenue upon the delivery of our products to the distributors, which can cause our quarterly net revenue to fluctuate significantly. Such revenue is reduced for estimated returns and distributor allowances.

Our software customers generally consist of large enterprises that have computing environments from multiple vendors and are highly complex. We believe our enterprise-wide license model will continue to offer our customers reduced complexity, more flexibility and an easier renewal process that will help drive revenue growth.

Costs and Expenses

Cost of products sold. Cost of products sold consists primarily of the costs for semiconductor wafers and other materials, as well as the costs of assembling and testing those products and materials. Such costs include personnel and overhead related to our manufacturing operations, which include stock-based compensation expense, related occupancy, computer services, equipment costs, manufacturing quality, order fulfillment, warranty adjustments, inventory adjustments including write-downs for inventory obsolescence, and acquisition costs, which include direct transaction costs and acquisition-related costs.

Although we outsource a significant portion of our manufacturing activities, we do have some proprietary semiconductor fabrication facilities. If we are unable to utilize our owned fabrication facilities at a desired level, the fixed costs associated with these facilities will not be fully absorbed, resulting in higher average unit costs and lower gross margins.

Cost of subscriptions and services. Cost of subscriptions and services consists of personnel, project costs associated with professional services or support of our subscriptions and services revenue, and allocated facilities costs and other corporate expenses. Personnel costs include stock-based compensation expense.

Total cost of revenue also includes amortization of acquisition-related intangible assets and restructuring charges.

Research and development. Research and development expense consists primarily of personnel costs for our engineers engaged in the design and development of our products and technologies, including stock-based compensation expense. These expenses also include project material costs, third-party fees paid to consultants, prototype development expense, allocated facilities costs and other corporate expenses and computer services costs related to supporting computer tools used in the engineering and design process.

Selling, general and administrative. Selling expense consists primarily of compensation and associated costs for sales and marketing personnel, including stock-based compensation expense, sales commissions paid to our independent sales representatives, advertising costs, trade shows, corporate marketing, promotion, travel related to our sales and marketing operations, related occupancy and equipment costs, and other marketing costs. General and administrative expense consists primarily of compensation and associated costs for executive management, finance, human resources and other administrative personnel, including stock-based compensation expense, outside professional fees, allocated facilities costs, acquisition-related costs and other corporate expenses.

Amortization of acquisition-related intangible assets. In connection with our acquisitions, we recognize intangible assets that are being amortized over their estimated useful lives. We also recognize goodwill, which is not amortized, and in-process research and development (“IPR&D”), which is initially capitalized as an indefinite-lived intangible asset, in connection with the acquisitions. Upon completion of each underlying project, IPR&D assets are reclassified as amortizable purchased intangible assets and amortized over their estimated useful lives.

Restructuring, impairment and disposal charges. Restructuring, impairment and disposal charges consist primarily of compensation costs associated with employee exit programs, alignment of our global manufacturing operations, rationalizing product development program costs, facility and lease abandonments, fixed asset impairment, IPR&D impairment, and other exit costs, including curtailment of service or supply agreements.

[Table of Contents](#)

Interest expense. Interest expense includes coupon interest, commitment fees, accretion of original issue discount, amortization of debt premiums and debt issuance costs, and expenses related to debt modifications or extinguishments.

Other income (expense), net. Other income (expense), net includes interest income, gains or losses on investments, foreign currency remeasurement, and other miscellaneous items.

Provision for income taxes. We have structured our operations to maximize the benefit from tax incentives extended to us in various jurisdictions to encourage investment or employment. Our tax incentives from the Singapore Economic Development Board provide that any qualifying income earned in Singapore is subject to tax incentives or reduced rates of Singapore income tax, subject to our compliance with the conditions specified in these incentives and legislative developments. These Singapore tax incentives are presently expected to expire in November 2025. The corporate income tax rate in Singapore that would otherwise apply to us would be 17%. We also have a tax holiday on our qualifying income in Malaysia, which is scheduled to expire in 2028.

Each tax incentive and tax holiday is also subject to our compliance with various operating and other conditions. If we cannot, or elect not to, comply with any such operating conditions specified, we could, in some instances, be required to refund previously realized material tax benefits, or if such tax incentive or tax holiday is terminated prior to its expiration absent a new incentive applying, we will lose the related tax benefits earlier than scheduled. We may elect to modify our operational structure and tax strategy, which may not be as beneficial to us as the benefits provided under the present tax concession arrangements. Before taking into consideration the effects of the U.S. Tax Cuts and Jobs Act and other indirect tax impacts, the effect of these tax incentives and tax holiday was to decrease the provision for income taxes by approximately \$1,821 million and \$1,156 million for fiscal years 2022 and 2021, respectively.

Our interpretations and conclusions regarding the tax incentives are not binding on any taxing authority, and if our assumptions about tax and other laws are incorrect or if these tax incentives are substantially modified or rescinded, we could suffer material adverse tax and other financial consequences, which would increase our expenses, reduce our profitability and adversely affect our cash flows. In addition, taxable income in any jurisdiction is dependent upon acceptance of our operational practices and intercompany transfer pricing by local tax authorities as being on an arm's length basis. Due to inconsistencies in application of the arm's length standard among taxing authorities, as well as lack of adequate treaty-based protection, transfer pricing challenges by tax authorities could, if successful, substantially increase our income tax expense.

Critical Accounting Estimates

The preparation of financial statements in accordance with generally accepted accounting principles in the United States ("GAAP") requires us to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent assets and liabilities at the date of the financial statements and the reported amounts of revenue and expenses during the reporting period. We base our estimates and assumptions on current facts, historical experience and various other factors that we believe to be reasonable under the circumstances, the results of which form the basis for making judgments about the carrying values of assets and liabilities and the accrual of costs and expenses that are not readily apparent from other sources. Our actual financial results may differ materially and adversely from our estimates. Our critical accounting policies are those that affect our historical financial statements materially and involve difficult, subjective or complex judgments by management. Those policies include revenue recognition, valuation of goodwill and long-lived assets, and income taxes. See Note 2. "Summary of Significant Accounting Policies" included in Part II, Item 8. of this Annual Report on Form 10-K for further information on our critical accounting policies and estimates.

Revenue recognition. We account for a contract with a customer when both parties have approved the contract and are committed to perform their respective obligations, each party's rights can be identified, payment terms can be identified, the contract has commercial substance, and it is probable we will collect substantially all of the consideration we are entitled to. Revenue is recognized when, or as, performance obligations are satisfied by transferring control of a promised product or service to a customer. Our products and services can be broadly categorized as sales of products and subscriptions and services.

We recognize products revenue from sales to direct customers and distributors when control transfers to the customer. An allowance for distributor credits covering price adjustments is made based on our estimate of historical experience rates as well as considering economic conditions and contractual terms. To date, actual distributor claims activity has been materially consistent with the provisions we have made based on our historical estimates. However, because of the inherent nature of estimates, there is always a risk that there could be significant differences between actual amounts and our estimates. Different judgments or estimates could result in variances that might be significant to reported operating results. We also record reductions of revenue for rebates in the same period that the related revenue is recorded. We accrue 100% of potential rebates at the time of sale. We reverse the accrual of unclaimed rebate amounts as specific rebate programs contractually end and when we believe unclaimed rebates are no longer subject to payment and will not be paid. Thus, the reversal of unclaimed rebates may have a positive impact on our net revenue and net income in subsequent periods.

[Table of Contents](#)

Valuation of goodwill and long-lived assets. We perform an annual impairment review of our goodwill during the fourth fiscal quarter of each year, and more frequently if we believe indicators of impairment exist. The process of evaluating the potential impairment of goodwill is highly subjective and requires significant judgment. To review for impairment, we first assess qualitative factors to determine whether events or circumstances lead to a determination that it is more likely than not that the fair value of any of our reporting units is less than its carrying amount. Our qualitative assessment of the recoverability of goodwill, whether performed annually or based on specific events or circumstances, considers various macroeconomic, industry-specific and company-specific factors. These factors include: (i) severe adverse industry or economic trends; (ii) significant company-specific actions, including exiting an activity in conjunction with restructuring of operations; (iii) current, historical or projected deterioration of our financial performance; or (iv) a sustained decrease in our market capitalization below our net book value. After assessing the totality of events and circumstances, if we determine that it is not more likely than not that the fair value of any of our reporting units is less than its carrying amount, no further assessment is performed. If we determine that it is more likely than not that the fair value of any of our reporting units is less than its carrying amount, we calculate the fair value of that reporting unit and compare the fair value to the reporting unit's net book value.

Determining the fair value of a reporting unit involves the use of significant estimates and assumptions. Our goodwill impairment test uses both the income approach and the market approach to estimate a reporting unit's fair value. The income approach is based on the discounted cash flow method that uses the reporting unit estimates for forecasted future financial performance, including revenues, operating expenses, and taxes, as well as working capital and capital asset requirements. These estimates are developed as part of our long-term planning process based on assumed market segment growth rates and our assumed market segment share, estimated costs based on historical data and various internal estimates. Projected cash flows are then discounted to a present value employing a discount rate that properly accounts for the estimated market weighted-average cost of capital, as well as any risk unique to the subject cash flows. The market approach is based on weighting the financial multiples of comparable companies and applying a control premium. A reporting unit's carrying value represents the assignment of various assets and liabilities, excluding certain corporate assets and liabilities, such as cash and debt.

We assess the impairment of long-lived assets, including purchased IPR&D, property, plant and equipment, and intangible assets, whenever events or changes in circumstances indicate that the carrying value of such assets may not be recoverable. Factors we consider important which could trigger an impairment review include: (i) significant under-performance relative to historical or projected future operating results, (ii) significant changes in the manner of our use of the acquired assets or the strategy for our overall business, or (iii) significant negative industry or economic trends. The process of evaluating the potential impairment of long-lived assets under the accounting guidance on property, plant and equipment, and intangible assets is also highly subjective and requires significant judgment. In order to estimate the fair value of long-lived assets, we typically make various assumptions about the future prospects of our business or the part of our business to which the long-lived assets relate. We also consider market factors specific to the business and estimate future cash flows to be generated by the business, which requires significant judgment as it is based on assumptions about market demand for our products over a number of future years. Based on these assumptions and estimates, we determine whether we need to take an impairment charge to reduce the value of the long-lived assets stated on our consolidated balance sheets to reflect their estimated fair value. Assumptions and estimates about future values and remaining useful lives are complex and often subjective. They can be affected by a variety of factors, including external factors, such as the real estate market, industry and economic trends, and internal factors, such as changes in our business strategy and our internal forecasts. Although we believe the assumptions and estimates we have made in the past have been reasonable and appropriate, changes in assumptions and estimates could materially impact our reported financial results.

Income taxes. Significant management judgment is required in developing our provision for or benefit from income taxes, including the determination of deferred tax assets and liabilities and any valuation allowances that might be required against the deferred tax assets. We have considered projected future taxable income and ongoing prudent and feasible tax planning strategies in assessing the need for valuation allowances. An adjustment to the valuation allowance will either increase or decrease our provision for or benefit from income taxes in the period such determination is made. In evaluating the exposure associated with various tax filing positions, we accrue an income tax liability when such positions do not meet the more-likely-than-not threshold for recognition.

The calculation of our tax liabilities involves dealing with uncertainties in the application of complex tax laws and regulations in a multitude of jurisdictions. We recognize potential liabilities for anticipated tax audit issues in the U.S. and other tax jurisdictions based on our estimate of whether, and the extent to which, additional taxes, interest, and penalties will be due. If our estimate of income tax liabilities proves to be less than the actual amount ultimately assessed, a further charge to tax expense would be required. If the payment of these amounts ultimately proves to be unnecessary, the reversal of the accrued liabilities would result in tax benefits being recognized in the period when we determine the liabilities no longer exist.

Fiscal Year Presentation

We operate on a 52- or 53-week fiscal year ending on the Sunday closest to October 31 in a 52-week year and the first Sunday in November in a 53-week year. Our fiscal years 2022, 2021 and 2020 consisted of 52 weeks.

The financial statements included in Part II, Item 8. of this Annual Report on Form 10-K are presented in accordance with GAAP and expressed in U.S. dollars.

Results of Operations

Fiscal Year 2022 Compared to Fiscal Year 2021

The following table sets forth our results of operations for the periods presented:

	Fiscal Year Ended			
	October 30, 2022	October 31, 2021	October 30, 2022	October 31, 2021
	(In millions)		(As a percentage of net revenue)	
Statements of Operations Data:				
Net revenue:				
Products	\$ 26,277	\$ 20,886	79 %	76 %
Subscriptions and services	6,926	6,564	21	24
Total net revenue	33,203	27,450	100	100
Cost of revenue:				
Cost of products sold	7,629	6,555	23	24
Cost of subscriptions and services	627	607	2	2
Amortization of acquisition-related intangible assets	2,847	3,427	8	13
Restructuring charges	5	17	—	—
Total cost of revenue	11,108	10,606	33	39
Gross margin	22,095	16,844	67	61
Research and development	4,919	4,854	15	18
Selling, general and administrative	1,382	1,347	4	5
Amortization of acquisition-related intangible assets	1,512	1,976	5	7
Restructuring, impairment and disposal charges	57	148	—	—
Total operating expenses	7,870	8,325	24	30
Operating income	\$ 14,225	\$ 8,519	43 %	31 %

Net Revenue

A relatively small number of customers account for a significant portion of our net revenue. Sales of products to distributors accounted for 56% and 53% of our net revenue for fiscal years 2022 and 2021, respectively. Direct sales to WT Microelectronics Co., Ltd., a distributor, accounted for 20% and 18% of our net revenue for fiscal years 2022 and 2021, respectively. We believe aggregate sales to our top five end customers, through all channels, accounted for approximately 35% of our net revenue for each of fiscal years 2022 and 2021. We believe aggregate sales to Apple Inc., through all channels, accounted for approximately 20% of our net revenue for each of fiscal years 2022 and 2021. We expect to continue to experience significant customer concentration in future periods. The loss of, or significant decrease in demand from, any of our top five end customers could have a material adverse effect on our business, results of operations and financial condition.

From time to time, some of our key semiconductor customers place large orders or delay orders, causing our quarterly net revenue to fluctuate significantly. This is particularly true of our wireless products as fluctuations may be magnified by the timing of launches, and seasonal variations in sales, of mobile devices. The COVID-19 pandemic and macroeconomic uncertainties may cause our net revenue to fluctuate significantly and impact our results of operations.

Although we recognize revenue for the majority of our products when title and control transfer in Penang, Malaysia, we disclose net revenue by country based primarily on the geographic shipment or delivery location specified by our distributors, OEMs, contract manufacturers, channel partners, or software customers. In each of fiscal years 2022 and 2021, approximately 35% of our net revenue came from shipments or deliveries to China (including Hong Kong). However, the end customers for either our products or for the end products into which our products are incorporated, are frequently located in countries other than China (including Hong Kong). As a result, we believe that a substantially smaller percentage of our net revenue is ultimately dependent on sales of either our product or our customers' product incorporating our product, to end customers located in China (including Hong Kong).

The following tables set forth net revenue by segment for the periods presented:

Net Revenue by Segment	Fiscal Year Ended		\$ Change	% Change
	October 30, 2022	October 31, 2021		
(In millions, except percentages)				
Semiconductor solutions	\$ 25,818	\$ 20,383	\$ 5,435	27 %
Infrastructure software	7,385	7,067	318	4 %
Total net revenue	<u>\$ 33,203</u>	<u>\$ 27,450</u>	<u>\$ 5,753</u>	21 %

Net Revenue by Segment	Fiscal Year Ended	
	October 30, 2022	October 31, 2021
(As a percentage of net revenue)		
Semiconductor solutions	78 %	74 %
Infrastructure software	22	26
Total net revenue	<u>100 %</u>	<u>100 %</u>

Net revenue from our semiconductor solutions segment increased due to strong product demand, primarily for networking, server storage and broadband products, as well as higher demand for our wireless content in mobile devices. Net revenue from our infrastructure software segment increased primarily due to higher demand for our mainframe solutions and FC SAN products.

Gross Margin

Gross margin was \$22,095 million, or 67% of net revenue, for fiscal year 2022, compared to \$16,844 million, or 61% of net revenue, for fiscal year 2021. The increase was primarily due to lower amortization of acquisition-related intangible assets, mainly from our 2016 acquisition of Broadcom Corporation and, to a lesser extent, favorable margin within our semiconductor solutions segment.

Research and Development Expense

Research and development expense increased \$65 million, or 1%, in fiscal year 2022, compared to the prior fiscal year. The increase was primarily due to higher variable employee compensation expense and engineering project costs, partially offset by lower stock-based compensation expense reflecting the full vesting of certain equity awards and the effects of forfeitures.

Selling, General and Administrative Expense

Selling, general and administrative expense increased \$35 million, or 3%, in fiscal year 2022, compared to the prior fiscal year. The increase was primarily due to higher variable employee compensation expense, offset in part by lower stock-based compensation expense.

Amortization of Acquisition-Related Intangible Assets

Amortization of acquisition-related intangible assets recognized in operating expenses decreased \$464 million, or 23%, in fiscal year 2022, compared to the prior fiscal year. The decrease was primarily due to lower amortization of certain intangible assets from our acquisition of CA, Inc.

Restructuring, Impairment and Disposal Charges

Restructuring, impairment and disposal charges recognized in operating expenses decreased \$91 million, or 61%, in fiscal year 2022, compared to the prior fiscal year. The decrease was primarily due to lower employee termination costs following the completion of key restructuring activities from acquisitions.

Stock-Based Compensation Expense

Total stock-based compensation expense was \$1,533 million and \$1,704 million for fiscal years 2022 and 2021, respectively. The decrease primarily reflects the full vesting of certain equity awards and the effect of forfeitures.

The following table sets forth the total unrecognized compensation cost related to unvested stock-based awards outstanding and expected to vest as of October 30, 2022, which we expect to recognize over the remaining weighted-average service period of 2.7 years.

Fiscal Year:	Unrecognized Compensation Cost, Net of Expected Forfeitures
	(In millions)
2023	\$ 1,221
2024	846
2025	507
2026	130
Total	\$ 2,704

During the first quarter of fiscal year ended November 3, 2019 (“fiscal year 2019”), our Compensation Committee approved a broad-based program of multi-year equity grants of time- and market-based RSUs (the “Multi-Year Equity Awards”) in lieu of our annual employee equity awards historically granted on March 15 of each year. Each Multi-Year Equity Award vests on the same basis as four annual grants made March 15 of each year, beginning in fiscal year 2019, with successive four-year vesting periods. We recognize stock-based compensation expense related to the Multi-Year Equity Awards from the grant date through their respective vesting date, ranging from 4 years to 7 years.

Segment Operating Results

Operating Income by Segment	Fiscal Year Ended		\$ Change	% Change
	October 30, 2022	October 31, 2021		
	(In millions, except percentages)			
Semiconductor solutions	\$ 15,075	\$ 10,976	\$ 4,099	37 %
Infrastructure software	5,219	4,936	283	6 %
Unallocated expenses	(6,069)	(7,393)	1,324	(18)%
Total operating income	\$ 14,225	\$ 8,519	\$ 5,706	67 %

Operating income from our semiconductor solutions segment increased primarily due to higher net revenue from networking, server storage, broadband, and wireless products, as well as higher gross margin. Operating income from our infrastructure software segment increased primarily due to higher demand for our mainframe solutions and FC SAN products.

Unallocated expenses include amortization of acquisition-related intangible assets; stock-based compensation expense; restructuring, impairment and disposal charges; acquisition-related costs; and other costs that are not used in evaluating the results of, or in allocating resources to, our segments. Unallocated expenses decreased 18% in fiscal year 2022, compared to the prior fiscal year, primarily due to lower amortization of acquisition-related intangible assets.

Non-Operating Income and Expenses

Interest expense. Interest expense was \$1,737 million and \$1,885 million for fiscal years 2022 and 2021, respectively. The decrease was primarily due to lower losses on extinguishment of debt. We expect to incur additional interest expense in future periods as a result of indebtedness associated with the pending VMware Merger.

Other income (expense), net. Other income (expense), net, includes interest income, gains or losses on investments, foreign currency remeasurement and other miscellaneous items. Other expense, net, was \$54 million for fiscal year 2022, compared to other income, net, of \$131 million for fiscal year 2021. The change was primarily due to changes in investment gains or losses.

Provision for income taxes. The provision for income taxes was \$939 million and \$29 million for fiscal years 2022 and 2021, respectively. The increase was primarily due to higher income from continuing operations before income taxes.

Liquidity and Capital Resources

The following section discusses our principal liquidity and capital resources as well as our primary liquidity requirements and uses of cash. Our cash and cash equivalents are maintained in highly liquid investments with remaining maturities of 90 days or less at the time of purchase. We believe our cash equivalents are liquid and accessible.

Our primary sources of liquidity as of October 30, 2022 consisted of: (i) \$12,416 million in cash and cash equivalents, (ii) cash we expect to generate from operations and (iii) available capacity under our \$7.5 billion unsecured revolving credit

[Table of Contents](#)

facility (the “Revolving Facility”). In addition, we may also generate cash from the sale of assets and debt or equity financing from time to time.

Our short-term and long-term liquidity requirements primarily arise from: (i) business acquisitions and investments we may make from time to time, including the pending VMware Merger, (ii) working capital requirements, (iii) research and development and capital expenditure needs, (iv) cash dividend payments (if and when declared by our Board of Directors), (v) interest and principal payments related to our \$41,218 million of outstanding indebtedness, (vi) share repurchases, and (vii) payment of income taxes. Our ability to fund these requirements will depend, in part, on our future cash flows, which are determined by our future operating performance and, therefore, subject to prevailing global macroeconomic conditions and financial, business and other factors, some of which are beyond our control. We expect capital expenditures to be higher in fiscal year 2023 as compared to fiscal year 2022. Our debt and liquidity needs will increase as a result of the pending VMware Merger, and we intend to fund the cash portion of the consideration with \$32 billion in new, fully committed debt financing.

We believe that our cash and cash equivalents on hand, cash flows from operations, and the Revolving Facility, as well as the committed debt funding related to the pending VMware Merger, will provide sufficient liquidity to operate our business and fund our current and assumed obligations for at least the next 12 months. For additional information regarding our cash requirement from contractual obligations, indebtedness and lease obligations, see Note 14. “Commitments and Contingencies”, Note 10. “Borrowings” and Note 6. “Leases” in Part II, Item 8 of this Annual Report on Form 10-K.

From time to time, we engage in discussions with third parties regarding potential acquisitions of, or investments in, businesses, technologies and product lines. Any such transaction, or evaluation of potential transactions, could require significant use of our cash and cash equivalents, or require us to increase our borrowings to fund such transactions. If we do not have sufficient cash to fund our operations or finance growth opportunities, including acquisitions, or unanticipated capital expenditures, our business and financial condition could suffer. In such circumstances, we may seek to obtain new debt or equity financing. However, we cannot assure you that such additional financing will be available on terms acceptable to us or at all. Our ability to service our senior unsecured notes and any other indebtedness we may incur will depend on our ability to generate cash in the future. We may also elect to sell additional debt or equity securities for reasons other than those specified above.

In addition, we may, at any time and from time to time, seek to retire or purchase our outstanding debt through cash tenders and/or exchanges for equity or debt, in open-market purchases, privately negotiated transactions or otherwise. Such tenders, exchanges or purchases, if any, will be upon such terms and at such prices as we may determine, and will depend on prevailing market conditions, our liquidity requirements, contractual restrictions and other factors. The amounts involved may be material.

Working Capital

Working capital increased to \$11,452 million at October 30, 2022 from \$10,305 million at October 31, 2021. The increase was attributable to the following:

- Accounts receivable increased to \$2,958 million at October 30, 2022 from \$2,071 million at October 31, 2021, primarily due to revenue linearity and less receivables sold through factoring arrangements.
- Inventory increased to \$1,925 million at October 30, 2022 from \$1,297 million at October 31, 2021, primarily to support customer demand and due to higher material costs.
- Cash and cash equivalents increased to \$12,416 million at October 30, 2022 from \$12,163 million at October 31, 2021, primarily due to \$16,736 million in net cash provided by operating activities and \$1,935 million in proceeds from long-term borrowings, partially offset by \$7,032 million of dividend payments, \$7,000 million of common stock repurchases, \$2,361 million of debt payments, and \$1,455 million of employee withholding tax payments related to net settled equity awards.
- Other current assets increased to \$1,205 million at October 30, 2022 from \$1,055 million at October 31, 2021, primarily due to an increase in prepaid taxes, offset in part by a decrease in short-term investments.

These increases in working capital were offset in part by the following:

- Other current liabilities increased to \$4,412 million at October 30, 2022 from \$3,839 million at October 31, 2021, primarily due to increases in contract liabilities, taxes payable and interest payable.
- Current portion of long-term debt increased to \$440 million at October 30, 2022 from \$290 million at October 31, 2021, primarily due to certain debt instruments becoming due within the next twelve months, offset in part by repayments.

- Employee compensation and benefits increased to \$1,202 million at October 30, 2022 from \$1,066 million at October 31, 2021, primarily due to higher variable compensation based on current fiscal year performance.

Capital Returns

Cash Dividends Declared and Paid	Fiscal Year Ended	
	October 30, 2022	October 31, 2021
	(In millions, except per share data)	
Dividends per share to common stockholders	\$ 16.40	\$ 14.40
Dividends to common stockholders	\$ 6,733	\$ 5,913
Dividends per share to preferred stockholders	\$ 80.00	\$ 80.00
Dividends to preferred stockholders	\$ 299	\$ 299

In December 2021, our Board of Directors authorized a stock repurchase program to repurchase up to \$10 billion of our common stock from time to time on or prior to December 31, 2022. During fiscal year 2022, we repurchased and retired approximately 12 million shares of our common stock for \$7 billion under this stock repurchase program. In May 2022, our Board of Directors authorized another stock repurchase program to repurchase up to an additional \$10 billion of our common stock from time to time through December 31, 2023.

Repurchases under our stock repurchase programs may be effected through a variety of methods, including open market or privately negotiated purchases. The timing and amount of shares repurchased will depend on the stock price, business and market conditions, corporate and regulatory requirements, alternative investment opportunities, acquisition opportunities, and other factors. We are not obligated to repurchase any specific amount of shares of common stock, and the stock repurchase programs may be suspended or terminated at any time.

During fiscal years 2022 and 2021, we paid approximately \$1,455 million and \$1,299 million, respectively, in employee withholding taxes due upon the vesting of net settled equity awards. We withheld approximately 3 million shares of common stock from employees in connection with such net share settlements during each of fiscal years 2022 and 2021.

Cash Flows

	Fiscal Year Ended	
	October 30, 2022	October 31, 2021
	(In millions)	
Net cash provided by operating activities	\$ 16,736	\$ 13,764
Net cash used in investing activities	(667)	(245)
Net cash used in financing activities	(15,816)	(8,974)
Net change in cash and cash equivalents	\$ 253	\$ 4,545

Operating Activities

Cash provided by operating activities consisted of net income adjusted for certain non-cash and other items and changes in assets and liabilities. The \$2,972 million increase in cash provided by operations during fiscal year 2022 compared to fiscal year 2021 was due to \$4,759 million higher net income and certain non-cash adjustments including deferred taxes and other non-cash taxes, offset by a decrease in amortization of intangible assets and stock-based compensation, as well as a \$1,527 million decrease resulting from changes in operating assets and liabilities.

Investing Activities

Cash flows from investing activities primarily consisted of cash used for acquisitions, capital expenditures, and sales and purchases of investments. The \$422 million increase in cash used in investing activities for fiscal year 2022 compared to fiscal year 2021 was primarily due to a \$238 million increase in cash paid for acquisitions and \$169 million lower net proceeds from sales of investments.

Financing Activities

Cash flows from financing activities primarily consisted of dividend payments, stock repurchases, proceeds and payments related to our long-term borrowings, and employee withholding tax payments related to net settled equity awards. The \$6,842 million increase in cash used in financing activities for fiscal year 2022 compared to fiscal year 2021 was primarily

due to \$7,000 million in common stock repurchases, a \$820 million increase in dividend payments, and a \$156 million increase in employee withholding tax payments related to net settled equity awards, offset in part by a \$1,165 million change in net borrowing activities.

Accounting Changes and Recent Accounting Standards

For a description of accounting changes and recent accounting standards, including the expected dates of adoption and estimated effects, if any, in our consolidated financial statements, see Note 2. "Summary of Significant Accounting Policies" included in Part II, Item 8. of this Annual Report on Form 10-K.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

Foreign Currency Exchange Risk

From time to time, we use foreign exchange forward contracts to hedge a portion of our exposures to changes in currency exchange rates, which result from our global operating and financing activities. We do not use derivative financial instruments for trading or speculative purposes. Gains and losses from foreign currency transactions, as well as foreign exchange forward contracts, were not significant for any period presented in the consolidated financial statements included in this Form 10-K. As of October 30, 2022, we did not have any outstanding foreign exchange forward contracts.

Interest Rate Risk

Changes in interest rates affect the fair value of our outstanding debt. As of October 30, 2022, we had \$41.2 billion in principal amount of debt outstanding. The carrying amount of the debt was \$39.5 billion, and the estimated aggregate fair value of debt was \$33.0 billion. As of October 30, 2022, a hypothetical 50 basis points increase or decrease in market interest rates would change the fair value of debt by a decrease or increase of approximately \$1.6 billion. However, this hypothetical change in interest rates would not impact the interest expense on our debt as we only had fixed rate senior notes outstanding. To hedge variability of cash flows due to changes in the benchmark interest rate of anticipated future debt issuances, we have entered, and in the future may enter, into treasury rate lock contracts.

ITEM 8. FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

BROADCOM INC.
INDEX TO CONSOLIDATED FINANCIAL STATEMENTS

	Page
Report of Independent Registered Public Accounting Firm (PCAOB ID 238)	48
Consolidated Balance Sheets	49
Consolidated Statements of Operations	50
Consolidated Statements of Comprehensive Income	51
Consolidated Statements of Cash Flows	52
Consolidated Statements of Stockholders' Equity	53
Notes to Consolidated Financial Statements	54
Schedule II — Valuation and Qualifying Accounts	89

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Board of Directors and Stockholders of Broadcom Inc.

Opinions on the Financial Statements and Internal Control over Financial Reporting

We have audited the accompanying consolidated balance sheets of Broadcom Inc. and its subsidiaries (the “Company”) as of October 30, 2022 and October 31, 2021, and the related consolidated statements of operations, of comprehensive income, of stockholders’ equity and of cash flows for each of the three years in the period ended October 30, 2022, including the related notes and financial statement schedule listed in the index appearing under Item 15(a)(2) (collectively referred to as the “consolidated financial statements”). We also have audited the Company's internal control over financial reporting as of October 30, 2022, based on criteria established in *Internal Control - Integrated Framework* (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Company as of October 30, 2022 and October 31, 2021, and the results of its operations and its cash flows for each of the three years in the period ended October 30, 2022 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of October 30, 2022, based on criteria established in *Internal Control - Integrated Framework* (2013) issued by the COSO.

Change in Accounting Principle

As discussed in Note 6 to the consolidated financial statements, the Company changed the manner in which it accounts for leases in fiscal 2020.

Basis for Opinions

The Company's management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in Management’s Report on Internal Control Over Financial Reporting appearing under Item 9A. Our responsibility is to express opinions on the Company’s consolidated financial statements and on the Company's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Definition and Limitations of Internal Control over Financial Reporting

A company’s internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally

accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matters

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that (i) relates to accounts or disclosures that are material to the consolidated financial statements and (ii) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Uncertain Tax Positions (UTPs)

As described in Notes 2 and 12 to the consolidated financial statements, the gross unrecognized tax benefits balance was \$5,117 million as of October 30, 2022. As management has disclosed, management evaluates the exposure associated with various tax filing positions and accrues an income tax liability when such positions do not meet the more-likely-than-not threshold for recognition. A tax benefit from an UTP may be recognized when it is more likely than not that the position will be sustained upon examination, including resolution of any related appeals or litigation processes, based on the technical merits.

The principal considerations for our determination that performing procedures relating to the UTPs is a critical audit matter are (i) the significant judgment by management when evaluating the technical merits of these tax positions, (ii) a high degree of auditor judgment, subjectivity, and effort in performing procedures and evaluating the technical merits of the tax positions, and (iii) the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to the identification and recognition of the income tax liability for UTPs, including controls addressing the completeness of the UTPs and the measurement of the income tax liability. These procedures also included, among others, (i) testing management's process for identifying potential new UTPs, (ii) for a selection of UTPs, evaluating possible outcomes, and (iii) for a selection of UTPs, testing the calculation of the income tax liability by jurisdiction, including management's assessment of the technical merits of tax positions and estimates of the amount of tax benefit expected to be sustained. Professionals with specialized skill and knowledge were used to assist in (i) the evaluation of the completeness of management's identification of the UTPs and (ii) for a selection of UTPs, the evaluation of the reasonableness of management's assessment of whether the tax positions are more-likely-than-not of being sustained, the amount of potential benefit to be realized, and the application of relevant tax laws.

/s/ PricewaterhouseCoopers LLP

San Jose, California
December 16, 2022

We have served as the Company's auditor since 2006.

**BROADCOM INC.
CONSOLIDATED BALANCE SHEETS**

	October 30, 2022	October 31, 2021
	(In millions, except par value)	
ASSETS		
Current assets:		
Cash and cash equivalents	\$ 12,416	\$ 12,163
Trade accounts receivable, net	2,958	2,071
Inventory	1,925	1,297
Other current assets	1,205	1,055
Total current assets	18,504	16,586
Long-term assets:		
Property, plant and equipment, net	2,223	2,348
Goodwill	43,614	43,450
Intangible assets, net	7,111	11,374
Other long-term assets	1,797	1,812
Total assets	<u>\$ 73,249</u>	<u>\$ 75,570</u>
LIABILITIES AND EQUITY		
Current liabilities:		
Accounts payable	\$ 998	\$ 1,086
Employee compensation and benefits	1,202	1,066
Current portion of long-term debt	440	290
Other current liabilities	4,412	3,839
Total current liabilities	7,052	6,281
Long-term liabilities:		
Long-term debt	39,075	39,440
Other long-term liabilities	4,413	4,860
Total liabilities	<u>50,540</u>	<u>50,581</u>
Commitments and contingencies (Note 14)		
Preferred stock dividend obligation	—	27
Stockholders' equity:		
Preferred stock, \$0.001 par value; 100 shares authorized; 8.00% Mandatory Convertible Preferred Stock, Series A, 0 and 4 shares issued and outstanding; aggregate liquidation value of \$0 and \$3,737 as of October 30, 2022 and October 31, 2021, respectively	—	—
Common stock, \$0.001 par value; 2,900 shares authorized; 418 and 413 shares issued and outstanding as of October 30, 2022 and October 31, 2021, respectively	—	—
Additional paid-in capital	21,159	24,330
Retained earnings	1,604	748
Accumulated other comprehensive loss	(54)	(116)
Total stockholders' equity	<u>22,709</u>	<u>24,962</u>
Total liabilities and equity	<u>\$ 73,249</u>	<u>\$ 75,570</u>

The accompanying notes are an integral part of these consolidated financial statements.

BROADCOM INC.
CONSOLIDATED STATEMENTS OF OPERATIONS

	Fiscal Year Ended		
	October 30, 2022	October 31, 2021	November 1, 2020
(In millions, except per share data)			
Net revenue:			
Products	\$ 26,277	\$ 20,886	\$ 17,435
Subscriptions and services	6,926	6,564	6,453
Total net revenue	33,203	27,450	23,888
Cost of revenue:			
Cost of products sold	7,629	6,555	5,892
Cost of subscriptions and services	627	607	626
Amortization of acquisition-related intangible assets	2,847	3,427	3,819
Restructuring charges	5	17	35
Total cost of revenue	11,108	10,606	10,372
Gross margin	22,095	16,844	13,516
Research and development	4,919	4,854	4,968
Selling, general and administrative	1,382	1,347	1,935
Amortization of acquisition-related intangible assets	1,512	1,976	2,401
Restructuring, impairment and disposal charges	57	148	198
Total operating expenses	7,870	8,325	9,502
Operating income	14,225	8,519	4,014
Interest expense	(1,737)	(1,885)	(1,777)
Other income (expense), net	(54)	131	206
Income from continuing operations before income taxes	12,434	6,765	2,443
Provision for (benefit from) income taxes	939	29	(518)
Income from continuing operations	11,495	6,736	2,961
Loss from discontinued operations, net of income taxes	—	—	(1)
Net income	11,495	6,736	2,960
Dividends on preferred stock	(272)	(299)	(297)
Net income attributable to common stock	\$ 11,223	\$ 6,437	\$ 2,663
Net income per share attributable to common stock:			
Basic	\$ 27.44	\$ 15.70	\$ 6.62
Diluted	\$ 26.53	\$ 15.00	\$ 6.33
Weighted-average shares used in per share calculations:			
Basic	409	410	402
Diluted	423	429	421

The accompanying notes are an integral part of these consolidated financial statements.

BROADCOM INC.
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME

	Fiscal Year Ended		
	October 30, 2022	October 31, 2021	November 1, 2020
	(In millions)		
Net income	\$ 11,495	\$ 6,736	\$ 2,960
Other comprehensive income (loss), net of tax:			
Change in unrealized gain on derivative instruments	37	—	—
Change in actuarial loss and prior service costs associated with defined benefit plans	25	(8)	24
Other comprehensive income (loss), net of tax	62	(8)	24
Comprehensive income	<u>\$ 11,557</u>	<u>\$ 6,728</u>	<u>\$ 2,984</u>

The accompanying notes are an integral part of these consolidated financial statements.

BROADCOM INC.
CONSOLIDATED STATEMENTS OF CASH FLOWS

	Fiscal Year Ended		
	October 30, 2022	October 31, 2021	November 1, 2020
	(In millions)		
Cash flows from operating activities:			
Net income	\$ 11,495	\$ 6,736	\$ 2,960
Adjustments to reconcile net income to net cash provided by operating activities:			
Amortization of intangible and right-of-use assets	4,455	5,502	6,335
Depreciation	529	539	570
Stock-based compensation	1,533	1,704	1,976
Deferred taxes and other non-cash taxes	(34)	(809)	(1,142)
Loss on debt extinguishment	100	198	169
Non-cash restructuring, impairment and disposal charges	13	38	44
Non-cash interest expense	129	96	108
Other	170	(113)	(52)
Changes in assets and liabilities, net of acquisitions and disposals:			
Trade accounts receivable, net	(870)	210	981
Inventory	(627)	(294)	(31)
Accounts payable	(79)	243	(3)
Employee compensation and benefits	136	186	217
Other current assets and current liabilities	222	(177)	331
Other long-term assets and long-term liabilities	(436)	(295)	(402)
Net cash provided by operating activities	16,736	13,764	12,061
Cash flows from investing activities:			
Acquisitions of businesses, net of cash acquired	(246)	(8)	(10,872)
Proceeds from sales of businesses	—	45	218
Purchases of property, plant and equipment	(424)	(443)	(463)
Purchases of investments	(200)	—	—
Sales of investments	200	169	—
Other	3	(8)	8
Net cash used in investing activities	(667)	(245)	(11,109)
Cash flows from financing activities:			
Proceeds from long-term borrowings	1,935	9,904	27,802
Payments on debt obligations	(2,361)	(11,495)	(18,814)
Other borrowings, net	—	—	(1,285)
Payments of dividends	(7,032)	(6,212)	(5,534)
Repurchases of common stock - repurchase program	(7,000)	—	—
Shares repurchased for tax withholdings on vesting of equity awards	(1,455)	(1,299)	(765)
Issuance of common stock	114	170	276
Other	(17)	(42)	(69)
Net cash provided by (used in) financing activities	(15,816)	(8,974)	1,611
Net change in cash and cash equivalents	253	4,545	2,563
Cash and cash equivalents at beginning of period	12,163	7,618	5,055
Cash and cash equivalents at end of period	\$ 12,416	\$ 12,163	\$ 7,618
Supplemental disclosure of cash flow information:			
Cash paid for interest	\$ 1,386	\$ 1,565	\$ 1,408
Cash paid for income taxes	\$ 908	\$ 775	\$ 501

The accompanying notes are an integral part of these consolidated financial statements.

BROADCOM INC.
CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY

	8.00% Mandatory Convertible Preferred Stock		Common Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Loss	Total Stockholders' Equity
	Shares	Par Value	Shares	Par Value				
	(In millions)							
Balance as of November 3, 2019	4	\$ —	398	\$ —	\$ 25,081	\$ —	\$ (140)	\$ 24,941
Net income	—	—	—	—	—	2,960	—	2,960
Other comprehensive income	—	—	—	—	—	—	24	24
Cumulative effect of accounting change	—	—	—	—	—	(10)	8	(2)
Fair value of partially vested equity awards assumed in connection with an acquisition	—	—	—	—	1	—	—	1
Dividends to common stockholders	—	—	—	—	(2,582)	(2,653)	—	(5,235)
Dividends to preferred stockholders	—	—	—	—	—	(297)	—	(297)
Common stock issued	—	—	12	—	276	—	—	276
Stock-based compensation	—	—	—	—	1,976	—	—	1,976
Shares repurchased for tax withholdings on vesting of equity awards	—	—	(3)	—	(770)	—	—	(770)
Balance as of November 1, 2020	4	—	407	—	23,982	—	(108)	23,874
Net income	—	—	—	—	—	6,736	—	6,736
Other comprehensive loss	—	—	—	—	—	—	(8)	(8)
Dividends to common stockholders	—	—	—	—	(224)	(5,689)	—	(5,913)
Dividends to preferred stockholders	—	—	—	—	—	(299)	—	(299)
Common stock issued	—	—	9	—	170	—	—	170
Stock-based compensation	—	—	—	—	1,704	—	—	1,704
Shares repurchased for tax withholdings on vesting of equity awards	—	—	(3)	—	(1,302)	—	—	(1,302)
Balance as of October 31, 2021	4	—	413	—	24,330	748	(116)	24,962
Net income	—	—	—	—	—	11,495	—	11,495
Other comprehensive income	—	—	—	—	—	—	62	62
Fair value of partially vested equity awards assumed in connection with an acquisition	—	—	—	—	4	—	—	4
Dividends to common stockholders	—	—	—	—	(50)	(6,683)	—	(6,733)
Dividends to preferred stockholders	—	—	—	—	—	(272)	—	(272)
Common stock issued	—	—	8	—	114	—	—	114
Stock-based compensation	—	—	—	—	1,533	—	—	1,533
Repurchases of common stock	—	—	(12)	—	(3,316)	(3,684)	—	(7,000)
Common stock issued in connection with Mandatory Convertible Preferred Stock conversion	(4)	—	12	—	—	—	—	—
Shares repurchased for tax withholdings on vesting of equity awards	—	—	(3)	—	(1,456)	—	—	(1,456)
Balance as of October 30, 2022	—	\$ —	418	\$ —	\$ 21,159	\$ 1,604	\$ (54)	\$ 22,709

The accompanying notes are an integral part of these consolidated financial statements.

BROADCOM INC.
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

1. Overview and Basis of Presentation

Overview

Broadcom Inc. ("Broadcom"), a Delaware corporation, is a global technology leader that designs, develops and supplies a broad range of semiconductor and infrastructure software solutions. We develop semiconductor devices with a focus on complex digital and mixed signal complementary metal oxide semiconductor based devices and analog III-V based products. We have a history of innovation in the semiconductor industry and offer thousands of products that are used in end products such as enterprise and data center networking, home connectivity, set-top boxes, broadband access, telecommunication equipment, smartphones and base stations, data center servers and storage systems, factory automation, power generation and alternative energy systems, and electronic displays. Our infrastructure software solutions enable customers to plan, develop, automate, manage and secure applications across mainframe, distributed, mobile and cloud platforms. Our portfolio of industry-leading infrastructure and security software is designed to modernize, optimize, and secure the most complex hybrid environments, enabling scalability, agility, automation, insights, resiliency and security. We also offer mission critical fibre channel storage area networking ("FC SAN") products and related software in the form of modules, switches and subsystems incorporating multiple semiconductor products. Unless stated otherwise or the context otherwise requires, references to "Broadcom," "we," "our," and "us" mean Broadcom and its consolidated subsidiaries.

Basis of Presentation

We operate on a 52- or 53-week fiscal year ending on the Sunday closest to October 31 in a 52-week year and the first Sunday in November in a 53-week year. Our fiscal year ended October 30, 2022 ("fiscal year 2022") was a 52-week fiscal year. The first quarter of our fiscal year 2022 ended on January 30, 2022, the second quarter ended on May 1, 2022 and the third quarter ended on July 31, 2022. Our fiscal year ended October 31, 2021 ("fiscal year 2021") and fiscal year ended November 1, 2020 ("fiscal year 2020") were both 52-week fiscal years.

On November 4, 2019, we completed the purchase of certain assets and assumption of certain liabilities of the Symantec Corporation Enterprise Security business (the "Symantec Business"). We have two reportable segments: semiconductor solutions and infrastructure software. See Note 13. "Segment Information" for additional information.

The accompanying consolidated financial statements include the accounts of Broadcom and its subsidiaries and have been prepared in accordance with generally accepted accounting principles in the United States ("GAAP"). All intercompany balances and transactions have been eliminated in consolidation.

2. Summary of Significant Accounting Policies

Foreign currency remeasurement. We operate in a U.S. dollar functional currency environment. Foreign currency assets and liabilities for monetary accounts are remeasured into U.S. dollars at current exchange rates. Non-monetary items such as inventory and property, plant and equipment, are measured and recorded at historical exchange rates. The effects of foreign currency remeasurement were not material for any period presented.

Use of estimates. The preparation of financial statements in conformity with GAAP requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent assets and liabilities at the date of the financial statements and the reported amounts of revenues and expenses during the reporting period. The inputs into certain of these estimates and assumptions include the consideration of the economic impact of the COVID-19 pandemic, and many of these estimates could require increased judgment and carry a higher degree of variability and volatility. Actual results could differ materially from these estimates, and such differences could affect the results of operations reported in future periods.

Cash and cash equivalents. We consider all highly liquid investment securities with original maturities of three months or less at the date of purchase to be cash equivalents. We determine the appropriate classification of our cash and cash equivalents at the time of purchase.

Trade accounts receivable, net. Trade accounts receivable are recognized at the invoiced amount and do not bear interest. Accounts receivable are reduced by an allowance for doubtful accounts, which is our best estimate of the expected credit losses in our existing accounts receivable. We determine the allowance based on historical experience, current economic conditions and certain forward-looking information, among other factors. Allowances for doubtful accounts were not material as of October 30, 2022 or October 31, 2021. Accounts receivable are also recognized net of sales returns and distributor credit allowances. These amounts are recognized when it is both probable and estimable that discounts will be granted or products will be returned. Allowances for sales returns and distributor credit allowances as of October 30, 2022 and October 31, 2021 were \$126 million and \$129 million, respectively.

[Table of Contents](#)

Concentrations of credit risk and significant customers. Our cash, cash equivalents and accounts receivable are potentially subject to concentration of credit risk. Cash and cash equivalents may be redeemable upon demand and are maintained with financial institutions that management believes are of high credit quality and therefore bear minimal credit risk. We seek to mitigate our credit risks by spreading such risks across multiple counterparties and monitoring the risk profile of these counterparties. Our accounts receivable are derived from revenue earned from customers located both within and outside the U.S. We mitigate collection risks from our customers by performing regular credit evaluations of our customers' financial conditions, and require collateral, such as letters of credit and bank guarantees, in certain circumstances.

Concentration of other risks. We operate in markets that are highly competitive and rapidly changing. Significant technological changes, shifting customer needs, the emergence of competitive products with new capabilities, general economic conditions worldwide, the ability to safeguard patents and other intellectual property ("IP") in a rapidly evolving market and reliance on assembly and test subcontractors, third-party wafer fabricators and independent distributors and other factors could affect our financial results.

Inventory. We value our inventory at the lower of actual cost or net realizable value of the inventory, with cost being determined under the first-in, first-out method. We record a provision for excess and obsolete inventory based primarily on our forecast of product demand and production requirements. The excess and obsolete balance determined by this analysis becomes the basis for our excess and obsolete inventory charge and the written-down value of the inventory becomes its new cost basis.

Retirement benefit plans. For defined benefit pension plans, we consider various factors in determining our respective benefit obligations and net periodic benefit (income) cost, including the number of employees that we expect to receive benefits, their salary levels and years of service, the expected return on plan assets, the discount rate, the timing of the payment of benefits, and other actuarial assumptions. If the actual results and events of the benefit plans differ from our current assumptions, the benefit obligations may be over- or under-valued.

The key assumptions are the discount rate and the expected rate of return on plan assets. The U.S. discount rates are based on a hypothetical yield curve constructed using high-quality corporate bonds selected to yield cash flows that match the expected timing and amount of the benefit payments. The U.S. expected rate of return on plan assets is set equal to the discount rate due to the implementation of our fully-matched, liability-driven investment strategy. We evaluate these assumptions at least annually. For the non-U.S. plans, we set assumptions specific to each country. We have elected to measure defined benefit pension plan assets and liabilities as of October 31, which is the month end that is closest to our fiscal year end.

Derivative instruments. We use derivative financial instruments to manage exposure to foreign exchange risk and interest rate risk. We do not use derivative financial instruments for speculative or trading purposes.

Outstanding derivatives are recognized as assets or liabilities at their fair values based on Level 2 inputs, as defined in the fair value hierarchy. For derivative instruments designated as cash flow hedges, the changes in fair value are initially recognized in other comprehensive income (loss), net of tax in the period of change, and are subsequently reclassified and recognized in the same line item as the hedged item when either the hedged transactions affect earnings or it becomes probable that the hedged transactions will not occur.

We use foreign exchange forward contracts to manage exposure to foreign exchange risk. These forward contracts are not designated as hedging instruments, and the changes in fair value are recognized in other income (expense), net in the period of change. We did not have any outstanding foreign exchange forward contracts as of October 30, 2022 or October 31, 2021. The gains and losses recorded in other income (expense), net for derivative instruments not designated as hedges were not material.

During fiscal year 2022, we entered into treasury rate lock contracts that mature in approximately one year to hedge variability of cash flows due to changes in the benchmark interest rate of anticipated future debt issuances. These treasury rate locks are designated and accounted for as cash flow hedging instruments. As of October 30, 2022, the total notional amount of these contracts was \$1.3 billion, and the fair value of these contracts was \$47 million, which was recorded as a derivative asset with the gains recorded net of tax as a component of accumulated other comprehensive loss on our consolidated balance sheet.

Property, plant and equipment. Property, plant and equipment are stated at cost less accumulated depreciation and amortization. Additions, improvements and major renewals are capitalized, and maintenance, repairs and minor renewals are expensed as incurred. Assets are held in construction in progress until placed in service, upon which date, we begin to depreciate these assets. When assets are retired or disposed of, the assets and related accumulated depreciation and amortization are removed from our property, plant and equipment balances and the resulting gain or loss is reflected in the consolidated statements of operations. Buildings and leasehold improvements are generally depreciated over 15 to 40 years,

or over the lease period, whichever is shorter, and machinery and equipment are generally depreciated over 3 to 10 years. We use the straight-line method of depreciation for all property, plant and equipment.

Leases. We determine if an arrangement is a lease, or contains a lease, at the inception of the arrangement and evaluate whether the lease is an operating lease or a finance lease at the commencement date. We recognize right-of-use ("ROU") assets and lease liabilities for operating and finance leases with terms greater than 12 months, and account for the lease and non-lease components as a single component. ROU assets represent our right to use an asset for the lease term, while lease liabilities represent our obligation to make lease payments. Operating and finance lease ROU assets and liabilities are recognized based on the present value of lease payments over the lease term at the lease commencement date. We use the implicit interest rate or, if not readily determinable, our incremental borrowing rate as of the lease commencement date to determine the present value of lease payments. The incremental borrowing rate is based on our unsecured borrowing rate, adjusted for the effects of collateral. Operating and finance lease ROU assets are recognized net of any lease prepayments and incentives. Lease terms may include options to extend or terminate the lease when it is reasonably certain that we will exercise that option. Operating lease expense is recognized on a straight-line basis over the lease term. Finance lease expense is recognized based on the effective-interest method over the lease term.

Fair value measurement. Fair value is defined as the price that would be received upon sale of an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. A three-level hierarchy is applied to prioritize the inputs to valuation techniques used to measure fair value. The hierarchy gives the highest priority to unadjusted quoted prices in active markets for identical assets or liabilities (Level 1 measurements) and the lowest priority to unobservable inputs (Level 3 measurements).

The three levels of the fair value hierarchy under the guidance for fair value measurements are described below:

Level 1 — Level 1 inputs are quoted prices (unadjusted) in active markets for identical assets or liabilities that the reporting entity has the ability to access at the measurement date. Our Level 1 assets include cash equivalents, banker's acceptances, trading securities investments and investment funds. We measure trading securities investments and investment funds at quoted market prices as they are traded in active markets with sufficient volume and frequency of transactions.

Level 2 — Level 2 inputs are inputs other than quoted prices included within Level 1 that are observable for the asset or liability, either directly or indirectly. If the asset or liability has a specified contractual term, a Level 2 input must be observable for substantially the full term of the asset or liability.

Level 3 — Level 3 inputs are unobservable inputs for the asset or liability in which there is little, if any, market activity for the asset or liability at the measurement date. Level 3 assets and liabilities include investment in equity securities without readily determinable fair values, goodwill, intangible assets, and property, plant and equipment, which are measured at fair value using a discounted cash flow approach when they are impaired. Quantitative information for Level 3 assets and liabilities reviewed at each reporting period includes indicators of significant deterioration in the earnings performance, credit rating, asset quality, business prospects of the investee, and financial indicators of the investee's ability to continue as a going concern.

Business combinations. We account for business combinations under the acquisition method of accounting, which requires us to recognize separately from goodwill the assets acquired and the liabilities assumed at their acquisition date fair values. While we use our best estimates and assumptions to accurately value assets acquired and liabilities assumed at the acquisition date as well as contingent consideration, where applicable, our estimates are inherently uncertain and subject to refinement. As a result, during the measurement period, which may be up to one year from the acquisition date, we record adjustments to the assets acquired and liabilities assumed with the corresponding offset to goodwill. Upon the conclusion of the measurement period or final determination of the values of assets acquired or liabilities assumed, whichever comes first, any subsequent adjustments are recognized in our consolidated statements of operations. Accounting for business combinations requires our management to make significant estimates and assumptions, especially at the acquisition date including our estimates for intangible assets, contractual obligations assumed, restructuring liabilities, pre-acquisition contingencies, and contingent consideration, where applicable. Although we believe the assumptions and estimates we have made in the past have been reasonable and appropriate, they are based, in part, on historical experience and information obtained from the management of the acquired companies and are inherently uncertain. Critical estimates in valuing certain acquired intangible assets under the income approach include growth in future expected cash flows from product sales, customer contracts and acquired technologies, revenue growth rate, customer ramp-up period, technology obsolescence rates, expected costs to develop in-process research and development ("IPR&D") into commercially viable products, estimated cash flows from the projects when completed and discount rates. Unanticipated events and circumstances may occur that may affect the accuracy or validity of such assumptions, estimates or actual results.

[Table of Contents](#)

Goodwill. Goodwill represents the excess of purchase price and related costs over the value assigned to the net tangible and identifiable intangible assets of businesses acquired. Goodwill is not amortized but is reviewed annually (or more frequently if impairment indicators arise) for impairment. To review for impairment we first assess qualitative factors to determine whether events or circumstances lead to a determination that it is more likely than not that the fair value of any of our reporting units is less than its carrying amount. Our qualitative assessment of the recoverability of goodwill, whether performed annually or based on specific events or circumstances, considers various macroeconomic, industry-specific and company-specific factors. Those factors include: (i) severe adverse industry or economic trends; (ii) significant company-specific actions, including exiting an activity in conjunction with restructuring of operations; (iii) current, historical or projected deterioration of our financial performance; or (iv) a sustained decrease in our market capitalization below our net book value. After assessing the totality of events and circumstances, if we determine that it is not more likely than not that the fair value of any of our reporting units is less than its carrying amount, no further assessment is performed. If we determine that it is more likely than not that the fair value of any of our reporting units is less than its carrying amount, we calculate the fair value of that reporting unit and compare the fair value to the reporting unit's net book value. If the fair value of the reporting unit is greater than its net book value, there is no impairment. Otherwise, we calculate the implied fair value of goodwill by deducting the fair value of all tangible and intangible assets, excluding goodwill, of the reporting unit from the fair value of the reporting unit. The implied fair value of goodwill is compared to the carrying value of goodwill. If the implied fair value of goodwill is less than the carrying value of goodwill, an impairment loss is recognized equal to the difference. Determining the fair value of a reporting unit involves the use of significant estimates and assumptions.

Long-lived assets. Purchased finite-lived intangible assets are carried at cost less accumulated amortization. Amortization is recognized over the periods during which the intangible assets are expected to contribute to our cash flows. Purchased IPR&D projects are capitalized at fair value as an indefinite-lived intangible asset and assessed for impairment thereafter. Upon completion of each underlying project, IPR&D assets are reclassified as amortizable purchased intangible assets and amortized over their estimated useful lives. If an IPR&D project is abandoned, we recognize the carrying value of the related intangible asset in our consolidated statements of operations in the period it is abandoned. On a quarterly basis, we monitor factors and changes in circumstances that could indicate carrying amounts of long-lived assets, including purchased intangible assets and property, plant and equipment, may not be recoverable. Factors we consider important which could trigger an impairment review include: (i) significant under-performance relative to historical or projected future operating results, (ii) significant changes in the manner of our use of the acquired assets or the strategy for our overall business, and (iii) significant negative industry or economic trends. An impairment loss must be measured if the sum of the expected future cash flows (undiscounted and before interest) from the use and eventual disposition of the asset (or asset group) is less than the net book value of the asset (or asset group). The amount of the impairment loss will generally be measured as the difference between the net book value of the asset (or asset group) and the estimated fair value.

Warranty. We accrue for the estimated costs of product warranties at the time revenue is recognized. Product warranty costs are estimated based upon our historical experience and specific identification of the product requirements, which may fluctuate based on product mix. Additionally, we accrue for warranty costs associated with occasional or unanticipated product quality issues if a loss is probable and can be reasonably estimated.

Revenue recognition. We account for a contract with a customer when both parties have approved the contract and are committed to perform their respective obligations, each party's rights can be identified, payment terms can be identified, the contract has commercial substance, and it is probable we will collect substantially all of the consideration we are entitled to. Revenue is recognized when, or as, performance obligations are satisfied by transferring control of a promised product or service to a customer.

Nature of Products and Services

Our products and services can be broadly categorized as sales of products and subscriptions and services. The following is a description of the principal activities from which we generate revenue.

Products. We recognize revenue from sales to direct customers and distributors when control transfers to the customer. Rebates and incentives offered to distributors, which are earned when sales to end customers are completed, are estimated at the point of revenue recognition. We have elected to exclude from the transaction price any taxes collected from a customer and to account for shipping and handling activities performed after a customer obtains control of the product as activities to fulfill the promise to transfer the product. From time to time, certain customers agree to pay us secure supply fees in exchange for prioritized fulfillment of product orders. Such fees are included in the transaction price of the product orders and are recognized as revenue in the period that control over the products is transferred to the customer.

Subscriptions and services. Our subscriptions and services revenue consists of sales and royalties from software arrangements, support services, professional services, transfer of IP, and non-recurring engineering ("NRE") arrangements.

Revenue from software arrangements primarily consists of fees, which may be paid either at contract inception or in

[Table of Contents](#)

installments over the contract term, that provide customers with a right to use the software, access general support and maintenance, and utilize our professional services.

Our software licenses have standalone functionality from which customers derive benefit, and the customer obtains control of the software when it is delivered or made available for download. We believe that for the majority of software arrangements, customers derive significant benefit from the ongoing support we provide. The majority of our subscriptions and services arrangements permit our customers to unilaterally terminate or cancel these arrangements at any time at the customer's convenience, referred to as termination for convenience provisions, without substantive termination penalty and receive a pro-rata refund of any prepaid fees. Accordingly, we account for arrangements with these termination for convenience provisions as a series of daily contracts, resulting in ratable revenue recognition of software revenue over the contractual period.

Support services consist primarily of telephone support and the provision of unspecified updates and upgrades on a when-and-if-available basis. Support services represent stand-ready obligations for which revenue is recognized ratably over the term of the arrangement.

Professional services consist of implementation, consulting, customer education and customer training services. The obligation to provide professional services is generally satisfied over time, with the customer simultaneously receiving and consuming the benefits as we satisfy our performance obligations.

Rights to our IP are either sold or licensed to a customer. IP revenue recognition is dependent on the nature and terms of each agreement. We recognize IP revenue upon delivery of the IP if there are no substantive future obligations to perform under the arrangement. Sales-based or usage-based royalties from the license of IP are recognized at the later of the period the sales or usages occur or the satisfaction of the performance obligation to which some or all of the sales-based or usage-based royalties have been allocated.

There are two main categories of NRE contracts that we enter into with our customers: (a) NRE contracts in which we develop a custom chip and (b) NRE contracts in which we accelerate our development of a new chip upon the customer's request. The majority of our NRE contract revenues meet the over time criteria. As such, revenue is recognized over the development period with the measure of progress using the input method based on costs incurred to total cost as the services are provided. For NRE contracts that do not meet the over time criteria, revenue is recognized at a point in time when the NRE services are complete.

Material rights. Contracts with customers may also include material rights that are also performance obligations. These include the right to renew or receive products or services at a discounted price in the future. Revenue allocated to material rights is recognized when the customer exercises the right or the right expires.

Arrangements with Multiple Performance Obligations

Our contracts may contain more than one of the products and services listed above, each of which is separately accounted for as a distinct performance obligation.

Allocation of consideration. We allocate total contract consideration to each distinct performance obligation in a bundled arrangement on a relative standalone selling price basis. The standalone selling price reflects the price we would charge for a specific product or service if it were sold separately in similar circumstances and to similar customers.

Standalone selling price. When available, we use directly observable transactions to determine the standalone selling prices for performance obligations. When directly observable transactions are not available, our estimates of standalone selling price for each performance obligation require judgment that considers multiple factors, including, but not limited to, historical discounting trends for products and services and pricing practices through different sales channels, gross margin objectives, internal costs, competitor pricing strategies, technology lifecycles and market conditions.

We separately determine the standalone selling prices by product or service type. Additionally, we segment the standalone selling prices for products where the pricing strategies differ, and where there are differences in customers and circumstances that warrant segmentation.

We also estimate the standalone selling price of our material rights. We estimate the value of the customer's option to purchase or receive additional products or services at a discounted price by estimating the incremental discount the customer would obtain when exercising the option and the likelihood that the option would be exercised.

Other Policies and Judgments

Contract modifications. We may modify contracts to offer customers additional products or services. Each of the additional products and services is generally considered distinct from those products or services transferred to the customer before the modification. We evaluate whether the contract price for the additional products and services reflects the

[Table of Contents](#)

standalone selling price as adjusted for facts and circumstances applicable to that contract. In these cases, we account for the additional products or services as a separate contract. In other cases where the pricing in the modification does not reflect the standalone selling price as adjusted for facts and circumstances applicable to that contract, we account for the additional products or services as part of the existing contract on a prospective basis, on a cumulative catch-up basis, or a combination of both based on the nature of the modification. In instances where the pricing in the modification offers the customer a credit for a prior arrangement, we adjust our variable consideration reserves for returns and other concessions.

Right of return. Certain contracts contain a right of return that allows the customer to cancel all or a portion of the product or service and receive a credit. We estimate returns based on historical returns data which is constrained to an amount for which a material revenue reversal is not probable. We do not recognize revenue for products or services that are expected to be returned.

Practical expedient elected. We do not disclose the value of unsatisfied performance obligations for (i) contracts with an original expected length of one year or less and (ii) contracts for which we recognize revenue at the amount to which we have the right to invoice for services performed. For contracts that were modified before the beginning of the earliest reporting period presented, we have not retrospectively restated the contract for those modifications. We have disclosed the aggregate effect of all modifications when identifying the satisfied and unsatisfied performance obligations for purposes of determining the transaction price and allocating the transaction price at transition.

Research and development. Research and development expense consists primarily of personnel costs for our engineers and third parties engaged in the design and development of our products, software and technologies, including salary, bonus and stock-based compensation expense, project material costs, services and depreciation. Such costs are charged to research and development expense as they are incurred.

Stock-based compensation expense. We recognize compensation expense for time-based restricted stock units ("RSUs") using the straight-line amortization method based on the fair value of RSUs on the date of grant. The fair value of RSUs is the closing market price of Broadcom common stock on the date of grant, reduced by the present value of dividends expected to be paid on Broadcom common stock prior to vesting. We recognize compensation expense for time-based stock options and employee stock purchase plan rights under the Broadcom Inc. Employee Stock Purchase Plan, as amended ("ESPP") based on the estimated grant-date fair value determined using the Black-Scholes valuation model with a straight-line amortization method.

Certain equity awards include both service and market conditions. The fair value of market-based awards is estimated on the date of grant using the Monte Carlo simulation technique. Compensation expense for market-based awards is amortized based upon a graded vesting method over the service period.

We estimate forfeitures expected to occur and recognize stock-based compensation expense for such awards expected to vest. Changes in the estimated forfeiture rates can have a significant effect on stock-based compensation expense since the effect of adjusting the rate is recognized in the period the forfeiture estimate is changed.

Shipping and handling costs. Our shipping and handling costs charged to customers are included in net revenue and the associated expense is included in cost of revenue for all periods presented.

Litigation and settlement cost. We are involved in legal actions and other matters arising in our recent business acquisitions and in the normal course of business. We recognize an estimated loss contingency when the outcome is probable prior to issuance of the consolidated financial statements and we are able to reasonably estimate the amount or range of any possible loss.

Income taxes. We account for income taxes under the asset and liability method, which requires the recognition of deferred tax assets and liabilities for the expected future tax consequences of events that have been included in the consolidated financial statements. Under this method, deferred tax assets and liabilities are determined based on the differences between the consolidated financial statements and tax basis of assets and liabilities using enacted tax rates in effect for the year in which the differences are expected to reverse. The effect of a change in tax rates on deferred tax assets and liabilities is recognized in income in the period that includes the enactment date.

We recognize net deferred tax assets to the extent we believe these assets will more likely than not be realized. In making such determination, we consider all available positive and negative evidence, including scheduled reversals of deferred tax liabilities, projected future taxable income, tax planning strategies and recent financial operations. If we determine that we are able to realize our deferred income tax assets in the future in excess of their net carrying values, we adjust the valuation allowance and reduce the provision for income taxes or increase the benefit from income taxes. Likewise, if we determine that we are not able to realize all or part of our net deferred tax assets, we increase the provision for income taxes or decrease the benefit from income taxes in the period such determination is made.

We account for uncertainty in income taxes in accordance with the applicable accounting guidance on income taxes. This guidance provides that a tax benefit from an uncertain tax position may be recognized when it is more likely than not that the position will be sustained upon examination, including resolutions of any related appeals or litigation processes, based on the technical merits.

Net income per share. Basic net income per share is computed by dividing net income attributable to common stock by the weighted-average number of shares of common stock outstanding during the period. Diluted net income per share is computed by dividing net income attributable to common stock by the weighted-average number of shares of common stock and potentially dilutive shares of common stock outstanding during the period. Diluted shares outstanding include the dilutive effect of unvested RSUs, in-the-money stock options, and ESPP rights (together referred to as “equity awards”), as well as convertible preferred stock. Potentially dilutive shares whose effect would have been antidilutive are excluded from the computation of diluted net income per share.

The dilutive effect of equity awards is calculated based on the average stock price for each fiscal period, using the treasury stock method. Under the treasury stock method, the amount the employee must pay for exercising stock options and purchasing shares under the ESPP and the amount of compensation cost for future service that we have not yet recognized are collectively assumed to be used to repurchase shares. The dilutive effect of convertible preferred stock is calculated using the if-converted method. The if-converted method assumes that these securities were converted at the beginning of the reporting period to the extent that the effect is dilutive.

Recently Adopted Accounting Guidance. In October 2021, the Financial Accounting Standards Board issued Accounting Standards Update (“ASU”) 2021-08, *Business Combinations (Topic 805): Accounting for Contract Assets and Contract Liabilities from Contracts with Customers*. The new guidance requires contract assets and contract liabilities acquired in a business combination to be recognized and measured by the acquirer on the acquisition date in accordance with Accounting Standards Codification 606, *Revenue from Contracts with Customers*, as if it had originated the contracts. We early adopted this guidance at the beginning of fiscal year 2022 and it did not materially impact our consolidated financial statements.

3. Revenue from Contracts with Customers

Disaggregation

We have considered (1) information that is regularly reviewed by our Chief Executive Officer, who has been identified as the chief operating decision maker (the “CODM”) as defined by the authoritative guidance on segment reporting, in evaluating financial performance and (2) disclosures presented outside of our financial statements in our earnings releases and used in investor presentations to disaggregate revenues. The principal category we use to disaggregate revenues is the nature of our products and subscriptions and services, as presented in our consolidated statements of operations. In addition, revenues by reportable segment are presented in Note 13. “Segment Information.”

The following tables present revenue disaggregated by type of revenue and by region for the periods presented:

	Fiscal Year 2022			
	Americas	Asia Pacific	Europe, the Middle East and Africa	Total
	(In millions)			
Products	\$ 2,371	\$ 21,761	\$ 2,145	\$ 26,277
Subscriptions and services ^(a)	4,573	744	1,609	6,926
Total	\$ 6,944	\$ 22,505	\$ 3,754	\$ 33,203

Fiscal Year 2021				
	Americas	Asia Pacific	Europe, the Middle East and Africa	Total
	(In millions)			
Products	\$ 1,809	\$ 17,258	\$ 1,819	\$ 20,886
Subscriptions and services ^(a)	4,290	720	1,554	6,564
Total	\$ 6,099	\$ 17,978	\$ 3,373	\$ 27,450

Fiscal Year 2020				
	Americas	Asia Pacific	Europe, the Middle East and Africa	Total
	(In millions)			
Products	\$ 1,775	\$ 14,442	\$ 1,218	\$ 17,435
Subscriptions and services ^(a)	4,059	881	1,513	6,453
Total	\$ 5,834	\$ 15,323	\$ 2,731	\$ 23,888

(a) Subscriptions and services predominantly includes software licenses with termination for convenience clauses.

Although we recognize revenue for the majority of our products when title and control transfer in Penang, Malaysia, we disclose net revenue by region based primarily on the geographic shipment location or delivery location specified by our distributors, original equipment manufacturer (“OEM”) customers, contract manufacturers, channel partners, or software customers.

Contract Balances

Contract assets and contract liabilities balances were as follows:

	October 30, 2022	October 31, 2021
	(In millions)	
Contract Assets	\$ 128	\$ 126
Contract Liabilities	\$ 3,341	\$ 3,185

Changes in our contract assets and contract liabilities primarily result from the timing difference between our performance and the customer’s payment. We fulfill our obligations under a contract with a customer by transferring products and services in exchange for consideration from the customer. We recognize a contract asset when we transfer products or services to a customer and the right to consideration is conditional on something other than the passage of time. Accounts receivable are recorded when the customer has been billed or the right to consideration is unconditional. We recognize contract liabilities when we have received consideration or an amount of consideration is due from the customer and we have a future obligation to transfer products or services. Contract liabilities include amounts billed or collected and advanced payments on contracts or arrangements which may include termination for convenience provisions. The amount of revenue recognized during fiscal year 2022 that was included in the contract liabilities balance as of October 31, 2021 was \$2,615 million. The amount of revenue recognized during fiscal year 2021 that was included in the contract liabilities balance as of November 1, 2020 was \$2,617 million.

Remaining Performance Obligations

Revenue allocated to remaining performance obligations represents the transaction price allocated to unsatisfied or partially unsatisfied performance obligations. Remaining performance obligations include unearned revenue and amounts that will be invoiced and recognized as revenue in future periods, but do not include contracts for software, subscriptions or services where the customer is not committed. The customer is not considered committed when termination for convenience without payment of a substantive penalty exists, either contractually or through customary business practice. The majority of our customer software contracts include termination for convenience clauses without a substantive penalty and are not considered committed. Additionally, as a practical expedient, we have not included contracts that have an original duration of one year or less, nor have we included contracts with sales-based or usage-based royalties promised in exchange for a license of IP.

Certain multi-year customer contracts in our semiconductor solutions segment contain firmly committed amounts and the remaining performance obligations under these contracts as of October 30, 2022 were approximately \$23.6 billion. We expect approximately 28% of this amount to be recognized as revenue over the next 12 months. Although the majority of our software contracts are not deemed to be committed, our customers generally do not exercise their termination for convenience rights. In addition, the majority of our contracts for products, subscriptions and services have a duration of one year or less. Accordingly, our remaining performance obligations disclosed above are not indicative of revenue for future periods.

4. Acquisitions

Pending Acquisition of VMware, Inc.

On May 26, 2022, we entered into an Agreement and Plan of Merger (the “VMware Merger Agreement”) to acquire all of the outstanding shares of VMware, Inc. (“VMware”) in a cash-and-stock transaction (the “VMware Merger”) that values VMware at approximately \$61 billion based on the closing price of Broadcom common stock on May 25, 2022. We will also assume VMware’s closing date outstanding debt, net of expected cash.

Under the terms of the VMware Merger Agreement, each share of VMware common stock issued and outstanding immediately prior to the effective time of the VMware Merger will be indirectly converted into the right to receive, at the election of the holder of such share of VMware common stock, either \$142.50 in cash, without interest, or 0.2520 shares of Broadcom common stock. The stockholder election will be subject to proration, such that the total number of shares of VMware common stock entitled to receive cash and the total number of shares of VMware common stock entitled to receive Broadcom common stock, will, in each case, be equal to 50% of the aggregate number of shares of VMware common stock issued and outstanding immediately prior to the effective time of the VMware Merger.

We will assume all outstanding VMware RSU awards and performance stock unit awards held by continuing employees. The assumed awards will be converted into RSU awards for shares of Broadcom common stock. All outstanding in-the-money VMware stock options and RSU awards held by non-employee directors will be accelerated and converted into the right to receive cash and shares of Broadcom common stock, in equal parts.

Effective upon the effective time of the VMware Merger, one member of the VMware Board of Directors, to be mutually agreed by us and VMware, will be added to our Board of Directors.

In connection with the execution of the VMware Merger Agreement, we entered into a commitment letter on May 26, 2022, with certain financial institutions that committed to provide, subject to the terms and conditions of the commitment letter, a senior unsecured bridge facility in an aggregate principal amount of \$32 billion.

The VMware Merger, which is expected to be completed in our fiscal year ending October 29, 2023 (“fiscal year 2023”), is subject to satisfaction or waiver of customary closing conditions, including the expiration or termination of the waiting period under the Hart-Scott-Rodino Antitrust Improvement Act of 1976 and clearance under the antitrust laws of the European Union and certain other jurisdictions. On October 3, 2022, we registered approximately 59 million shares of our common stock. On November 4, 2022, VMware stockholders adopted the VMware Merger Agreement. We and VMware each have termination rights under the VMware Merger Agreement and, under specified circumstances, upon termination of the agreement, we and VMware would be required to pay the other a termination fee of \$1.5 billion.

Acquisition of the Symantec Corporation Enterprise Security Business

On November 4, 2019 (the “Symantec Acquisition Date”), we completed the purchase of the Symantec Business, which was an established leader in cyber security, for \$10.7 billion in cash. We acquired the Symantec Business to expand our footprint of mission critical infrastructure software with our existing customer base. The Symantec Business includes a deep and broad mix of products, services and solutions, unifying cloud and on-premises security to provide advanced threat protection and information protection across endpoints, network, email and cloud applications. We financed this acquisition with borrowings.

The following table presents our allocation of the total purchase price:

	Fair Value (In millions)
Current assets	\$ 273
Goodwill	6,638
Intangible assets	5,411
Other long-term assets	92
Total assets acquired	12,414
Current liabilities	(1,127)
Other long-term liabilities	(587)
Total liabilities assumed	(1,714)
Fair value of net assets acquired	\$ 10,700

Goodwill is primarily attributable to the assembled workforce and anticipated synergies and economies of scale expected from the integration of the Symantec Business. The synergies include certain cost savings, operating efficiencies, and other strategic benefits projected to be achieved resulting from the acquisition of the Symantec Business. Substantially all goodwill is deductible for tax purposes.

Current assets and current liabilities included amounts held-for-sale related to the acquired Symantec Cyber Security Services business, which was not aligned with our acquisition-date strategic objectives and was sold on April 30, 2020. We do not have any material continuing involvement with this business and have presented its results in discontinued operations.

Our results of continuing operations for fiscal year 2020 included \$1,610 million of net revenue attributable to the Symantec Business. It was impracticable to determine the effect on net income attributable to the Symantec Business as we had integrated the Symantec Business into our ongoing operations during the year. The results of operations of the Symantec Business were included in our infrastructure software segment. Transaction costs related to the acquisition of the Symantec Business of \$110 million were included in selling, general and administrative expense for fiscal year 2020.

Intangible Assets

	Fair Value (In millions)	Weighted-Average Amortization Periods (In years)
Developed technology	\$ 2,900	5
Customer contracts and related relationships	2,410	5
Trade name	90	6
Order backlog	11	3
Total identified finite-lived intangible assets	\$ 5,411	

Developed technology relates to products used for cyber security solutions, including data loss prevention, endpoint protection, and web, email and cloud security solutions. We valued the developed technology using the multi-period excess earnings method under the income approach. This method reflects the present value of the projected cash flows that are expected to be generated by the developed technology less charges representing the contribution of other assets to those cash flows. The economic useful life was determined based on the technology cycle related to each developed technology, as well as the cash flows over the forecast period.

[Table of Contents](#)

Customer contracts and related relationships represent the fair value of future projected revenue that will be derived from sales of products to existing customers of the Symantec Business. Customer contracts and related relationships were valued using the with-and-without-method under the income approach. In the with-and-without method, the fair value was measured by the difference between the present values of the cash flows with and without the existing customers in place over the period of time necessary to reacquire the customers. The economic useful life was determined by evaluating many factors, including the useful life of other intangible assets, the length of time remaining on the acquired contracts and the historical customer turnover rates.

Trade name relates to the “Symantec” trade name. The fair value was determined by applying the relief-from-royalty method under the income approach. This method is based on the application of a royalty rate to forecasted revenue under the trade name. The economic useful life was determined based on the expected life of the trade name and the cash flows anticipated over the forecast period.

Order backlog represents business under existing contractual obligations. The fair value of backlog was determined using the multi-period excess earnings method under the income approach based on expected operating cash flows from future contractual revenue. The economic useful life was determined based on the expected life of the backlog and the cash flows over the forecast period.

We believe the amounts of purchased intangible assets recorded above represent the fair values of, and approximate the amounts a market participant would pay for, these intangible assets as of the Symantec Acquisition Date.

Unaudited Pro Forma Information

The following unaudited pro forma financial information presents combined results of operations for the period presented, as if we had completed the acquisition of the Symantec Business as of the beginning of our fiscal year ended November 3, 2019 (“fiscal year 2019”). The unaudited pro forma information includes adjustments to amortization and depreciation for intangible assets and property, plant and equipment acquired, adjustments to interest expense for the additional indebtedness incurred to complete the acquisition, restructuring charges related to the acquisition and transaction costs. The unaudited pro forma information presented below is for informational purposes only and is not necessarily indicative of our consolidated results of operations of the combined business had the acquisition actually occurred at the beginning of fiscal year 2019 or of the results of our future operations of the combined business.

	Fiscal Year 2020
	(In millions)
Pro forma net revenue	\$ 23,264
Pro forma net income attributable to common stock	\$ 2,368

Other Acquisitions

During fiscal year 2022, we completed four acquisitions qualifying as business combinations for total consideration of \$245 million. For these acquisitions, \$164 million was allocated to goodwill and \$110 million was allocated to intangible assets, with additional amounts allocated to tangible assets and liabilities, primarily within our infrastructure software segment.

During fiscal year 2020, we completed three other acquisitions qualifying as business combinations for total consideration of \$201 million. For these acquisitions, \$109 million was allocated to goodwill and \$46 million was allocated to intangible assets, with additional amounts allocated to tangible assets and liabilities, primarily within our infrastructure software segment.

5. Supplemental Financial Information

Cash Equivalents

Cash equivalents included \$3,915 million and \$4,668 million of time deposits and \$2,365 million and \$1,607 million of money-market funds as of October 30, 2022 and October 31, 2021, respectively. For time deposits, carrying value approximates fair value due to the short-term nature of the instruments. The fair value of money-market funds, which was consistent with their carrying value, was determined using unadjusted prices in active, accessible markets for identical assets, and as such, they were classified as Level 1 assets in the fair value hierarchy.

Accounts Receivable Factoring

We sell certain of our trade accounts receivable on a non-recourse basis to third-party financial institutions pursuant to factoring arrangements. We account for these transactions as sales of receivables and present cash proceeds as cash provided

by operating activities in the consolidated statements of cash flows. Total trade accounts receivable sold under the factoring arrangements were \$3,700 million, \$4,027 million and \$3,723 million during fiscal years 2022, 2021 and 2020, respectively. Factoring fees for the sales of receivables were recorded in other income (expense), net and were not material for any of the periods presented.

Inventory

	October 30, 2022	October 31, 2021
	(In millions)	
Finished goods	\$ 780	\$ 423
Work-in-process	966	680
Raw materials	179	194
Total inventory	<u>\$ 1,925</u>	<u>\$ 1,297</u>

Property, Plant and Equipment, Net

	October 30, 2022	October 31, 2021
	(In millions)	
Land	\$ 195	\$ 195
Construction in progress	63	38
Buildings and leasehold improvements	1,156	1,150
Machinery and equipment	4,413	4,161
Total property, plant and equipment	5,827	5,544
Accumulated depreciation and amortization	(3,604)	(3,196)
Total property, plant and equipment, net	<u>\$ 2,223</u>	<u>\$ 2,348</u>

Depreciation expense was \$529 million, \$539 million and \$570 million for fiscal years 2022, 2021, and 2020, respectively.

Other Current Assets

	October 30, 2022	October 31, 2021
	(In millions)	
Prepaid expenses	\$ 864	\$ 539
Other	341	516
Total other current assets	<u>\$ 1,205</u>	<u>\$ 1,055</u>

Other Current Liabilities

	October 30, 2022	October 31, 2021
	(In millions)	
Contract liabilities	\$ 2,931	\$ 2,619
Tax liabilities	680	541
Interest payable	393	282
Other	408	397
Total other current liabilities	<u>\$ 4,412</u>	<u>\$ 3,839</u>

Other Long-Term Liabilities

	October 30, 2022	October 31, 2021
	(In millions)	
Unrecognized tax benefits, interest and penalties	\$ 3,229	\$ 3,407
Contract liabilities	410	566
Other	774	887
Total other long-term liabilities	<u>\$ 4,413</u>	<u>\$ 4,860</u>

Other Income (Expense), Net

	Fiscal Year		
	2022	2021	2020
	(In millions)		
Gain (loss) on investments	\$ (169)	\$ 99	\$ 31
Other income	30	26	56
Interest income	100	16	53
Other expense	(15)	(10)	(50)
Gain from lapse of indemnification	—	—	116
Other income (expense), net	<u>\$ (54)</u>	<u>\$ 131</u>	<u>\$ 206</u>

Other income includes foreign exchange gains, dividends, and other miscellaneous items.

6. Leases

At the beginning of fiscal year 2020, we adopted ASU 2016-02, *Leases* ("Topic 842") using the optional adoption method, whereby no adjustment to the financial statements of comparative periods was required. We have operating and finance leases for our facilities, data centers and certain equipment. Operating lease expense was \$98 million, \$102 million and \$106 million for fiscal years 2022, 2021 and 2020, respectively. Finance lease expense was \$18 million, \$16 million and \$14 million for fiscal years 2022, 2021 and 2020 respectively.

Other information related to leases was as follows:

	Fiscal Year		
	2022	2021	2020
	(In millions)		
Cash paid for operating leases included in operating cash flows	\$ 103	\$ 140	\$ 125
ROU assets obtained in exchange for operating lease liabilities	\$ 16	\$ 92	\$ 682
ROU assets obtained in exchange for finance lease liabilities	\$ 1	\$ 15	\$ 74

	October 30, 2022	October 31, 2021
Weighted-average remaining lease term – operating leases (In years)	10	10
Weighted-average remaining lease term – finance leases (In years)	2	3
Weighted-average discount rate – operating leases	3.60 %	3.78 %
Weighted-average discount rate – finance leases	3.05 %	3.11 %

Supplemental balance sheet information related to leases was as follows:

	Classification on the Consolidated Balance Sheets	October 30, 2022	October 31, 2021
(In millions)			
ROU assets - operating leases	Other long-term assets	\$ 517	\$ 588
ROU assets - finance leases	Property, plant and equipment, net	\$ 40	\$ 55
Short-term lease liabilities - operating leases	Other current liabilities	\$ 74	\$ 83
Long-term lease liabilities - operating leases	Other long-term liabilities	\$ 389	\$ 460
Short-term lease liabilities - finance leases	Current portion of long-term debt	\$ 37	\$ 26
Long-term lease liabilities - finance leases	Long-term debt	\$ 22	\$ 39

Future minimum lease payments under non-cancelable leases as of October 30, 2022 were as follows:

	October 30, 2022	
	Operating Leases	Finance Leases
(In millions)		
2023	\$ 89	\$ 38
2024	69	18
2025	58	2
2026	45	2
2027	40	—
Thereafter	267	—
Total undiscounted liabilities	568	60
Less: interest	(105)	(1)
Present value of lease liabilities	\$ 463	\$ 59

7. Goodwill and Intangible Assets

Goodwill

	Semiconductor Solutions	Infrastructure Software	Total
(In millions)			
Balance as of November 1, 2020	\$ 25,959	\$ 17,488	\$ 43,447
Acquisition	—	10	10
Sale of business	—	(7)	(7)
Balance as of October 31, 2021	25,959	17,491	43,450
Acquisitions	8	156	164
Balance as of October 30, 2022	\$ 25,967	\$ 17,647	\$ 43,614

During the fourth quarter of fiscal years 2022, 2021 and 2020, we completed our annual impairment assessments and concluded that goodwill was not impaired in any of these years.

Intangible Assets

	Gross Carrying Amount	Accumulated Amortization	Net Book Value
		(In millions)	
As of October 30, 2022:			
Purchased technology	\$ 19,450	\$ (15,422)	\$ 4,028
Customer contracts and related relationships	7,066	(4,535)	2,531
Order backlog	484	(382)	102
Trade names	700	(372)	328
Other	174	(81)	93
Intangible assets subject to amortization	27,874	(20,792)	7,082
IPR&D	29	—	29
Total	<u>\$ 27,903</u>	<u>\$ (20,792)</u>	<u>\$ 7,111</u>

As of October 31, 2021:			
Purchased technology	\$ 23,932	\$ (17,148)	\$ 6,784
Customer contracts and related relationships	8,356	(4,533)	3,823
Order backlog	2,579	(2,352)	227
Trade names	787	(386)	401
Other	239	(127)	112
Intangible assets subject to amortization	35,893	(24,546)	11,347
IPR&D	27	—	27
Total	<u>\$ 35,920</u>	<u>\$ (24,546)</u>	<u>\$ 11,374</u>

Based on the amount of intangible assets subject to amortization at October 30, 2022, the expected amortization expense for each of the next five fiscal years and thereafter was as follows:

Fiscal Year:	Expected Amortization Expense
	(In millions)
2023	\$ 3,255
2024	2,388
2025	681
2026	344
2027	216
Thereafter	198
Total	<u>\$ 7,082</u>

The weighted-average amortization periods remaining by intangible asset category were as follows:

Amortizable intangible assets:	October 30, 2022	October 31, 2021
	(In years)	
Purchased technology	3	4
Customer contracts and related relationships	2	3
Order backlog	1	2
Trade names	8	8
Other	8	9

8. Net Income Per Share

	Fiscal Year		
	2022	2021	2020
(In millions, except per share data)			
Numerator:			
Income from continuing operations	\$ 11,495	\$ 6,736	\$ 2,961
Dividends on preferred stock	(272)	(299)	(297)
Income from continuing operations attributable to common stock	11,223	6,437	2,664
Loss from discontinued operations, net of income taxes, attributable to common stock	—	—	(1)
Net income attributable to common stock	<u>\$ 11,223</u>	<u>\$ 6,437</u>	<u>\$ 2,663</u>
Denominator:			
Weighted-average shares outstanding - basic	409	410	402
Dilutive effect of equity awards	14	19	19
Weighted-average shares outstanding - diluted	<u>423</u>	<u>429</u>	<u>421</u>
Net income per share attributable to common stock:			
Basic	\$ 27.44	\$ 15.70	\$ 6.62
Diluted	\$ 26.53	\$ 15.00	\$ 6.33

For fiscal years 2022, 2021 and 2020, diluted net income per share excluded the potentially dilutive effect of 10 million, 12 million and 12 million shares of common stock, respectively, issuable upon the conversion of Mandatory Convertible Preferred Stock, as defined in Note 11. "Stockholders' Equity," as their effect was antidilutive.

9. Retirement Plans

Defined Benefit Pension Plans

The U.S. defined benefit pension plans primarily consist of a qualified pension plan. Benefits of the qualified pension plan are provided under an adjusted career-average-pay program, a cash-balance program or a dollar-per-month program. Benefit accruals under this plan were frozen in 2009. Participants in the adjusted career-average-pay program no longer earn service accruals. Participants in the cash-balance program no longer earn service accruals, but continue to earn 4% interest per year on their cash-balance accounts. There are no active participants under the dollar-per-month program. We also have a non-qualified supplemental pension plan in the United States that principally provides benefits based on compensation in excess of amounts that can be considered under the qualified pension plan.

We also have defined benefit pension plans for certain employees in Austria, France, Germany, India, Israel, Italy, Japan and Taiwan. Eligibility is generally determined based on the terms of our plans and local statutory requirements.

Net Periodic Benefit Cost

	Fiscal Year		
	2022	2021	2020
(In millions)			
Service cost	\$ 8	\$ 11	\$ 12
Interest cost	39	39	45
Expected return on plan assets	(39)	(40)	(46)
Other	1	1	(3)
Net periodic benefit cost	<u>\$ 9</u>	<u>\$ 11</u>	<u>\$ 8</u>
Net actuarial (gain) loss	\$ (17)	\$ 8	\$ (28)

The components of net periodic benefit cost other than the service cost are included in other income (expense), net. Service cost is recognized in operating expenses.

Benefit Obligations and Plan Assets

	October 30, 2022	October 31, 2021
	(In millions)	
Change in plan assets:		
Fair value of plan assets — beginning of period	\$ 1,521	\$ 1,593
Actual return on plan assets	(279)	20
Employer contributions	10	9
Plan participants' contributions	1	—
Payments from plan assets	(95)	(102)
Foreign currency impact	2	1
Fair value of plan assets — end of period	<u>1,160</u>	<u>1,521</u>
Change in benefit obligations:		
Benefit obligations — beginning of period	1,526	1,588
Service cost	8	11
Interest cost	39	39
Actuarial gain ^(a)	(336)	(11)
Plan participants' contributions	1	—
Benefit payments	(95)	(102)
Curtailments	—	(1)
Foreign currency impact	—	2
Benefit obligations — end of period	<u>1,143</u>	<u>1,526</u>
Overfunded (underfunded) status of benefit obligations ^(b)	<u>\$ 17</u>	<u>\$ (5)</u>
Actuarial losses and prior service costs recognized in accumulated other comprehensive loss, net of taxes	\$ (82)	\$ (100)

(a) The actuarial gain in fiscal year 2022 was primarily due to an increase in discount rates experienced by the majority of our plans.

(b) Substantially all amounts recognized on the consolidated balance sheets were recorded in other long-term assets and other long-term liabilities for all periods presented.

Plans with benefit obligations in excess of plan assets:

	October 30, 2022	October 31, 2021
	(In millions)	
Projected benefit obligations	\$ 71	\$ 83
Accumulated benefit obligations	\$ 55	\$ 65
Fair value of plan assets	\$ 12	\$ 13

Plans with benefit obligations less than plan assets:

	October 30, 2022	October 31, 2021
	(In millions)	
Projected benefit obligations	\$ 1,072	\$ 1,443
Accumulated benefit obligations	\$ 1,070	\$ 1,442
Fair value of plan assets	\$ 1,148	\$ 1,508

The fair value of pension plan assets as of October 30, 2022 and October 31, 2021 included \$184 million and \$174 million, respectively, of assets for our non-U.S. pension plans.

The projected benefit obligations as of October 30, 2022 and October 31, 2021 included \$185 million and \$217 million, respectively, of obligations related to our non-U.S. pension plans. The accumulated benefit obligations as of October 30, 2022 and October 31, 2021 included \$168 million and \$199 million, respectively, of obligations related to our non-U.S. pension plans.

Expected Future Benefit Payments

Fiscal Years:	Expected Benefit Payments
	(In millions)
2023	\$ 95
2024	\$ 95
2025	\$ 94
2026	\$ 94
2027	\$ 93
2028-2032	\$ 444

Investment Policy

Plan assets of the U.S. qualified pension plan, which represent substantially all of the plan assets, are generally invested in funds held by third-party fund managers. Our benefit plan investment committee has set the investment strategy to fully match the liability. We direct the overall portfolio allocation and use a third-party investment consultant that has the discretion to structure portfolios and select the investment managers within those allocation parameters. Multiple investment managers are utilized, including both active and passive management approaches. The plan assets are invested using the liability-driven investment strategy intended to minimize market and interest rate risks, and those assets are periodically rebalanced toward asset allocation targets.

The target asset allocation for the U.S. qualified pension plan reflects a risk/return profile that we believe is appropriate relative to the liability structure and return goals for the plan. We periodically review the allocation of plan assets relative to alternative allocation models to evaluate the need for adjustments based on forecasted liabilities and plan liquidity needs. For both fiscal years 2022 and 2021, 100% of the U.S. qualified pension plan assets were allocated to fixed income, in line with the target allocation. The fixed income allocation is primarily directed toward long-term core bond investments, with smaller allocations to Treasury Inflation-Protected Securities and high-yield bonds.

Fair Value Measurement of Plan Assets

	October 30, 2022		
	Fair Value Measurements at Reporting Date		
	Using		
	Level 1	Level 2	Total
	(In millions)		
Cash equivalents	\$ 19 ^(a)	\$ —	\$ 19
Equity securities:			
Non-U.S. equity securities	46 ^(b)	—	46
Fixed-income securities:			
U.S. treasuries	—	147 ^(c)	147
Corporate bonds	—	901 ^(c)	901
Municipal bonds	—	20 ^(c)	20
Government bonds	—	25 ^(c)	25
Asset-backed securities	—	2 ^(c)	2
Total plan assets	\$ 65	\$ 1,095	\$ 1,160

	October 31, 2021		
	Fair Value Measurements at Reporting Date		
	Using		
	Level 1	Level 2	Total
	(In millions)		
Cash equivalents	\$ 24 ^(a)	\$ —	\$ 24
Equity securities:			
Non-U.S. equity securities	28 ^(b)	—	28
Fixed-income securities:			
U.S. treasuries	—	186 ^(c)	186
Corporate bonds	—	1,222 ^(c)	1,222
Municipal bonds	—	24 ^(c)	24
Government bonds	—	34 ^(c)	34
Asset-backed securities	—	3 ^(c)	3
Total plan assets	\$ 52	\$ 1,469	\$ 1,521

(a) Cash equivalents primarily included short-term investment funds which consisted of short-term money market instruments that were valued based on quoted prices in active markets.

(b) These equity securities were valued based on quoted prices in active markets.

(c) These amounts consisted of investments that were traded less frequently than Level 1 securities and were valued using inputs that included quoted prices for similar assets in active markets and inputs other than quoted prices that were observable for the assets, such as interest rates, yield curves, prepayment speeds, collateral performance, broker/dealer quotes and indices that were observable at commonly quoted intervals.

Assumptions

The assumptions used to determine the benefit obligations and net periodic benefit cost for our defined benefit pension plans are presented in the table below. The expected long-term return on assets shown in the table below represents an estimate of long-term returns on investment portfolios primarily consisting of combinations of debt, equity and other investments, depending on the plan. The long-term rates of return are then weighted based on the asset classes in which the pension funds are invested. Discount rates reflect the current rate at which defined benefit pension obligations could be settled based on the measurement dates of the plans, which is October 31, the month end closest to our fiscal year end. The range of assumptions reflects the different economic environments within various countries.

	Assumptions for Benefit Obligations as of		Assumptions for Net Periodic Benefit Cost Fiscal Year		
	October 30, 2022	October 31, 2021	2022	2021	2020
Discount rate	1.25%-7.25%	0.75%-6.50%	0.75%-6.50%	0.61%-6.54%	0.47%-7.00%
Average increase in compensation levels	2.00%-10.00%	2.00%-10.00%	2.00%-10.00%	2.00%-10.00%	2.00%-10.00%
Expected long-term return on assets	N/A	N/A	1.50%-7.25%	1.00%-8.00%	1.50%-7.80%

Defined Contribution Plans

Our eligible U.S. employees participate in a company-sponsored 401(k) plan. Under the plan, we match employee contributions dollar for dollar up to 6% of their eligible earnings. All matching contributions vest immediately. During fiscal years 2022, 2021 and 2020, we made contributions of \$96 million, \$94 million and \$99 million, respectively, to the 401(k) plan.

In addition, other eligible employees outside of the U.S. receive retirement benefits under various defined contribution retirement plans.

10. Borrowings

	Effective Interest Rate	October 30, 2022	October 31, 2021
(In millions, except percentages)			
<u>April 2022 Senior Notes - fixed rate</u>			
4.000% notes due April 2029	4.17 %	\$ 750	\$ —
4.150% notes due April 2032	4.30 %	1,200	—
4.926% notes due May 2037	5.33 %	2,500	—
		4,450	—
<u>September 2021 Senior Notes - fixed rate</u>			
3.137% notes due November 2035	4.23 %	3,250	3,250
3.187% notes due November 2036	4.79 %	2,750	2,750
		6,000	6,000
<u>March 2021 Senior Notes - fixed rate</u>			
3.419% notes due April 2033	4.66 %	2,250	2,250
3.469% notes due April 2034	4.63 %	3,250	3,250
		5,500	5,500
<u>January 2021 Senior Notes - fixed rate</u>			
1.950% notes due February 2028	2.10 %	750	750
2.450% notes due February 2031	2.56 %	2,750	2,750
2.600% notes due February 2033	2.70 %	1,750	1,750
3.500% notes due February 2041	3.60 %	3,000	3,000
3.750% notes due February 2051	3.84 %	1,750	1,750
		10,000	10,000
<u>June 2020 Senior Notes - fixed rate</u>			
3.459% notes due September 2026	4.19 %	752	752
4.110% notes due September 2028	5.02 %	1,118	1,965
		1,870	2,717
<u>May 2020 Senior Notes - fixed rate</u>			
2.250% notes due November 2023	2.40 %	105	105
3.150% notes due November 2025	3.29 %	900	900
4.150% notes due November 2030	4.27 %	1,856	2,679
4.300% notes due November 2032	4.39 %	2,000	2,000
		4,861	5,684
<u>April 2020 Senior Notes - fixed rate</u>			
4.700% notes due April 2025	4.88 %	—	1,020
5.000% notes due April 2030	5.18 %	606	1,086
		606	2,106
<u>April 2019 Senior Notes - fixed rate</u>			
3.625% notes due October 2024	3.98 %	622	622
4.250% notes due April 2026	4.54 %	—	944
4.750% notes due April 2029	4.95 %	1,655	1,958
		2,277	3,524
<u>2017 Senior Notes - fixed rate</u>			
3.000% notes due January 2022	3.21 %	—	255
2.650% notes due January 2023	2.78 %	260	260
3.625% notes due January 2024	3.74 %	829	829

	Effective Interest Rate	October 30, 2022	October 31, 2021
(In millions, except percentages)			
3.125% notes due January 2025	3.23 %	495	495
3.875% notes due January 2027	4.02 %	2,922	2,922
3.500% notes due January 2028	3.60 %	777	777
		5,283	5,538
Assumed CA Senior Notes - fixed rate			
4.500% notes due August 2023	4.10 %	143	143
4.700% notes due March 2027	5.15 %	215	265
		358	408
Other senior notes - fixed rate			
2.500% notes due August 2022	2.59 %	—	9
3.500% notes due August 2024	3.55 %	7	7
4.500% notes due August 2034	4.55 %	6	6
		13	22
Total principal amount outstanding		\$ 41,218	\$ 41,499
Current portion of principal amount outstanding		\$ 403	\$ 264
Short-term finance lease liabilities		37	26
Total current portion of long-term debt		\$ 440	\$ 290
Non-current portion of principal amount outstanding		\$ 40,815	\$ 41,235
Long-term finance lease liabilities		22	39
Unamortized discount and issuance costs		(1,762)	(1,834)
Total long-term debt		\$ 39,075	\$ 39,440

The senior notes are recorded net of discount and issuance costs, which are amortized to interest expense over the respective terms of such senior notes.

April 2022 Senior Notes

In April 2022, we issued \$750 million of 4.000% senior unsecured notes due April 2029 and \$1,200 million of 4.150% senior unsecured notes due April 2032. Using the net proceeds, we redeemed the outstanding balance of \$1,020 million of our 4.700% notes due 2025 and \$944 million of our 4.250% notes due 2026. As a result of these redemptions, we incurred premiums of \$85 million and wrote off \$15 million of unamortized discount and issuance costs, both of which were included in interest expense.

In April 2022, we issued \$2,500 million of 4.926% senior unsecured notes due May 2037 in exchange for \$2,502 million of certain of our outstanding notes maturing between 2027 and 2030. As a result of this exchange, we paid premiums of \$47 million, which were included in unamortized discount and issuance costs. The 4.926% notes due 2037, the 4.000% notes due 2029 and the 4.150% notes due 2032 are collectively referred as the “April 2022 Senior Notes.”

We may redeem or purchase, in whole or in part, any of the April 2022 Senior Notes prior to their respective maturities, subject to a specified make-whole premium determined in accordance with the indentures governing the April 2022 Senior Notes, plus accrued and unpaid interest. In the event of a change in control, note holders will have the right to require us to repurchase their notes at a price equal to 101% of the principal amount of such notes, plus accrued and unpaid interest.

September 2021 Senior Notes

In September 2021, we completed our private offers to exchange \$6.0 billion of certain of our outstanding notes maturing between 2025 and 2030 for \$3,250 million of 3.137% senior unsecured notes due November 2035 and \$2,750 million of 3.187% senior unsecured notes due November 2036 (collectively, the “September 2021 Senior Notes”). As a result of this exchange, we paid premiums of \$762 million, which were included in unamortized discount and issuance costs. We

may redeem or purchase, in whole or in part, any of the September 2021 Senior Notes prior to their respective maturities, subject to a specified make-whole premium determined in accordance with the indenture governing the September 2021 Senior Notes, plus accrued and unpaid interest. In the event of a change in control, note holders will have the right to require us to repurchase their notes at a price equal to 101% of the principal amount of such notes, plus accrued and unpaid interest.

March 2021 Senior Notes

In March 2021, we completed our private offers to exchange \$5.5 billion of certain of our outstanding notes maturing between 2024 and 2027 (the “March 2021 Exchange Offer”) for \$2,250 million of 3.419% senior unsecured notes due April 2033 and \$3,250 million of 3.469% senior unsecured notes due April 2034 (collectively, the “March 2021 Senior Notes”). As a result of this exchange, we paid premiums of \$581 million, which were included in unamortized discount and issuance costs. We may redeem or purchase, in whole or in part, any of the March 2021 Senior Notes prior to their respective maturities, subject to a specified make-whole premium determined in accordance with the indenture governing the March 2021 Senior Notes, plus accrued and unpaid interest. In the event of a change in control, note holders will have the right to require us to repurchase their notes at a price equal to 101% of the principal amount of such notes, plus accrued and unpaid interest.

In connection with the March 2021 Exchange Offer, Broadcom Corporation (“BRCM”) and Broadcom Technologies Inc. (“BTI”) were automatically and unconditionally released from their guarantees in accordance with the respective indentures governing the January 2021 Senior Notes, June 2020 Senior Notes, May 2020 Senior Notes, April 2020 Senior Notes, and April 2019 Senior Notes, as defined below respectively.

January 2021 Senior Notes

In January 2021, we issued \$10 billion of senior unsecured notes (the “January 2021 Senior Notes”). We may redeem or purchase, in whole or in part, any of the January 2021 Senior Notes prior to their respective maturities, subject to a specified make-whole premium determined in accordance with the indenture governing the January 2021 Senior Notes, plus accrued and unpaid interest. In the event of a change in control, note holders will have the right to require us to repurchase their notes at a price equal to 101% of the principal amount of such notes, plus accrued and unpaid interest.

Using the net proceeds from the January 2021 Senior Notes, we repaid the outstanding balance of \$5,888 million of our unsecured term A-3 facility and unsecured term A-5 facility under the credit agreement entered into on November 4, 2019 (the “November 2019 Credit Agreement”), repurchased \$3,830 million of certain of our outstanding notes maturing between 2021 and 2023 through a cash tender offer and redemption, and repaid \$282 million of our 2.200% notes upon maturity in January 2021. As a result of these repayments and repurchases, we incurred premiums of \$151 million and wrote off \$47 million of unamortized discount and issuance costs, both of which were included in interest expense.

January 2021 Credit Agreement

In January 2021, we entered into a credit agreement (the “January 2021 Credit Agreement”), which provides for a five-year \$7.5 billion unsecured revolving credit facility (the “Revolving Facility”), of which \$500 million is available for the issuance of multi-currency letters of credit. The issuance of letters of credit and certain other instruments would reduce the aggregate amount otherwise available under the Revolving Facility for revolving loans. Subject to the terms of the January 2021 Credit Agreement, we are permitted to borrow, repay and reborrow revolving loans at any time prior to the earlier of (a) January 19, 2026 and (b) the date of termination in whole of the revolving lenders’ commitments under the January 2021 Credit Agreement. In connection with the January 2021 Credit Agreement, we terminated the credit agreement entered into on May 7, 2019 (the “May 2019 Credit Agreement”), which provided for a five-year \$5 billion unsecured revolving credit facility, and the November 2019 Credit Agreement. We had no borrowings outstanding under the Revolving Facility at either October 30, 2022 or October 31, 2021.

June 2020 Senior Notes

In June 2020, we completed our private offers to exchange \$3,742 million of certain series of our outstanding notes maturing between 2021 and 2024 for \$1,695 million of senior notes due 2026 and \$2,222 million of senior notes due 2028 (collectively, the “June 2020 Senior Notes”). As a result of this exchange, we paid premiums of \$177 million, which were included in unamortized discount and issuance costs. We may redeem or purchase, in whole or in part, any of the June 2020 Senior Notes prior to their respective maturities, subject to a specified make-whole premium determined in accordance with the indenture governing the June 2020 Senior Notes, plus accrued and unpaid interest. In the event of a change in control, note holders will have the right to require us to repurchase their notes at a price equal to 101% of the principal amount of such notes plus accrued and unpaid interest.

May 2020 Senior Notes

In May 2020, we issued \$8 billion of senior unsecured notes (the “May 2020 Senior Notes”). We may redeem or purchase, in whole or in part, any of the May 2020 Senior Notes prior to their respective maturities, subject to a specified make-whole premium determined in accordance with the indenture governing the May 2020 Senior Notes, plus accrued and

unpaid interest. In the event of a change in control, note holders will have the right to require us to repurchase their notes at a price equal to 101% of the principal amount of such notes plus accrued and unpaid interest.

The net proceeds from this issuance, together with the remaining net proceeds from the issuance of the April 2020 Senior Notes, as defined below, were used to repay an aggregate of \$5,424 million of term loans outstanding under the November 2019 Credit Agreement, consisting of repayments of \$2,712 million of each of our unsecured term A-3 and A-5 facilities and \$3 billion of borrowings outstanding under the unsecured revolving credit facility provided by the May 2019 Credit Agreement. During fiscal year 2020, we wrote off \$60 million of unamortized discount and issuance costs as a result of repayments of term loans outstanding under the November 2019 Credit Agreement, which were included in interest expense.

April 2020 Senior Notes

In April 2020, we issued \$4.5 billion of senior unsecured notes (the “April 2020 Senior Notes”). We may redeem or purchase, in whole or in part, any of the April 2020 Senior Notes prior to their respective maturities, subject to a specified make-whole premium determined in accordance with the indenture governing the April 2020 Senior Notes, plus accrued and unpaid interest. In the event of a change in control, note holders will have the right to require us to repurchase their notes at a price equal to 101% of the principal amount of such notes plus accrued and unpaid interest.

Pursuant to a cash tender offer that we completed in April 2020, we repurchased certain of outstanding notes maturing between 2021 and 2022 with the net proceeds from the April 2020 Senior Notes. As a result of these repurchases, we incurred premiums of \$78 million and wrote off \$15 million of unamortized discount and issuance costs, both of which were included in interest expense.

April 2019 Senior Notes

In April 2019, we issued \$11 billion of senior unsecured notes (the “April 2019 Senior Notes”). We may redeem or purchase, in whole or in part, any of the April 2019 Senior Notes prior to their respective maturities, subject to a make-whole premium determined in accordance with the indenture governing the April 2019 Senior Notes, plus accrued and unpaid interest.

Registered Exchange Offer

In connection with the issuance of the June 2020 Senior Notes, the May 2020 Senior Notes, the April 2020 Senior Notes (collectively, the “2020 Senior Notes”) and the April 2019 Senior Notes, we entered into registration rights agreements, pursuant to which we were obligated to use commercially reasonable efforts to file with the Securities and Exchange Commission (the “SEC”), and cause to be declared effective, a registration statement with respect to an offer to exchange (the “Registered Exchange Offer”) each series of the 2020 Senior Notes and the April 2019 Senior Notes for notes that are registered with the SEC (the “Registered Notes”), with substantially identical terms. We completed the Registered Exchange Offer on August 10, 2020. Substantially all of our 2020 Senior Notes and April 2019 Senior Notes were tendered and exchanged for the corresponding Registered Notes in the Registered Exchange Offer.

Commercial Paper

In February 2019, we established a commercial paper program pursuant to which we may issue unsecured commercial paper notes (“Commercial Paper”) in principal amount of up to \$2 billion outstanding at any time with maturities of up to 397 days from the date of issue. Commercial Paper is sold under customary terms in the commercial paper market and may be issued at a discount from par or, alternatively, may be sold at par and bear interest at rates dictated by market conditions at the time of their issuance. The discount associated with the Commercial Paper is amortized to interest expense over its term. Outstanding Commercial Paper reduces the amount that would otherwise be available to borrow for general corporate purposes under our revolving credit facility. We had no Commercial Paper outstanding at either October 30, 2022 or October 31, 2021.

2017 Senior Notes

During the fiscal year ended October 29, 2017, Broadcom Cayman Finance Limited, which subsequently merged into BTI during fiscal year 2019 with BTI remaining as the surviving entity, and BRCM issued \$17,550 million of senior unsecured notes (the “2017 Senior Notes”). Our 2017 Senior Notes are fully and unconditionally guaranteed, jointly and severally, on an unsecured, unsubordinated basis by Broadcom and BTI. We may redeem or purchase, in whole or in part, any of the 2017 Senior Notes prior to their respective maturities, subject to a make-whole premium determined in accordance with the indenture governing the 2017 Senior Notes, plus accrued and unpaid interest. In the event of a change in control, note holders will have the right to require us to repurchase their notes at a price equal to 101% of the principal amount of such notes plus accrued and unpaid interest.

During the fiscal year ended November 4, 2018, substantially all of the 2017 Senior Notes were tendered and exchanged for notes registered with the SEC, with substantially identical terms.

Assumed CA Senior Notes

In connection with our acquisition of CA, Inc. (“CA”) during fiscal year 2019, we assumed \$2.25 billion of CA’s outstanding senior unsecured notes (the “Assumed CA Senior Notes”). CA remains the sole obligor under the Assumed CA Senior Notes. We may redeem all or a portion of the Assumed CA Senior Notes at any time, subject to a specified make-whole premium as set forth with the indenture governing the Assumed CA Senior Notes. In the event of a change in control, note holders will have the right to require us to repurchase their notes at a price equal to 101% of the principal amount of such notes plus accrued and unpaid interest.

Fair Value of Debt

As of October 30, 2022, the estimated aggregate fair value of our debt was \$33,014 million. The fair value of our senior notes was determined using quoted prices from less active markets. All of our debt obligations are categorized as Level 2 instruments.

Future Principal Payments of Debt

The future scheduled principal payments of debt as of October 30, 2022 were as follows:

Fiscal Year:	Future Scheduled Principal Payments
	(In millions)
2023	\$ 403
2024	1,563
2025	495
2026	1,652
2027	3,137
Thereafter	33,968
Total	<u>\$ 41,218</u>

As of October 30, 2022 and October 31, 2021, we were in compliance with all debt covenants.

11. Stockholders’ Equity

Mandatory Convertible Preferred Stock

On September 30, 2019, we completed an offering of approximately 4 million shares of 8.00% Mandatory Convertible Preferred Stock, Series A, \$0.001 par value per share (“Mandatory Convertible Preferred Stock”), which generated net proceeds of approximately \$3,679 million and would automatically convert into shares of our common stock on September 30, 2022. At any time prior to September 30, 2022, holders could elect to convert each share of Mandatory Convertible Preferred Stock at the then minimum conversion rate. The conversion rates were subject to anti-dilution adjustments.

The holders of Mandatory Convertible Preferred Stock were entitled to receive, when, as and if declared by our Board of Directors, or an authorized committee thereof, out of funds legally available for payment, cumulative dividends at the annual rate of 8.00% of the liquidation preference of \$1,000 per share (equivalent to \$80 annually per share), payable in cash or, subject to certain limitations, by delivery of shares of our common stock or any combination of cash and shares of our common stock, at our election; provided, however, that any undeclared and unpaid dividends will continue to accumulate.

Subject to limited exceptions, no dividends may be declared or paid on shares of our common stock, unless all accumulated dividends have been paid or set aside for payment on all outstanding shares of our Mandatory Convertible Preferred Stock for all past completed dividend periods. In the event of our voluntary or involuntary liquidation, dissolution or winding-up, no distribution of our assets may be made to holders of our common stock until we have paid to holders of our Mandatory Convertible Preferred Stock a liquidation preference equal to \$1,000 per share plus accumulated and unpaid dividends.

During fiscal year 2022, outstanding shares of our Mandatory Convertible Preferred Stock converted into an aggregate of approximately 12 million shares of our common stock at conversion rates ranging between 3.0894 and 3.1149 common shares per share of Mandatory Convertible Preferred Stock. We paid cash in lieu of fractional shares of common stock upon conversion.

As of October 31, 2021, we recognized \$27 million of accrued preferred stock dividends, which was presented as temporary equity on our consolidated balance sheet.

Cash Dividends Declared and Paid

	Fiscal Year		
	2022	2021	2020
	(In millions, except per share data)		
Dividends per share to common stockholders	\$ 16.40	\$ 14.40	\$ 13.00
Dividends to common stockholders	\$ 6,733	\$ 5,913	\$ 5,235
Dividends per share to preferred stockholders	\$ 80.00	\$ 80.00	\$ 80.00
Dividends to preferred stockholders	\$ 299	\$ 299	\$ 299

Stock Repurchase Program

In December 2021, our Board of Directors authorized a stock repurchase program to repurchase up to \$10 billion of our common stock from time to time on or prior to December 31, 2022. During fiscal year 2022, we repurchased and retired approximately 12 million shares of our common stock for \$7 billion under this stock repurchase program.

In May 2022, our Board of Directors authorized another stock repurchase program to repurchase up to an additional \$10 billion of our common stock from time to time through December 31, 2023.

Repurchases under our stock repurchase programs may be effected through a variety of methods, including open market or privately negotiated purchases. The timing and amount of shares repurchased will depend on the stock price, business and market conditions, corporate and regulatory requirements, alternative investment opportunities, acquisition opportunities, and other factors. We are not obligated to repurchase any specific amount of shares of common stock, and the stock repurchase programs may be suspended or terminated at any time.

Equity Incentive Award Plan

2012 Plan

In connection with the acquisition of BRCM, we assumed the BRCM 2012 Stock Incentive Plan (the “Original 2012 Plan”) and outstanding unvested RSUs originally granted by BRCM under the Original 2012 Plan that were held by continuing employees. During the second quarter of fiscal year 2021, our stockholders approved the amendment and restatement of the Original 2012 Plan, now called Broadcom Inc. 2012 Stock Incentive Plan (the “Amended 2012 Plan”). Under the Amended 2012 Plan, we may grant to employees stock options and stock appreciation rights with an exercise price that is no less than the fair market value on the date of grant, restricted stock awards and RSUs. No participant may be granted such awards for more than an aggregate of 4 million shares in any fiscal year. Equity awards granted under the Amended 2012 Plan generally vest over four years. The Amended 2012 Plan reduced the number of shares available for new equity award grants to 20 million shares and removed the annual share replenishment provision provided under the Original 2012 Plan. Awards cancelled or forfeited and shares withheld to satisfy tax withholding obligations become available for future issuance. As of October 30, 2022, 21 million shares remained available for issuance under the Amended 2012 Plan.

We may grant market-based RSUs with both a service condition and a market condition as part of our equity compensation programs. The market-based RSUs generally vest over four years, subject to satisfaction of market conditions. During fiscal years 2022, 2021 and 2020, we granted market-based RSUs under which grantees may receive the number of shares ranging from 0% to 300% of the original grant at vesting based upon the total stockholder return (“TSR”) on our common stock on an absolute basis and as compared to the TSR of an index group of companies.

Employee Stock Purchase Plan

The ESPP provides eligible employees with the opportunity to acquire an ownership interest in us through periodic payroll deductions, based on a 6-month look-back period, at a price equal to the lesser of 85% of the fair market value of our common stock at either the beginning or the end of the relevant offering period. The ESPP is structured as a qualified employee stock purchase plan under Section 423 of the Internal Revenue Code of 1986. However, the ESPP is not intended to be a qualified pension, profit sharing or stock bonus plan under Section 401(a) of the Internal Revenue Code of 1986 and is not subject to the provisions of the Employee Retirement Income Security Act of 1974.

Stock-Based Compensation Expense

	Fiscal Year		
	2022	2021	2020
	(In millions)		
Cost of products sold	\$ 65	\$ 78	\$ 109
Cost of subscriptions and services	82	65	50
Research and development	1,048	1,199	1,419
Selling, general and administrative	338	362	398
Total stock-based compensation expense	\$ 1,533	\$ 1,704	\$ 1,976
Estimated income tax benefits for stock-based compensation	\$ 255	\$ 283	\$ 345
Excess income tax benefits for stock-based awards exercised or released	\$ 375	\$ 310	\$ 147

We have assumed an annualized forfeiture rate for RSUs of 5%. We will recognize additional expense if actual forfeitures are lower than we estimated, and will recognize a benefit if actual forfeitures are higher than we estimated.

During the first quarter of fiscal year 2019, the Compensation Committee of our Board of Directors approved a broad-based program of multi-year equity grants of time- and market-based RSUs (the “Multi-Year Equity Awards”) in lieu of our annual employee equity awards historically granted on March 15 of each year. Each Multi-Year Equity Award vests on the same basis as four annual grants made March 15 of each year, beginning in fiscal year 2019, with successive four-year vesting periods. Stock-based compensation expense related to the Multi-Year Equity Awards was \$794 million, \$816 million and \$902 million for fiscal years 2022, 2021 and 2020, respectively.

As of October 30, 2022, the total unrecognized compensation cost related to unvested stock-based awards was \$2,704 million, which is expected to be recognized over the remaining weighted-average service period of 2.7 years.

The following table summarizes the weighted-average assumptions utilized to calculate the fair value of market-based awards granted in the periods presented:

	Fiscal Year		
	2022	2021	2020
Risk-free interest rate	1.4 %	0.3 %	1.2 %
Dividend yield	2.7 %	3.0 %	4.7 %
Volatility	37.1 %	39.0 %	31.2 %
Expected term (in years)	3.4	3.4	4.0

The risk-free interest rate was derived from the average U.S. Treasury Strips rate, which approximated the rate in effect appropriate for the term at the time of grant.

The dividend yield was based on the historical and expected dividend payouts as of the respective award grant dates.

The volatility was based on our own historical stock price volatility over the period commensurate with the expected life of the awards and the implied volatility of a 180-day call option on our own common stock measured at a specific date.

The expected term was commensurate with the awards’ contractual terms.

Restricted Stock Unit Awards

A summary of time- and market-based RSU activity is as follows:

	Number of RSUs Outstanding	Weighted-Average Grant Date Fair Value Per Share
	(In millions, except per share data)	
Balance as of November 3, 2019	40	\$ 188.52
Granted	3	\$ 252.36
Vested	(8)	\$ 210.84
Forfeited	(3)	\$ 198.17
Balance as of November 1, 2020	32	\$ 188.35
Granted	2	\$ 408.69
Vested	(8)	\$ 214.15
Forfeited	(3)	\$ 189.84
Balance as of October 31, 2021	23	\$ 200.38
Granted	3	\$ 527.69
Vested	(7)	\$ 225.52
Forfeited	(1)	\$ 242.82
Balance as of October 30, 2022	18	\$ 238.49

The aggregate fair value of time- and market-based RSUs that vested in fiscal years 2022, 2021 and 2020 was \$4,207 million, \$3,715 million, and \$2,254 million, respectively, which represented the market value of our common stock on the date that the RSUs vested. The number of RSUs vested included shares of common stock that we withheld for settlement of employees' tax obligations due upon the vesting of RSUs.

Stock Option Awards

As of October 30, 2022, our stock options outstanding were not material. The aggregate intrinsic value of stock options exercised in fiscal years 2022, 2021 and 2020 was \$3 million, \$339 million, and \$917 million, respectively.

12. Income Taxes

The components of income from continuing operations before income taxes by U.S. and foreign jurisdictions were as follows:

	Fiscal Year		
	2022	2021	2020
	(In millions)		
Domestic loss	\$ (2,020)	\$ (3,103)	\$ (4,221)
Foreign income	14,454	9,868	6,664
Income from continuing operations before income taxes	\$ 12,434	\$ 6,765	\$ 2,443

The components of the provision for and benefit from income taxes were as follows:

	Fiscal Year		
	2022	2021	2020
	(In millions)		
Current tax expense:			
Federal	\$ 174	\$ 446	\$ 7
State	48	46	51
Foreign	762	534	506
Total	984	1,026	564
Deferred tax expense (benefit from):			
Federal	68	(876)	(627)
State	(15)	(114)	(161)
Foreign	(98)	(7)	(294)
Total	(45)	(997)	(1,082)
Total provision for (benefit from) income taxes	\$ 939	\$ 29	\$ (518)

The following is a reconciliation of our effective tax rate to the statutory federal tax rate:

	Fiscal Year		
	2022	2021	2020
Statutory tax rate	21.0 %	21.0 %	21.0 %
State, net of federal benefit	0.2	(0.8)	(3.6)
Foreign income taxed at different rates	(19.1)	(22.8)	(48.6)
Deemed inclusion of foreign earnings	9.4	12.7	23.3
Foreign-derived intangible income deduction	—	(3.1)	(1.5)
Deferred taxes on unremitted foreign earnings	0.1	(0.7)	(1.1)
Excess tax benefits from stock-based compensation	(3.0)	(4.6)	(6.0)
Research and development credit	(1.4)	(2.3)	(4.3)
Other, net	0.3	1.0	(0.4)
Effective tax rate on income before income taxes	7.5 %	0.4 %	(21.2)%

The increase in provision for income taxes in fiscal year 2022 compared to fiscal year 2021 was primarily due to higher income from continuing operations.

The provision for income taxes in fiscal year 2021 compared to the benefit from income taxes in fiscal year 2020 was primarily due to higher income from continuing operations, offset in part by higher excess tax benefits from stock-based awards. The benefit from income taxes in fiscal year 2020 was primarily due to jurisdictional mix of income and expenses, discrete benefits from the remeasurement of certain deferred tax assets and liabilities in a foreign jurisdiction, and excess tax benefits from stock-based awards.

Our tax incentives from the Singapore Economic Development Board provide that any qualifying income earned in Singapore is subject to tax incentives or reduced rates of Singapore income tax, subject to our compliance with the conditions specified in these incentives and legislative developments. These Singapore tax incentives are expected to expire in November 2025.

We have also obtained a tax holiday on our qualifying income in Malaysia, which is scheduled to expire in fiscal year 2028. The tax holiday that we negotiated in Malaysia is also subject to our compliance with various operating and other conditions. If we cannot, or elect not to, comply with any such conditions specified, we will lose the related tax benefits and we could be required to refund previously realized material tax benefits.

Before taking into consideration the effects of the U.S. Tax Cuts and Jobs Act and other indirect tax impacts, the effect of these tax incentives and tax holiday was to decrease the provision for income taxes by approximately \$1,821 million and \$1,156 million for fiscal years 2022 and 2021, respectively, and increase the benefit from income taxes by approximately \$833 million for fiscal year 2020.

Significant components of our deferred tax assets and liabilities consisted of the following:

	October 30, 2022	October 31, 2021
	(In millions)	
Deferred income tax assets:		
Net operating loss, credit and other carryforwards	\$ 1,808	\$ 1,774
Deferred revenue	645	1,332
Employee stock awards	183	192
Other deferred income tax assets	499	446
Gross deferred income tax assets	3,135	3,744
Less: valuation allowance	(1,777)	(1,782)
Deferred income tax assets	1,358	1,962
Deferred income tax liabilities:		
Depreciation and amortization	341	847
Unamortized discount and issuance costs	322	374
Foreign earnings not indefinitely reinvested	86	73
Other deferred income tax liabilities	36	—
Deferred income tax liabilities	785	1,294
Net deferred income tax assets	\$ 573	\$ 668

We continue to indefinitely reinvest \$2,112 million of certain accumulated foreign earnings. The unrecognized deferred income tax liability related to these earnings is estimated to be \$222 million. All other current and future earnings of all our foreign subsidiaries are not considered permanently reinvested.

As of October 30, 2022, we had tax effected U.S. state net operating loss carryforwards of \$200 million and foreign net operating loss carryforwards of \$190 million, all of which expire in various years beginning fiscal year 2023. We had \$1,334 million of state research and development tax credits which begin to expire in fiscal year 2023. We have provided a valuation allowance on substantially all state tax credits and state and foreign net operating loss carryforwards as we do not expect them to be realized.

Uncertain Tax Positions

The following table reconciles the beginning and ending balance of gross unrecognized tax benefits:

	Fiscal Year		
	2022	2021	2020
	(In millions)		
Beginning balance	\$ 5,030	\$ 4,748	\$ 4,422
Lapses of statutes of limitations	(50)	(58)	(95)
Increases in balances related to tax positions taken during prior periods (including those related to acquisitions made during the year)	—	41	98
Decreases in balances related to tax positions taken during prior periods	(113)	—	(14)
Increases in balances related to tax positions taken during current period	288	337	379
Decreases in balances related to settlements with taxing authorities	(38)	(38)	(42)
Ending balance	<u>\$ 5,117</u>	<u>\$ 5,030</u>	<u>\$ 4,748</u>

We recognize interest and penalties related to unrecognized tax benefits within the provision for (benefit from) income taxes. Accrued interest and penalties were included within other long-term liabilities. During fiscal years 2022 and 2021, we recognized interest and penalties of \$25 million and \$46 million, respectively, within the provision for income taxes. During fiscal year 2020, we recognized interest and penalties of \$37 million within the benefit from income taxes. As of October 30, 2022 and October 31, 2021, the combined amount of cumulative accrued interest and penalties was approximately \$411 million and \$386 million, respectively.

As of October 30, 2022 and October 31, 2021, approximately \$5,528 million and \$5,416 million, respectively, of the unrecognized tax benefits and accrued interest and penalties would, if recognized, benefit our effective income tax rate. We are subject to U.S. income tax examination for fiscal years 2015 and later. Certain of our acquired companies are subject to tax examinations in major jurisdictions outside of the U.S. for fiscal years 2008 and later. It is possible that our existing unrecognized tax benefits may change up to \$163 million as a result of lapses of the statute of limitations for certain audit periods and/or audit examinations expected to be completed within the next 12 months.

13. Segment Information

Reportable Segments

We have two reportable segments: semiconductor solutions and infrastructure software. Each segment has separate financial information that is utilized on a regular basis by the CODM in determining how to allocate resources and evaluate performance. The reportable segments are determined based on several factors including, but not limited to, customer base, homogeneity of products, technology, delivery channels and similar economic characteristics.

Semiconductor solutions. We provide semiconductor solutions for managing the movement of data in data center, service provider, enterprise and embedded networking applications. We provide a broad variety of radio frequency semiconductor devices, wireless connectivity solutions, custom touch controllers, and inductive charging solutions for mobile applications. We also provide semiconductor solutions for enabling the set-top box and broadband access markets and for enabling secure movement of digital data to and from host machines, such as servers, personal computers and storage systems, to the underlying storage devices, such as hard disk drives and solid state drives. We also provide a broad variety of products for the general industrial and automotive markets. Our semiconductor solutions segment also includes our IP licensing.

Infrastructure software. We provide a portfolio of software solutions that enables customers to plan, develop, automate, manage and secure applications across mainframe, distributed, mobile and cloud platforms. Our portfolio of infrastructure and security software is designed to modernize, optimize, and secure the most complex hybrid environments, enabling scalability, agility, automation, insights, resiliency and security. We also offer mission critical FC SAN products and related software.

Our CODM assesses the performance of each segment and allocates resources to each segment based on net revenue and operating results and does not evaluate each segment using discrete asset information. Operating results by segment include items that are directly attributable to each segment and also include shared expenses such as marketing, general and administrative activities, facilities and information technology ("IT") expenses. Shared expenses are primarily allocated based on revenue, headcount or evenly between the segments.

Unallocated Expenses

Unallocated expenses include amortization of acquisition-related intangible assets, stock-based compensation expense, restructuring, impairment and disposal charges, acquisition-related costs, and other costs, which are not used in evaluating the results of, or in allocating resources to, our segments. Acquisition-related costs include transaction costs and any costs directly related to the acquisition and integration of acquired businesses.

Depreciation expense directly attributable to each reportable segment is included in operating results for each segment. However, the CODM does not evaluate depreciation expense by operating segment and, therefore, it is not separately presented. There was no inter-segment revenue for any of the periods presented. The accounting policies of the segments are the same as those described in the summary of significant accounting policies.

	Fiscal Year		
	2022	2021	2020
	(In millions)		
Net revenue:			
Semiconductor solutions	\$ 25,818	\$ 20,383	\$ 17,267
Infrastructure software	7,385	7,067	6,621
Total net revenue	<u>\$ 33,203</u>	<u>\$ 27,450</u>	<u>\$ 23,888</u>
Operating income:			
Semiconductor solutions	\$ 15,075	\$ 10,976	\$ 8,576
Infrastructure software	5,219	4,936	4,363
Unallocated expenses	(6,069)	(7,393)	(8,925)
Total operating income	<u>\$ 14,225</u>	<u>\$ 8,519</u>	<u>\$ 4,014</u>

Geographic Information

Net revenue by country is based primarily on the geographic shipment or delivery location as specified by the distributors, OEMs, contract manufacturers, channel partners, or software customers who purchased our products or services. For the majority of our products, title and control transfer to our customers in Penang, Malaysia. The products are then transported to the customer specific locations. Net revenue from the United States for fiscal years 2022, 2021 and 2020 was \$5,915 million, \$5,285 million and \$4,778 million, respectively. Net revenue from China (including Hong Kong) for fiscal years 2022, 2021 and 2020 was \$11,637 million, \$9,752 million and \$7,808 million, respectively. Net revenue from Singapore for fiscal years 2022 and 2021 was \$4,003 million and \$2,754 million, respectively (the amount was less than 10% for fiscal year 2020). Net revenue from other foreign countries for fiscal years 2022, 2021 and 2020 was \$11,648 million, \$9,659 million and \$11,302 million, respectively. These geographic delivery locations are not necessarily indicative of the geographic location of our end customers or the country in which our end customers sell devices containing our products. For example, we believe a substantial portion of our products shipped or delivered to China (including Hong Kong) is included in devices sold by our end customers in the United States and Europe.

Long-lived assets include property, plant and equipment and are based on the physical location of the assets.

	October 30, 2022	October 31, 2021
	(In millions)	
Long-lived assets:		
United States	\$ 1,441	\$ 1,540
Taiwan	318	313
Other	464	495
Total long-lived assets	<u>\$ 2,223</u>	<u>\$ 2,348</u>

Significant Customer Information

We sell our products through our direct sales force and a select network of distributors and channel partners globally. Two customers accounted for 15% and 11% of our net accounts receivable balance as of October 30, 2022. No customer accounted for more than 10% of our net accounts receivable balance as of October 31, 2021. During fiscal years 2022, 2021 and 2020, one customer accounted for 20%, 18% and 13% of our net revenue, respectively. Revenue from this customer was included in our semiconductor solutions segment.

14. Commitments and Contingencies

Commitments

The following table summarizes contractual obligations and commitments as of October 30, 2022:

Fiscal Year:	Purchase Commitments	Other Contractual Commitments
	(In millions)	
2023	\$ 199	\$ 304
2024	211	261
2025	90	147
2026	11	162
2027	7	112
Thereafter	7	447
Total	\$ 525	\$ 1,433

Purchase Commitments. Represents unconditional purchase obligations that include agreements to purchase goods or services, primarily inventory, that are enforceable and legally binding on us and that specify all significant terms, including fixed or minimum quantities to be purchased, fixed, minimum or variable price provisions, and the approximate timing of the transaction. Purchase obligations exclude agreements that are cancelable without penalty.

Other Contractual Commitments. Represents amounts payable pursuant to agreements related to IT, human resources, and other service agreements.

Due to the inherent uncertainty with respect to the timing of future cash outflows associated with our unrecognized tax benefits at October 30, 2022, we are unable to reliably estimate the timing of cash settlement with the respective taxing authorities. Therefore, \$3,229 million of unrecognized tax benefits and accrued interest and penalties classified within other long-term liabilities on our consolidated balance sheet as of October 30, 2022 have been excluded from the table above.

Contingencies

From time to time, we are involved in litigation that we believe is of the type common to companies engaged in our lines of business, including commercial disputes, employment issues, tax disputes and disputes involving claims by third parties that our activities infringe their patent, copyright, trademark or other IP rights, as well as regulatory investigations or inquiries. Legal proceedings and regulatory investigations or inquiries are often complex, may require the expenditure of significant funds and other resources, and the outcome of such proceedings is inherently uncertain, with material adverse outcomes possible. IP property claims generally involve the demand by a third-party that we cease the manufacture, use or sale of the allegedly infringing products, processes or technologies and/or pay substantial damages or royalties for past, present and future use of the allegedly infringing IP. Claims that our products or processes infringe or misappropriate any third-party IP rights (including claims arising through our contractual indemnification of our customers) often involve highly complex, technical issues, the outcome of which is inherently uncertain. Moreover, from time to time, we pursue litigation to assert our IP rights. Regardless of the merit or resolution of any such litigation, complex IP litigation is generally costly and diverts the efforts and attention of our management and technical personnel.

Lawsuits Relating to California Institute of Technology

California Institute of Technology ("Caltech") filed a complaint against Broadcom and Apple Inc. on May 26, 2016 in the United States District Court for the Central District of California (the "U.S. Central District Court"), and an amended complaint adding Cypress Semiconductor Corporation as a defendant on August 15, 2016. The amended complaint alleged that chips that support certain error correction codes as specified in IEEE Standards 802.11n and 802.11ac willfully infringed four patents related to error correction coding: U.S. Patent Nos. 7,116,710; 7,421,032; 7,916,781; and 8,284,833 ("833 patent"). Prior to trial, Caltech dismissed its claims against Cypress and withdrew its infringement allegations as to 833 patent. The complaint sought a preliminary and permanent injunction, damages, pre- and post-judgment interest, as well as attorneys' fees, costs, and expenses. The trial was held in January 2020, and on January 29, 2020, the jury issued its verdict finding infringement and awarding Caltech past damages of \$270.2 million from Broadcom and \$837.8 million from Apple, for which Apple is seeking indemnification from Broadcom. On August 3, 2020, the U.S. Central District Court issued its judgment, awarding Caltech past damages in the amounts awarded by the jury, as well as pre- and post-judgment interest. Additionally, the U.S. Central District Court awarded Caltech an unspecified amount of ongoing royalties to be determined after the anticipated appeals process is resolved. Neither the jury nor the U.S. Central District Court found willful infringement, which if it had, could have resulted in enhanced damages up to three times the amount awarded. Broadcom and Apple appealed to the United States Court of

Appeals for the Federal Circuit (the “Federal Circuit Court”). In February 2022, the Federal Circuit Court affirmed infringement of two patents, both of which expired in August 2020, but it did not address all issues and ordered a new trial on damages and on the infringement of the 7,916,781 patent, which also expired in August 2020. In May 2022, the Federal Circuit Court denied the petition for rehearing filed by Broadcom and Apple, and remanded the case to the U.S. Central District Court. Subsequently, Caltech withdrew its infringement allegations as to the 7,916,781 patent.

We believe that the evidence and the law do not support the U.S. Central District Court’s findings of infringement. We cannot reasonably estimate the ultimate outcome as the Federal Circuit Court vacated the above damages, and a number of factors (including a retrial at the lower court and further appeals) could significantly change the assessment of damages. As a result, we have not recorded a reserve with respect to this litigation, in accordance with the applicable accounting standards.

Other Matters

In addition to the matters discussed above, we are currently engaged in a number of legal actions in the ordinary course of our business.

Contingency Assessment

We do not believe, based on currently available facts and circumstances, that the final outcome of any pending legal proceedings or ongoing regulatory investigations, taken individually or as a whole, will have a material adverse effect on our consolidated financial statements. However, lawsuits may involve complex questions of fact and law and may require the expenditure of significant funds and other resources to defend. The results of litigation or regulatory investigations are inherently uncertain, and material adverse outcomes are possible. From time to time, we may enter into confidential discussions regarding the potential settlement of such lawsuits. Any settlement of pending litigation could require us to incur substantial costs and other ongoing expenses, such as future royalty payments in the case of an IP dispute.

During the periods presented, no material amounts have been accrued or disclosed in the accompanying consolidated financial statements with respect to loss contingencies associated with any other legal proceedings or regulatory investigations, as potential losses for such matters are not considered probable and ranges of losses are not reasonably estimable. These matters are subject to many uncertainties and the ultimate outcomes are not predictable. There can be no assurances that the actual amounts required to satisfy any liabilities arising from the matters described above will not have a material adverse effect on our consolidated financial statements.

Other Indemnifications

As is customary in our industry and as provided for in local law in the U.S. and other jurisdictions, many of our standard contracts provide remedies to our customers and others with whom we enter into contracts, such as defense, settlement, or payment of judgment for IP claims related to the use of our products. From time to time, we indemnify customers, as well as our suppliers, contractors, lessors, lessees, companies that purchase our businesses or assets and others with whom we enter into contracts, against combinations of loss, expense, or liability arising from various triggering events related to the sale and the use of our products, the use of their goods and services, the use of facilities and state of our owned facilities, the state of the assets and businesses that we sell and other matters covered by such contracts, usually up to a specified maximum amount. In addition, from time to time we also provide protection to these parties against claims related to undiscovered liabilities, additional product liabilities or environmental obligations. In our experience, claims made under such indemnifications are rare and the associated estimated fair value of the liability is not material.

15. Restructuring, Impairment and Disposal Charges

Restructuring Charges

From time to time, we initiate cost reduction activities to integrate acquired businesses, to align our workforce with strategic business activities, or to improve efficiencies in our operations. We recognized charges of \$55 million, \$149 million and \$233 million during fiscal years 2022, 2021 and 2020, respectively. These charges were primarily recognized in operating expenses.

The following table summarizes the significant activities within, and components of, the restructuring liabilities:

	Employee Termination Costs	Other Exit Costs ^(a)	Total
	(In millions)		
Balance as of November 3, 2019	\$ 69	\$ 39	\$ 108
Restructuring charges ^(b)	186	47	233
Utilization	(221)	(50)	(271)
Effect of adoption of Topic 842 ^(c)	—	(36)	(36)
Balance as of November 1, 2020	34	—	34
Restructuring charges	100	13	113
Utilization	(130)	(13)	(143)
Balance as of October 31, 2021	4	—	4
Restructuring charges	24	6	30
Utilization	(24)	(4)	(28)
Balance as of October 30, 2022	\$ 4	\$ 2	\$ 6

- (a) Included \$30 million of restructuring expense related to the write-down of certain lease-related ROU assets and other lease-related charges during fiscal year 2020.
- (b) Included \$19 million of restructuring expense related to discontinued operations recognized during fiscal year 2020, which was included in loss from discontinued operations.
- (c) Upon adoption of Topic 842, certain restructuring lease liabilities were required to be recognized as a reduction to the corresponding ROU assets.

Restructuring, impairment and disposal charges in our consolidated statement of operations for the fiscal years 2022 and 2021 included \$25 million and \$36 million, respectively, for the write-down of certain lease-related ROU assets and other lease-related charges. As of each October 30, 2022 and October 31, 2021, short-term and long-term lease liabilities included \$52 million of liabilities related to restructuring activities.

Impairment and Disposal Charges

During fiscal years 2022, 2021 and 2020, impairment and disposal charges of \$7 million, \$16 million and \$19 million, respectively, primarily related to leasehold improvements.

16. Subsequent Events

Cash Dividends Declared

On December 6, 2022, our Board of Directors declared a quarterly cash dividend of \$4.60 per share on our common stock, payable on December 30, 2022 to stockholders of record on December 20, 2022.

Schedule II — Valuation and Qualifying Accounts

	Balance at Beginning of Period	Additions to Allowances	Charges Utilized/ Write-offs	Balance at End of Period
(In millions)				
Accounts receivable allowances:				
Distributor credit allowances ⁽¹⁾				
Fiscal year ended October 30, 2022	\$ 128	\$ 484	\$ (487)	\$ 125
Fiscal year ended October 31, 2021	\$ 149	\$ 756	\$ (777)	\$ 128
Fiscal year ended November 1, 2020	\$ 153	\$ 696	\$ (700)	\$ 149
Other accounts receivable allowances ⁽²⁾				
Fiscal year ended October 30, 2022	\$ 2	\$ 10	\$ (11)	\$ 1
Fiscal year ended October 31, 2021	\$ 28	\$ 14	\$ (40)	\$ 2
Fiscal year ended November 1, 2020	\$ 38	\$ 84	\$ (94)	\$ 28
Income tax valuation allowances:				
Fiscal year ended October 30, 2022	\$ 1,782	\$ 118	\$ (123)	\$ 1,777
Fiscal year ended October 31, 2021	\$ 1,707	\$ 121	\$ (46)	\$ 1,782
Fiscal year ended November 1, 2020	\$ 1,563	\$ 149	\$ (5)	\$ 1,707

(1) Distributor credit allowances relate to price adjustments and other allowances.

(2) Other accounts receivable allowances primarily include sales returns and allowance for doubtful accounts.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

Not applicable.

ITEM 9A. CONTROLS AND PROCEDURES

Evaluation of Disclosure Controls and Procedures

Our management, with the participation of our Chief Executive Officer (“CEO”) and Chief Financial Officer (“CFO”), evaluated the effectiveness of our disclosure controls and procedures as of October 30, 2022. The term “disclosure controls and procedures,” as defined in Rules 13a-15(e) and 15d-15(e) under the Exchange Act, means controls and other procedures of a company that are designed to ensure that information required to be disclosed by a company in the reports that it files or submits under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the SEC’s rules and forms. Disclosure controls and procedures include, without limitation, controls and procedures designed to ensure that information required to be disclosed by a company in the reports that it files or submits under the Exchange Act is accumulated and communicated to our management, including its principal executive and principal financial officers, as appropriate to allow timely decisions regarding required disclosure. Management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving their objectives and management necessarily applies its judgment in evaluating the cost-benefit relationship of possible controls and procedures. Based on the evaluation of our disclosure controls and procedures as of October 30, 2022, our CEO and CFO concluded that, as of such date, our disclosure controls and procedures were effective at the reasonable assurance level.

Management's Report on Internal Control Over Financial Reporting

Our management is responsible for establishing and maintaining adequate internal control over financial reporting. Internal control over financial reporting is defined in Rules 13a-15(f) and 15d-15(f) promulgated under the Exchange Act as a process designed by, or under the supervision of, our principal executive and principal financial officers and effected by the Board, management and other personnel, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with GAAP and includes those policies and procedures that:

- pertain to the maintenance of records that in reasonable detail accurately and fairly reflect the transactions and dispositions of the assets;
- provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with GAAP, and that receipts and expenditures of us are being made only in accordance with authorizations of management and directors; and
- provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of our assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Our management assessed the effectiveness of our internal control over financial reporting as of October 30, 2022. In making this assessment, our management used the criteria set forth by the Committee of Sponsoring Organizations of the Treadway Commission in *Internal Control-Integrated Framework* (2013). Based on this assessment, our management concluded that, as of October 30, 2022, our internal control over financial reporting is effective based on those criteria.

The effectiveness of our internal control over financial reporting, as of October 30, 2022 has been audited by PricewaterhouseCoopers LLP, an independent registered public accounting firm, as stated in their report which is included in Part II, Item 8. of this Annual Report on Form 10-K.

Changes in Internal Control over Financial Reporting

No change in our internal control over financial reporting (as defined in Rules 13a-15(f) and 15d-15(f) under the Exchange Act) occurred during the fourth quarter ended October 30, 2022 that has materially affected, or is reasonably likely to materially affect, our internal control over financial reporting. Although we have modified our workplace practices globally due to the COVID-19 pandemic, resulting in some of our employees working remotely, this has not meaningfully affected our internal controls over financial reporting. We are continually monitoring and assessing the COVID-19 situation on our internal controls to minimize the impact on their design and operating effectiveness.

ITEM 9B. OTHER INFORMATION

None.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not applicable.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE

The information required by Item 10 is incorporated herein by reference from sections entitled “Board of Directors,” “Corporate Governance” and “Proposal 1 — Election of Directors” in our definitive Proxy Statement for our 2023 Annual Meeting of Stockholders. Our executive officers are listed at the end of Item 1 of this Annual Report on Form 10-K.

ITEM 11. EXECUTIVE COMPENSATION

The information required by Item 11 is incorporated herein by reference from sections entitled “Board of Directors — Director Compensation,” “Board of Directors — Board Committees — Compensation Committee — Compensation Committee Interlocks and Insider Participation,” “Compensation Discussion and Analysis,” “Compensation Committee Report” and “Executive Compensation” in our definitive Proxy Statement for our 2023 Annual Meeting of Stockholders.

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

The information required by Item 12 is incorporated herein by reference from sections entitled “Stockholder Information — Security Ownership of Certain Beneficial Owners, Directors and Executive Officers” and “Equity Compensation Plan Information” in our definitive Proxy Statement for our 2023 Annual Meeting of Stockholders.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE

The information required by Item 13 is incorporated herein by reference from sections entitled “Board of Directors” and “Certain Relationships and Related Party Transactions” in our definitive Proxy Statement for our 2023 Annual Meeting of Stockholders.

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES

The information required by Item 14 is incorporated herein by reference from the section entitled “Proposal 2 — Ratification of Appointment of Independent Registered Public Accounting Firm” in our definitive Proxy Statement for our 2023 Annual Meeting of Stockholders.

PART IV

ITEM 15. EXHIBITS AND FINANCIAL STATEMENT SCHEDULES

(a) The following are filed as part of this Annual Report on Form 10-K:

1. Financial Statements

The following consolidated financial statements are included in Item 8 of this Annual Report on Form 10-K:

	Page
Reports of Independent Registered Public Accounting Firm	48
Consolidated Balance Sheets	49
Consolidated Statements of Operations	50
Consolidated Statements of Comprehensive Income	51
Consolidated Statements of Cash Flows	52
Consolidated Statements of Stockholders' Equity	53
Notes to Consolidated Financial Statements	54

2. Financial Statement Schedules

The financial statement schedule of the Registrant and its subsidiaries for fiscal years 2022, 2021 and 2020 required by Item 15(a) (Schedule II, Valuation and Qualifying Accounts) is included in Item 8 of this Annual Report on Form 10-K:

	Page
Schedule II - Valuation and Qualifying Accounts	89

Schedules not filed have been omitted because they are not applicable, are not required or the information required to be set forth therein is included in the financial statements or notes thereto.

3. Exhibits

The documents set forth below are filed herewith or incorporated by reference to the location indicated.

Exhibit Number	Description	Incorporated by Reference		Filed Herewith
		Form (File No.)	Filing Date	
2.1#	Agreement and Plan of Merger, dated as of July 11, 2018, by and among Broadcom Inc., Collie Acquisition Corp. and CA, Inc.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	July 12, 2018	
2.2#	Asset Purchase Agreement, dated August 8, 2019, by and between Broadcom Inc. and Symantec Corporation.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	August 9, 2019	
2.3#	APA Letter Agreement, dated as of October 1, 2020, by and between Broadcom Inc. and NortonLifeLock Inc.	Broadcom Inc. Annual Report on Form 10-K (Commission File No. 001-38449)	December 18, 2020	
2.4	Agreement and Plan of Merger, dated as of May 26, 2022, by and among Broadcom Inc., VMware, Inc., Verona Holdco, Inc., Verona Merger Sub, Inc., Barcelona Merger Sub 2, Inc. and Barcelona Merger Sub 3, LLC.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 26, 2022	
3.1	Amended and Restated Certificate of Incorporation.	Broadcom Inc. Current Report on Form 8-K12B (Commission File No.001-38449)	April 4, 2018	

Table of Contents

Exhibit Number	Description	Incorporated by Reference		Filed Herewith
		Form (File No.)	Filing Date	
3.2	Certificate of Designation of the 8.00% Mandatory Convertible Preferred Stock, Series A.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	September 30, 2019	
3.3	Amended and Restated Bylaws.	Broadcom Inc. Current Report on Form 8-K12B (Commission File No. 001-38449)	April 4, 2018	
4.1	Form of Common Stock Certificate.	Broadcom Inc. Quarterly Report on Form 10-Q (Commission File No. 001-38449)	June 14, 2018	
4.2	Form of Certificate of the 8.00% Mandatory Convertible Preferred Stock, Series A (included in the Exhibit 3.2).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	September 30, 2019	
4.3	Description of Common Stock.	Broadcom Inc. Annual Report on Form 10-K (Commission File No. 001-38449)	December 20, 2019	
4.4	Description of 8.00% Mandatory Convertible Preferred Stock, Series A.	Broadcom Inc. Annual Report on Form 10-K (Commission File No. 001-38449)	December 20, 2019	
4.5	Indenture, dated as of January 19, 2017, by and among the Broadcom Corporation and Broadcom Cayman Finance Limited (the “Co-Issuers”), the guarantors and Wilmington Trust, National Association, as trustee.	Broadcom Limited Current Report on Form 8-K (Commission File No. 001-37690)	January 20, 2017	
4.6	Supplement Indenture to the January 2017 Indenture, dated as of April 9, 2018.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 9, 2018	
4.7	Second Supplement Indenture to the January 2017 Indenture, dated as of January 25, 2019.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	January 25, 2019	
4.8	Form of 3.000% Senior Notes due 2022 (included in Exhibit 4.5).	Broadcom Limited Current Report on Form 8-K (Commission File No. 001-37690)	January 20, 2017	
4.9	Form of 3.625% Senior Notes due 2024 (included in Exhibit 4.5).	Broadcom Limited Current Report on Form 8-K (Commission File No. 001-37690)	January 20, 2017	
4.10	Form of 3.875% Senior Notes due 2027 (included in Exhibit 4.5).	Broadcom Limited Current Report on Form 8-K (Commission File No. 001-37690)	January 20, 2017	
4.11	Indenture, dated as of October 17, 2017, by and among the Co-Issuers, the guarantors and Wilmington Trust, National Association, as trustee.	Broadcom Limited Current Report on Form 8-K (Commission File No. 001-37690)	October 17, 2017	
4.12	Supplemental Indenture to October 2017 Indenture, dated as of April 9, 2018.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 9, 2018	

Table of Contents

Exhibit Number	Description	Incorporated by Reference		Filed Herewith
		Form (File No.)	Filing Date	
4.13	Second Supplemental Indenture to October 2017 Indenture, dated as of January 25, 2019.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	January 25, 2019	
4.14	Form of 2.650% Senior Notes due 2023 (included in Exhibit 4.11).	Broadcom Limited Current Report on Form 8-K (Commission File No. 001-37690)	October 17, 2017	
4.15	Form of 3.125% Senior Notes due 2025 (included in Exhibit 4.11).	Broadcom Limited Current Report on Form 8-K (Commission File No. 001-37690)	October 17, 2017	
4.16	Form of 3.500% Senior Notes due 2028 (included in Exhibit 4.11).	Broadcom Limited Current Report on Form 8-K (Commission File No. 001-37690)	October 17, 2017	
4.17	Indenture, dated as of April 5, 2019, by and among the Company, as Issuer, Broadcom Technologies Inc., Broadcom Corporation and Broadcom Cayman Finance Limited (the "2019 Guarantors"), and Wilmington Trust, National Association, as trustee.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 5, 2019	
4.18	Form of 3.625% Senior Notes due 2024 (included in Exhibit 4.17).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 5, 2019	
4.19	Form of 4.250% Senior Notes due 2026 (included in Exhibit 4.17).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 5, 2019	
4.20	Form of 4.750% Senior Notes due 2029 (included in Exhibit 4.17).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 5, 2019	
4.21	Indenture, dated as of April 9, 2020, by and among the Company, as Issuer, Broadcom Technologies Inc. and Broadcom Corporation (the "2020 Guarantors"), and Wilmington Trust, National Association, as trustee.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 9, 2020	
4.22	Form of 4.700% Senior Notes due 2025 (included in Exhibit 4.21).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 9, 2020	
4.23	Form of 5.000% Senior Notes due 2030 (included in Exhibit 4.21).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 9, 2020	
4.24	Indenture, dated as of May 8, 2020, by and among the Company, as Issuer, the 2020 Guarantors, and Wilmington Trust, National Association, as trustee.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 8, 2020	
4.25	Form of 2.250% Senior Notes due 2023 (included in Exhibit 4.24).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 8, 2020	

[Table of Contents](#)

Exhibit Number	Description	Incorporated by Reference		Filed Herewith
		Form (File No.)	Filing Date	
4.26	Form of 3.150% Senior Notes due 2025 (included in Exhibit 4.24).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 8, 2020	
4.27	Form of 4.150% Senior Notes due 2030 (included in Exhibit 4.24).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 8, 2020	
4.28	Form of 4.300% Senior Notes due 2032 (included in Exhibit 4.24).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 8, 2020	
4.29	Indenture, dated as of May 21, 2020, by and among the Company, the 2020 Guarantors and Wilmington Trust, National Association, as trustee.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 21, 2020	
4.30	Form of 3.459% Senior Notes due 2026 (included in Exhibit 4.29).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 21, 2020	
4.31	Form of 4.110% Senior Notes due 2028 (included in Exhibit 4.29).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 21, 2020	
4.32	Indenture, dated as of January 19, 2021, by and among the Company, the 2020 Guarantors and Wilmington Trust, National Association, as Trustee.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	January 19, 2021	
4.33	Form of 1.950% Senior Notes due 2028 (included in Exhibit 4.32).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	January 19, 2021	
4.34	Form of 2.450% Senior Notes due 2031 (included in Exhibit 4.32).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	January 19, 2021	
4.35	Form of 2.600% Senior Notes due 2033 (included in Exhibit 4.32).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	January 19, 2021	
4.36	Form of 3.500% Senior Notes due 2041 (included in Exhibit 4.32).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	January 19, 2021	
4.37	Form of 3.750% Senior Notes due 2051 (included in Exhibit 4.32).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	January 19, 2021	
4.38	Registration Rights Agreement, dated as of January 19, 2021, by and among the Company, the 2020 Guarantors and Morgan Stanley & Co. LLC, BNP Paribas Securities Corp., RBC Capital Markets, LLC, SMBC Nikko Securities America, Inc., and Truist Securities, Inc., as representatives of the several initial purchasers of the January 2021 Senior Notes.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	January 19, 2021	

Table of Contents

Exhibit Number	Description	Incorporated by Reference		Filed Herewith
		Form (File No.)	Filing Date	
4.39	<u>Indenture, dated as of March 31, 2021, by and between the Company and Wilmington Trust, National Association, as Trustee.</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	March 31, 2021	
4.40	<u>Form of 3.419% Senior Notes due 2033 (included in Exhibit 4.39).</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	March 31, 2021	
4.41	<u>Form of 3.469% Senior Notes due 2034 (included in Exhibit 4.39).</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	March 31, 2021	
4.42	<u>Registration Rights Agreement, dated as of March 31, 2021, by and among the Company and BofA Securities, Inc. and HSBC Securities (USA) Inc., as dealer-managers in connection with the March 2021 Exchange Offer.</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	March 31, 2021	
4.43	<u>Indenture, dated as of September 30, 2021, by and between the Company and Wilmington Trust, National Association, as Trustee.</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	September 30, 2021	
4.44	<u>Form of 3.137% Senior Notes due 2035 (included in Exhibit 4.43).</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	September 30, 2021	
4.45	<u>Form of 3.187% Senior Notes due 2036 (included in Exhibit 4.43).</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	September 30, 2021	
4.46	<u>Registration Rights Agreement, dated as of September 30, 2021, by and among the Company and BNP Paribas Securities Corp., J.P. Morgan Securities LLC and TD Securities (USA) LLC, as dealer-managers in connection with the September 2021 exchange offer.</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	September 30, 2021	
4.47	<u>Indenture, dated April 14, 2022, between the Company and Wilmington Trust, National Association, as trustee.</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 15, 2022	
4.48	<u>Form of 4.00% Senior Notes due 2029 (included in Exhibit 4.47).</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 15, 2022	
4.49	<u>Form of 4.15% Senior Notes due 2032 (included in Exhibit 4.47).</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 15, 2022	
4.50	<u>Registration Rights Agreement, dated as of April 14, 2022, between the Company and BofA Securities, Inc., HSBC Securities (USA) Inc., and RBC Capital Markets, LLC, as representatives of the several initial purchasers of the April 2022 Senior Notes.</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 15, 2022	
4.51	<u>Indenture, dated April 18, 2022, between the Company and Wilmington Trust, National Association, as trustee.</u>	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 18, 2022	

Table of Contents

Exhibit Number	Description	Incorporated by Reference		Filed Herewith
		Form (File No.)	Filing Date	
4.52	Form of 4.926% Senior Notes due 2037 (included in Exhibit 4.51).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 18, 2022	
4.53	Registration Rights Agreement, dated April 18, 2022, between the Company and Barclays Capital Inc., BBVA Securities Inc., BNP Paribas Securities Corp. and J.P. Morgan Securities LLC, as dealer-managers in connection with the April 2022 Exchange Offer.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	April 18, 2022	
10.1	Form of Indemnification and Advancement Agreement (effective April 4, 2018).	Broadcom Inc. Current Report on Form 8-K12B (Commission File No. 001-38449)	April 4, 2018	
10.2	Credit Agreement, dated as of May 7, 2019, among Broadcom Inc., the lenders and other parties party thereto, and Bank of America, N.A., as Administrative Agent.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 7, 2019	
10.3	Credit Agreement, dated as of November 4, 2019, among Broadcom Inc., the lenders and other parties party thereto, and Bank of America, N.A., as Administrative Agent.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	November 4, 2019	
10.4	Credit Agreement, dated as of January 19, 2021, among the Company, the lenders and other parties party thereto, and Bank of America, N.A., as Administrative Agent.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	January 19, 2021	
10.5	Lease Agreement dated August 10, 2017 between Five Point Office Venture I, LLC and Broadcom Corporation.	Broadcom Limited Annual Report on Form 10-K (Commission File No. 001-37690)	December 21, 2017	
10.6	First Amendment to Lease Agreement by and between Five Point Office Venture 1, LLC and Broadcom Corporation.	Broadcom Inc. Annual Report on Form 10-K (Commission File No. 001-38449)	December 18, 2020	
10.7*	Settlement and Patent License and Non-Assert Agreement by and between Qualcomm Incorporated and Broadcom Corporation.	Broadcom Corporation Current Report on Form 8-K/A (Commission File No. 000-23993)	July 23, 2009	
10.8+	Avago Technologies Limited 2009 Equity Incentive Award Plan.	Amendment No. 5 to Avago Technologies Limited Registration Statement on Form S-1 (Commission File No. 333-153127)	July 27, 2009	
10.9+	Broadcom Inc. Employee Stock Purchase Plan (as amended and restated on April 1, 2019).	Broadcom Inc. Definitive Proxy Statement on Schedule 14A (Commission File No. 001-38449)	February 19, 2019	
10.10+	LSI Corporation 2003 Equity Incentive Plan, as amended.	Avago Technologies Limited Registration Statement on Form S-8 (Commission File No. 333-195741)	May 6, 2014	
10.11+	Amendment to the LSI Corporation 2003 Equity Incentive Plan (effective February 1, 2016).	Broadcom Limited Annual Report on Form 10-K (Commission File No. 001-37690)	December 23, 2016	

[Table of Contents](#)

Exhibit Number	Description	Incorporated by Reference		
		Form (File No.)	Filing Date	Filed Herewith
10.12+	Amendment to the LSI Corporation 2003 Equity Incentive Plan (effective April 4, 2018).	Broadcom Inc. Current Report on Form 8-K12B (Commission File No. 001-38449)	April 4, 2018	
10.13+	Broadcom Inc. 2012 Stock Incentive Plan (as amended and restated on April 5, 2021).	Broadcom Inc. Quarterly Report on Form 10-Q (Commission File No. 001-38449)	June 11, 2021	
10.14+	Form of Annual Bonus Plan for Executive Employees.	Broadcom Limited Annual Report on Form 10-K (Commission File No. 001-37690)	December 23, 2016	
10.15+	Form of Option Agreement under Avago Technologies Limited 2009 Equity Incentive Plan.	Amendment No. 5 to Avago Technologies Limited Registration Statement on Form S-1 (Commission File No. 333-153127)	July 27, 2009	
10.16+	Form of Restricted Stock Unit Agreement (Sell to Cover) Under Avago Technologies Limited 2009 Equity Incentive Award Plan (effective December 5, 2017).	Broadcom Limited Annual Report on Form 10-K (Commission File No. 001-37690)	December 21, 2017	
10.17+	Form of Agreement for Multi-Year Equity Award of Restricted Stock Unit Award under the Avago Technologies Limited 2009 Equity Incentive Award Plan).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	December 6, 2018	
10.18+	Form of Performance Share Unit Agreement (Relative TSR) under Avago Technologies Limited 2009 Equity Incentive Plan (effective March 13, 2018).	Broadcom Limited Quarterly Report on Form 10-Q (Commission File No. 001-37690)	March 15, 2018	
10.19+	Form of Agreement for Multi-Year Equity Award of Performance Stock Units under the Avago Technologies Limited 2009 Equity Incentive Award Plan).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	December 6, 2018	
10.20+	Form of Restricted Stock Unit Award Agreement under LSI Corporation 2003 Equity Incentive Plan, as amended (effective December 8, 2020).	Broadcom Inc. Annual Report on Form 10-K (Commission File No. 001-38449)	December 18, 2020	
10.21+	Form of Performance Stock Unit Agreement (Relative TSR) under LSI Corporation 2003 Equity Incentive Plan, as amended (effective December 8, 2020).	Broadcom Inc. Annual Report on Form 10-K (Commission File No. 001-38449)	December 18, 2020	
10.22+	Form of Restricted Stock Unit Award Agreement under Broadcom Corporation 2012 Stock Incentive Plan (effective December 5, 2017).	Broadcom Limited Annual Report on Form 10-K (Commission File No. 001-37690)	December 21, 2017	
10.23+	Form of Restricted Stock Unit Award Agreement under Broadcom Inc. 2012 Stock Incentive Plan (effective April 5, 2021).	Broadcom Inc. Quarterly Report on Form 10-Q (Commission File No. 001-38449)	June 11, 2021	
10.24+	Form of Performance Share Unit Agreement (Relative TSR) under Broadcom Corporation 2012 Stock Incentive Plan (effective March 15, 2018).	Broadcom Limited Quarterly Report on Form 10-Q (Commission File No. 001-37690)	March 15, 2018	
10.25+	Form of Performance Stock Unit Award Agreement under the Broadcom Inc. 2012 Stock Incentive Plan (effective April 5, 2021).	Broadcom Inc. Quarterly Report on Form 10-Q (Commission File No. 001-38449)	June 11, 2021	

Table of Contents

Exhibit Number	Description	Incorporated by Reference		Filed Herewith
		Form (File No.)	Filing Date	
10.26+	Form of Performance Stock Unit Award Agreement (Price Contingency) under Broadcom Inc. 2012 Stock Incentive Plan.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	November 2, 2022	
10.27+	Performance Stock Unit Award Agreement, dated April 5, 2021, between Broadcom Inc. and Hock E. Tan.	Broadcom Inc. Quarterly Report on Form 10-Q (Commission File No. 001-38449)	June 11, 2021	
10.28+	Policy on Acceleration of Executive Staff Equity Awards in the Event of Permanent Disability (as amended June 2, 2021).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	June 3, 2021	
10.29+	Policy on Acceleration of Equity Awards in the Event of Death (as amended June 2, 2021).	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	June 3, 2021	
10.30+	Amended and Restated Severance Benefits Agreement, dated December 10, 2020, between Broadcom Inc. and Hock E. Tan.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	December 10, 2020	
10.31+	Amended and Restated Severance Benefits Agreement, dated December 10, 2020, between Broadcom Inc. and Thomas H. Krause, Jr.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	December 10, 2020	
10.32+	Amended and Restated Severance Benefits Agreement, dated December 10, 2020, between Broadcom Inc. and Charlie B. Kawwas.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	December 10, 2020	
10.33+	Severance Benefits Agreement, dated September 26, 2017, between Broadcom Limited and Mark Brazeal.	Broadcom Inc. Quarterly Report on Form 10-Q (Commission File No. 001-38449)	June 16, 2018	
10.34+	Severance Benefits Agreement, dated December 10, 2020, between Broadcom Inc. and Kirsten M. Spears.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	December 10, 2020	
21.1	List of Subsidiaries.			X
23.1	Consent of PricewaterhouseCoopers LLP, independent registered public accounting firm.			X
24.1	Power of Attorney (see signature page to this Form 10-K).			X
31.1	Certification of Principal Executive Officer of Broadcom Inc. Pursuant to Rule 13a-14 of the Securities Exchange Act of 1934, As Adopted Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002.			X
31.2	Certification of Principal Financial Officer of Broadcom Inc. Pursuant to Rule 13a-14 of the Securities Exchange Act of 1934, As Adopted Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002.			X
32.1	Certification of Principal Executive Officer of Broadcom Inc. Pursuant to 18 U.S.C. Section 1350, As Adopted Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002.			X

Table of Contents

Exhibit Number	Description	Incorporated by Reference		Filed Herewith
		Form (File No.)	Filing Date	
32.2	Certification of Principal Financial Officer of Broadcom Inc. Pursuant to 18 U.S.C. Section 1350, As Adopted Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002.			X
99.1	Voting Agreement, dated as of May 26, 2022, by and among Broadcom Inc., Michael S. Dell and Susan Lieberman Dell Separate Property Trust.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 26, 2022	
99.2	Voting Agreement, dated as of May 26, 2022, by and among Broadcom Inc., Silver Lake Partners IV, L.P., Silver Lake Technology Investors IV, L.P., Silver Lake Partners V DE (AIV), L.P., Silver Lake Technology Investors V, L.P., SL SPV-2, L.P. and Silver Lake Group, L.L.C.	Broadcom Inc. Current Report on Form 8-K (Commission File No. 001-38449)	May 26, 2022	
101.INS	XBRL Instance Document - the instance document does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document.			X
101.SCH	XBRL Schema Document			X
101.CAL	XBRL Calculation Linkbase Document			X
101.DEF	XBRL Definition Linkbase Document			X
101.LAB	XBRL Labels Linkbase Document			X
101.PRE	XBRL Presentation Linkbase Document			X
104	Cover Page Interactive Data File - the cover page interactive data file does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document.			X

Notes:

- + Indicates a management contract or compensatory plan or arrangement.
- # Schedules have been omitted pursuant to Item 601(b)(2) of Regulation S-K. Broadcom Inc. hereby undertakes to furnish supplementally copies of any omitted schedules upon request by the SEC.
- * Certain information omitted pursuant to a request for confidential treatment filed with the SEC.

ITEM 16. FORM 10-K SUMMARY

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

BROADCOM INC.

By: /s/ Hock E. Tan

Name: Hock E. Tan

Title: President and Chief Executive Officer

Date: December 16, 2022

POWER OF ATTORNEY

Each person whose individual signature appears below hereby authorizes and appoints Hock E. Tan, Kirsten M. Spears and Mark D. Brazeal, and each of them, with full power of substitution and resubstitution and full power to act without the other, as his or her true and lawful attorney-in-fact and agent to act in his or her name, place and stead and to execute in the name and on behalf of each person, individually and in each capacity stated below, and to file any and all amendments to this Annual Report on Form 10-K, and to file the same, with all exhibits thereto, and other documents in connection therewith, with the Securities and Exchange Commission, granting unto said attorneys-in-fact and agents, and each of them, full power and authority to do and perform each and every act and thing, ratifying and confirming all that said attorneys-in-fact and agents or any of them or their or his substitute or substitutes may lawfully do or cause to be done by virtue thereof.

[Table of Contents](#)

Pursuant to the requirements of the Securities Exchange Act of 1934, this Annual Report on Form 10-K has been signed by the following persons on behalf of the Registrant in the capacities indicated and on the dates indicated.

<u>Signature</u>	<u>Title</u>	<u>Date</u>
<u>/s/ Hock E. Tan</u> Hock E. Tan	President and Chief Executive Officer and Director (Principal Executive Officer)	December 16, 2022
<u>/s/ Kirsten M. Spears</u> Kirsten M. Spears	Chief Financial Officer (Principal Financial Officer and Principal Accounting Officer)	December 16, 2022
<u>/s/ Henry Samuelli</u> Henry Samuelli	Chairman of the Board of Directors	December 16, 2022
<u>/s/ Eddy W. Hartenstein</u> Eddy W. Hartenstein	Lead Independent Director	December 16, 2022
<u>/s/ Diane M. Bryant</u> Diane M. Bryant	Director	December 16, 2022
<u>/s/ Gayla J. Delly</u> Gayla J. Delly	Director	December 16, 2022
<u>/s/ Raul F. Fernandez</u> Raul F. Fernandez	Director	December 16, 2022
<u>/s/ Check Kian Low</u> Check Kian Low	Director	December 16, 2022
<u>/s/ Justine F. Page</u> Justine F. Page	Director	December 16, 2022
<u>/s/ Harry L. You</u> Harry L. You	Director	December 16, 2022